

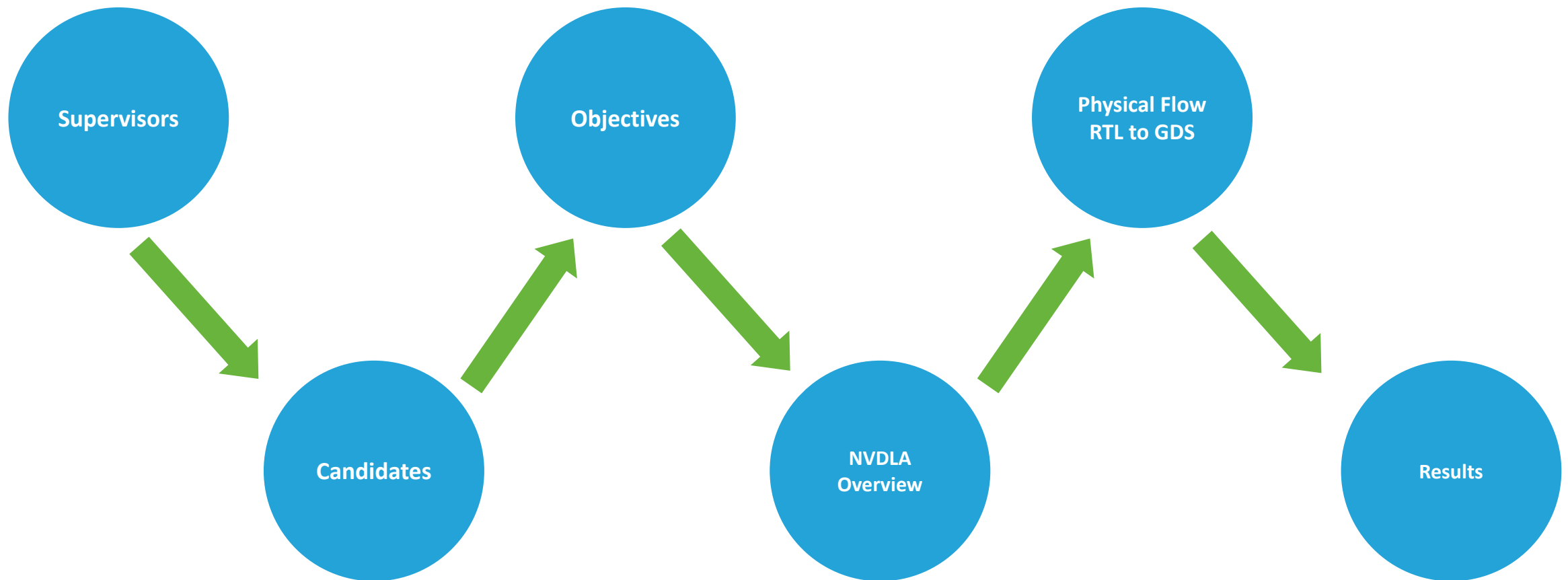


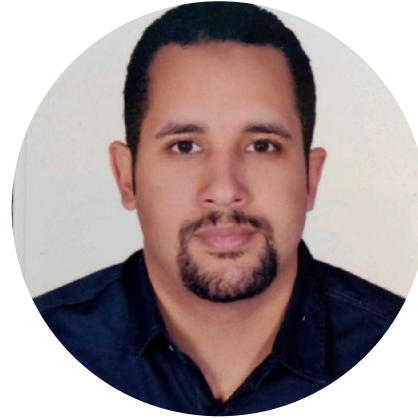
**NVIDIA®**

# NVIDIA Deep Learning Accelerator Physical Implementation

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Eng. Islam Samir



Eng. Mariam



Eng. Bassem



Eng. Momen

# Candidates

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Eng.Ahmed El-Gammal



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Eng.Momen Ashraf



Eng.Mohamed Emam



Eng.Asmaa Anwar



Eng.Ahmed Abdullah



## 1. Large-Scale Design

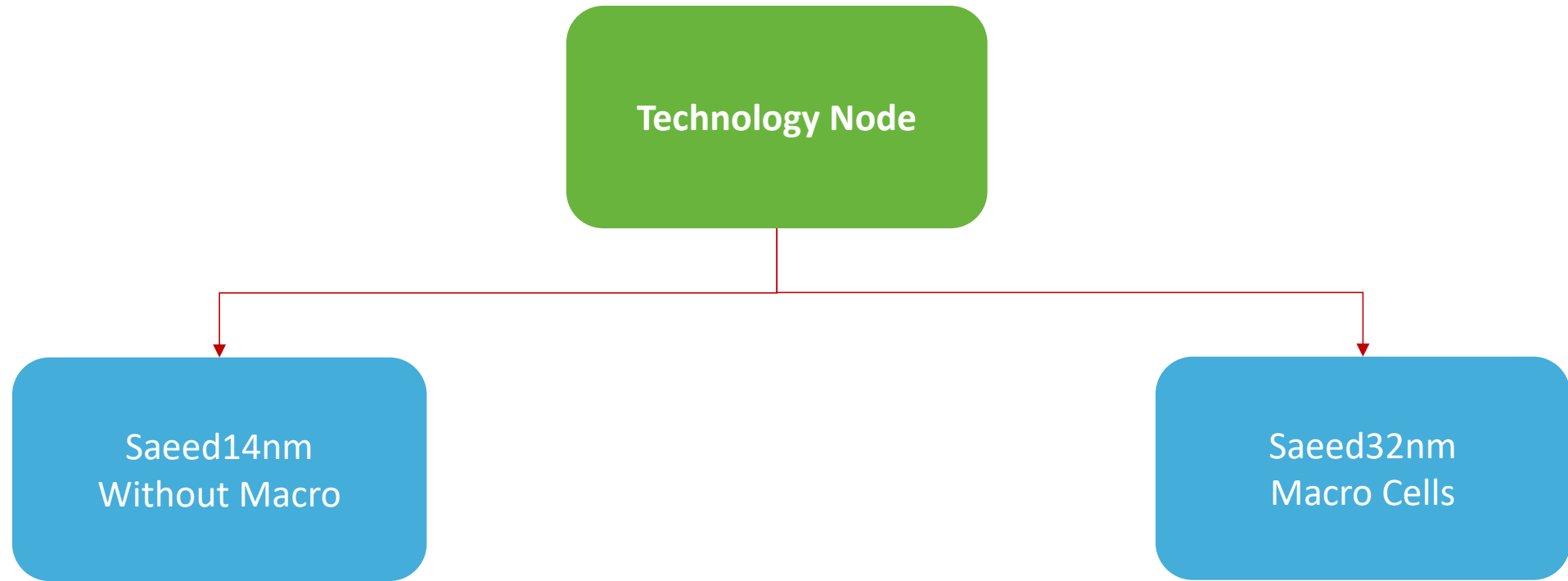
NVDLA provides a large and complex design, allowing us to identify and overcome the challenges that typically arise during a complete RTL-to-GDS physical implementation flow.

## 2. Macro Integration

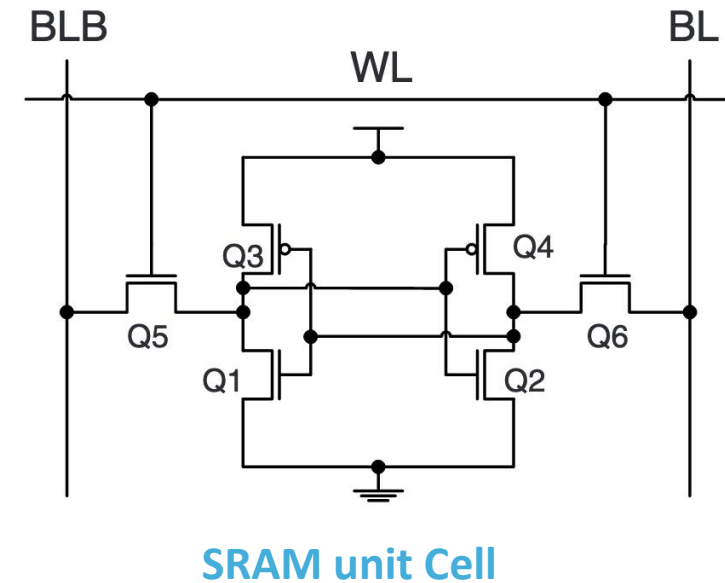
The design contains multiple macros, providing practical experience in macro placement, power connectivity, routing, congestion, timing, and other macro-related physical implementation challenges.

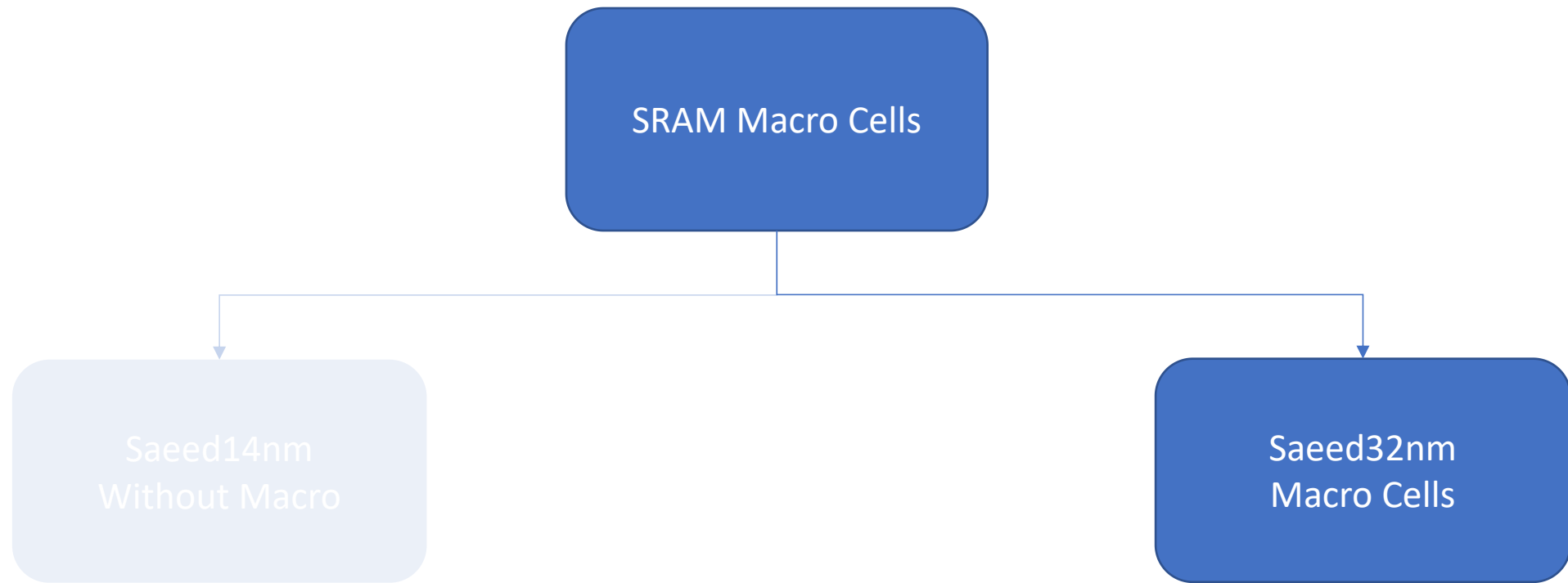
## 3. Real-World Design:

NVDLA is an actual **Deep Learning Accelerator** architecture designed for AI workloads and has been integrated into real-world applications.

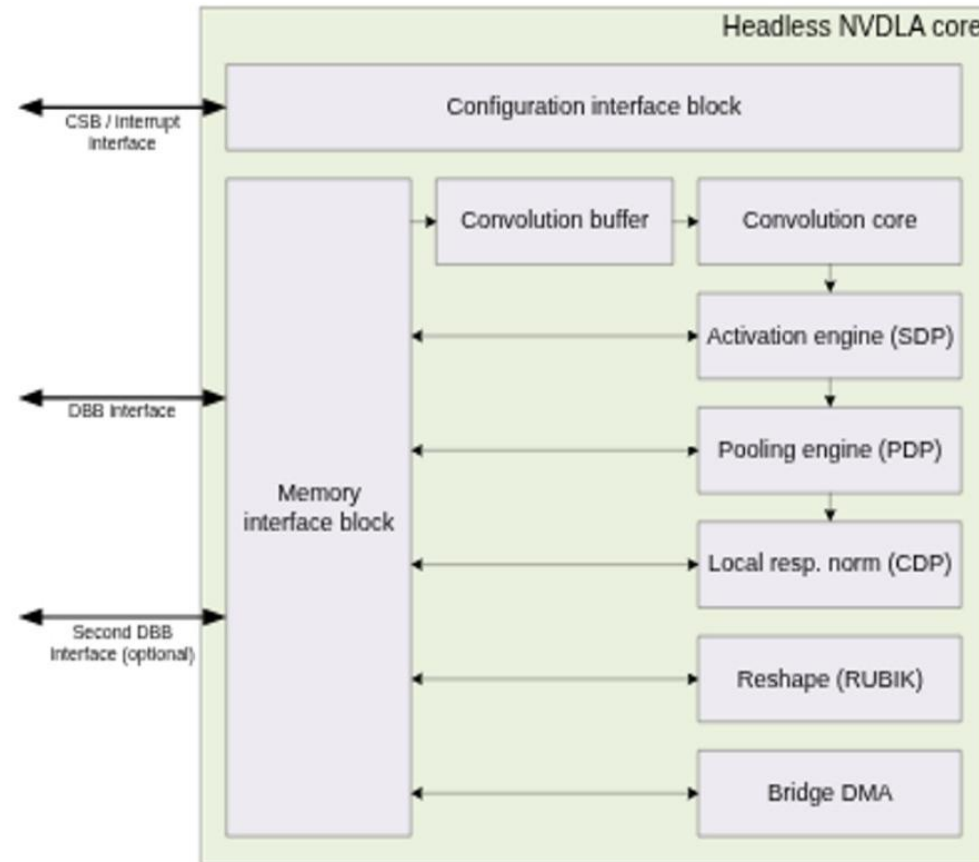


- **Open-source SRAM** compiler framework that generates layout, netlists, timing/power models, and P&R abstracts.
- Works with both open-source PDKs (e.g., SkyWater 130, GF 180) and commercial PDKs if you configure the tech rules

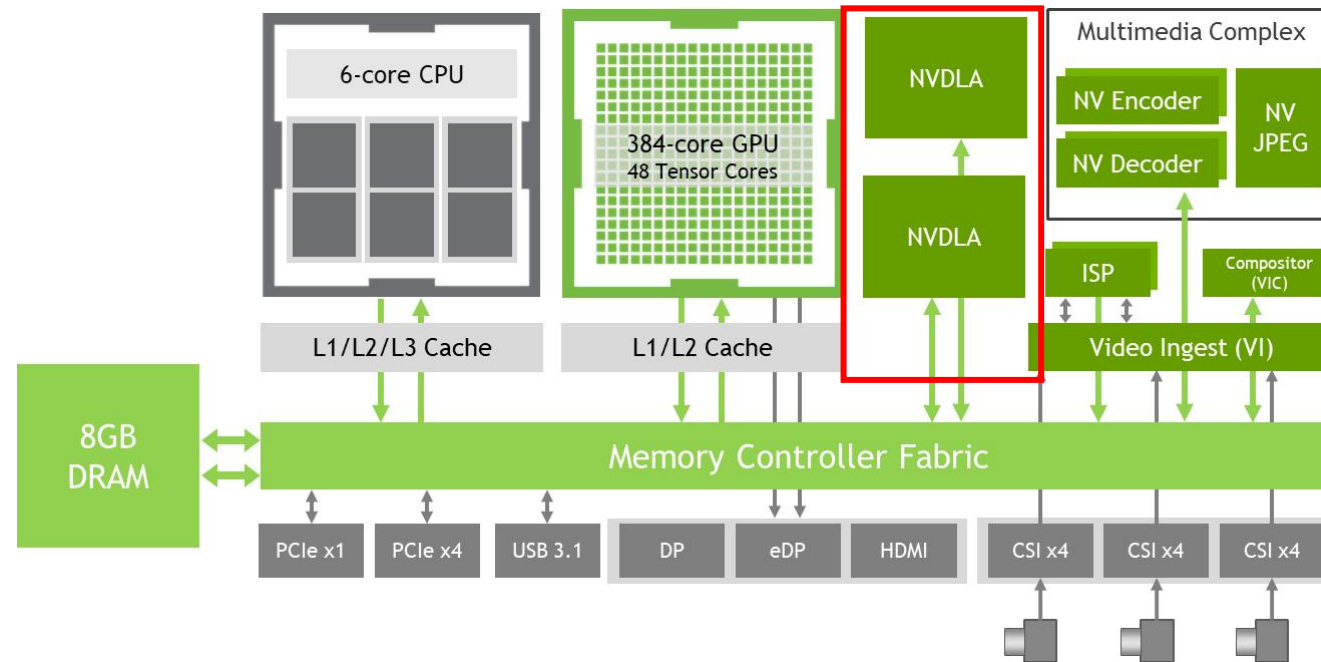




- **NVDLA** is an open-source, scalable AI accelerator architecture designed for neural network inference. It provides dedicated hardware to accelerate key operations such as **convolution, pooling, activation, and data reshaping**, enabling faster and more efficient AI data processing.

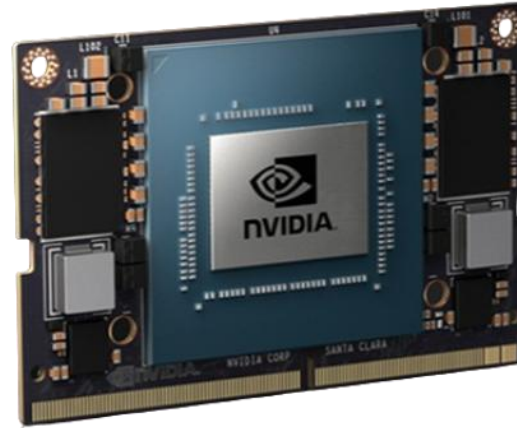


- **NVDLA** hardware-accelerated operations reduce the computational workload on the CPU/GPU and enable faster and more efficient processing of AI workloads.



*NVIDIA Jetson Xavier*

**NVIDIA Jetson AGX Xavier** integrates NVDLA as a dedicated deep-learning accelerator and is used in commercial and real-world applications, including drones, robotics, autonomous machines, and other AI-enabled systems



*NVIDIA Jetson Xavier*



*Drones*

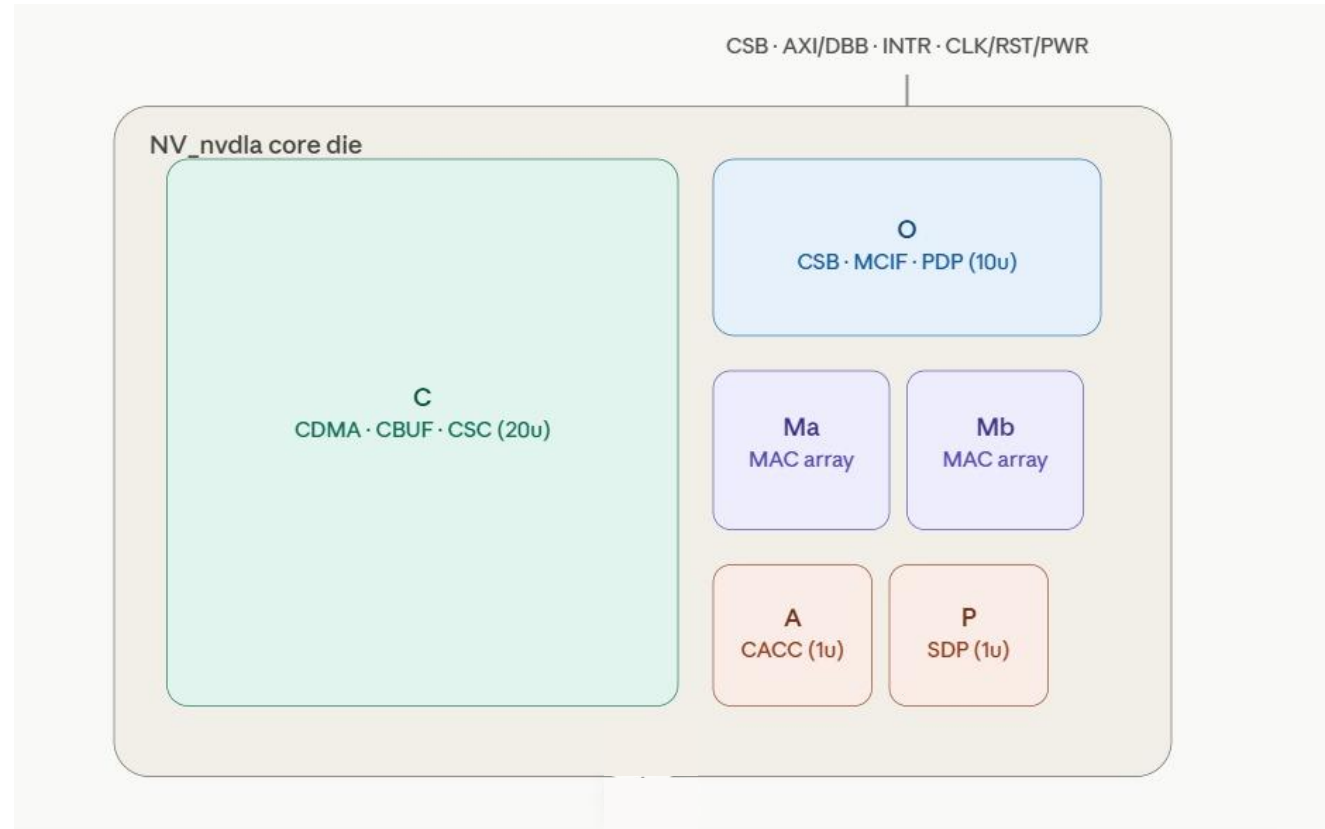


*Medical*

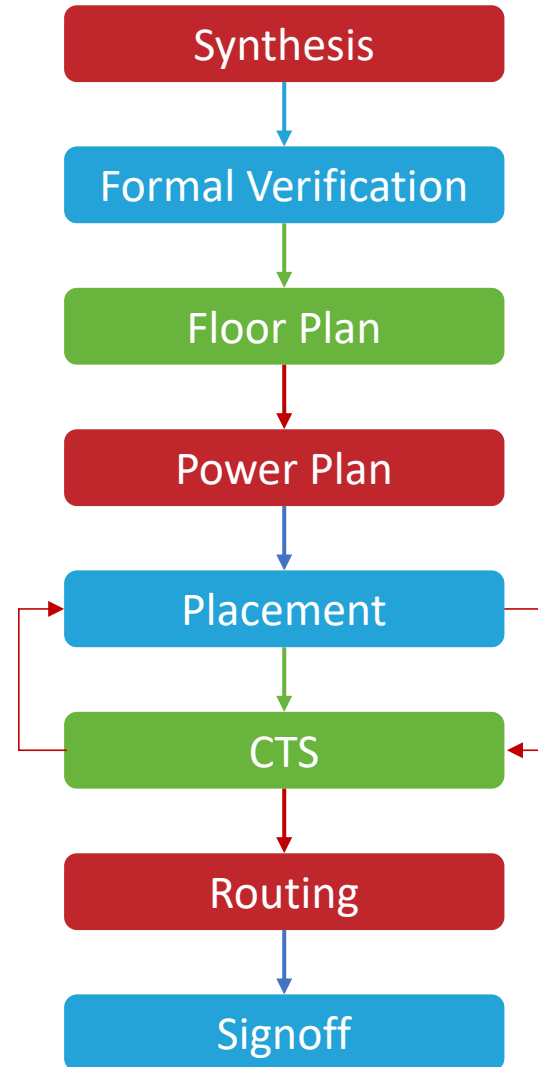


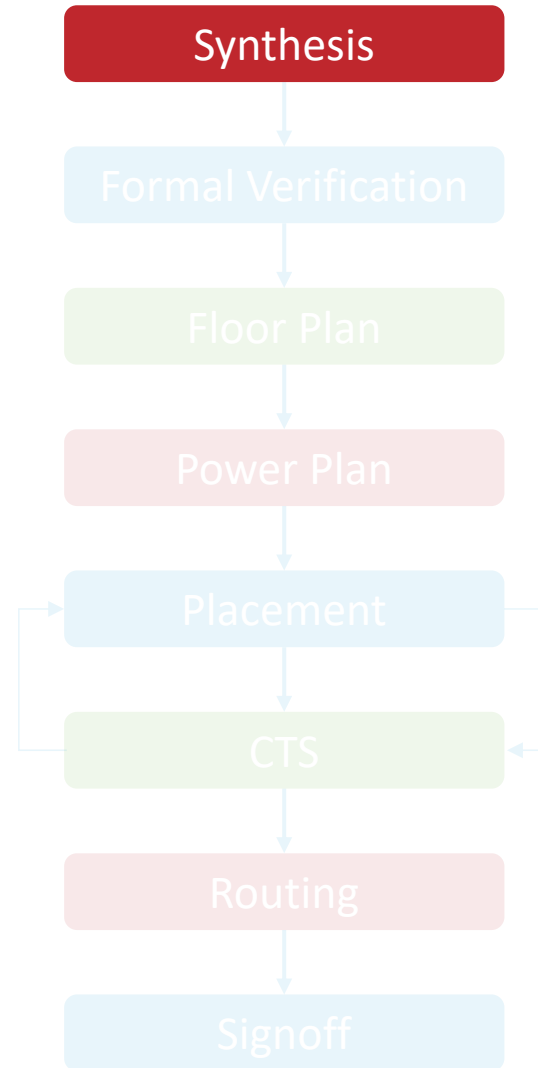
*Robotics*

The **NVDLA open-source** project is divided into **six partitions**, with each member responsible for implementing one partition through the complete RTL-to-GDS physical implementation flow.



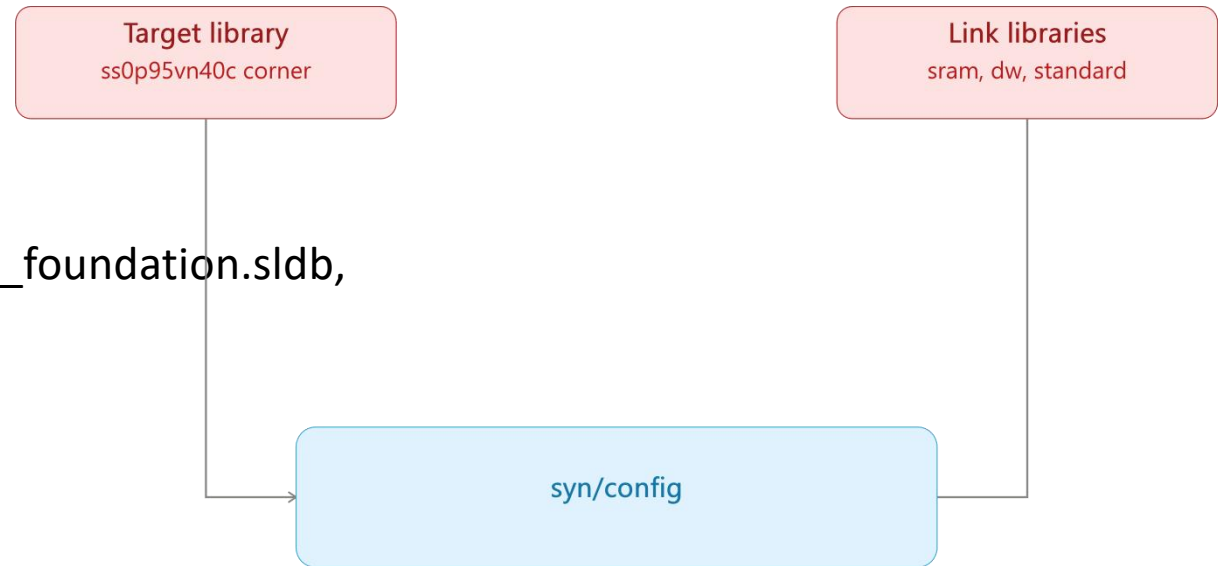






1. **Libs**
2. Memory mapping
3. Constraints
4. Formality
5. MMMC and library creation

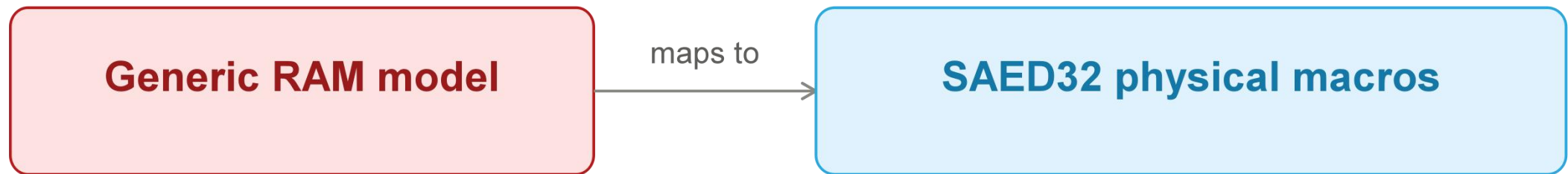
- **Target library** : saed32rvt\_ss0p95vn40c.db
- **Link libraries** : saed32sram\_ss0p95vn40c.db, dw\_foundation.sldb,  
standard.sldb



- Libs
- **Memory mapping**
- Constraints
- Formality
- MMMC and library creation

## Why we needed it ?

- The original open-source NVDLA RTL uses generic/behavioral RAM models (vmod/rams/synth/), which are vendor-agnostic wrappers. However, real ASIC manufacturing requires actual physical SRAM macro blocks from a foundry/library (Synopsys SAED32 32nm SRAMs).

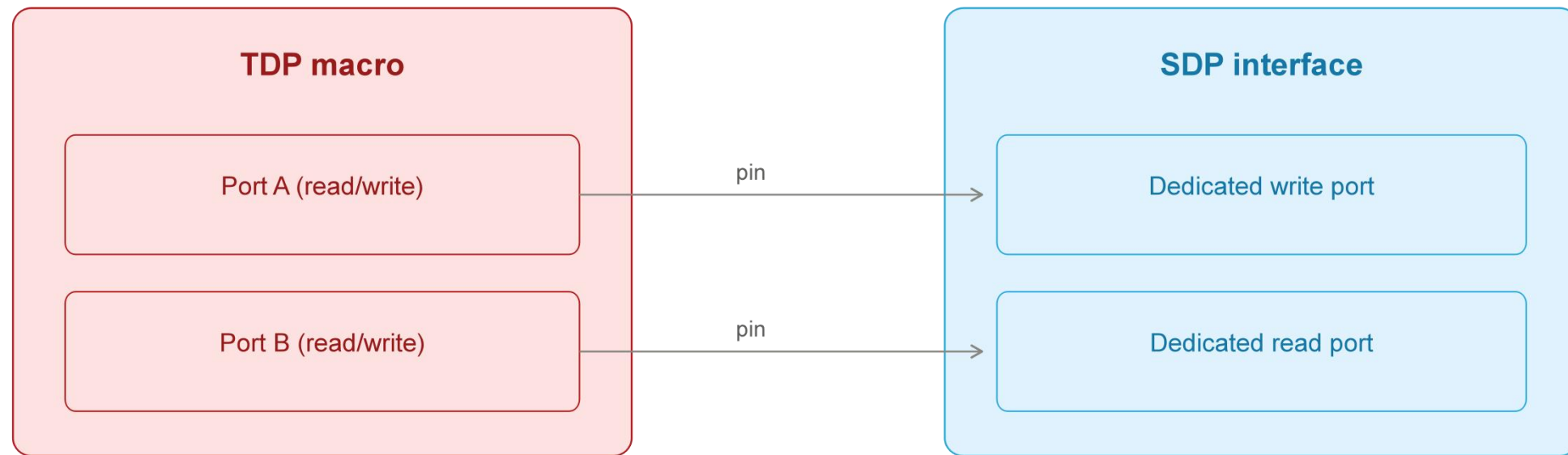


## Techniques used:

### Technique A: True Dual Port (TDP) → Simple Dual Port (SDP) Adaptation

The physical SAED32 PDK library only provides True Dual Port (SRAM2RW) macros. However, NVDLA functional memories operate as Simple Dual Port (1 Dedicated Write Port + 1 Dedicated Read Port).

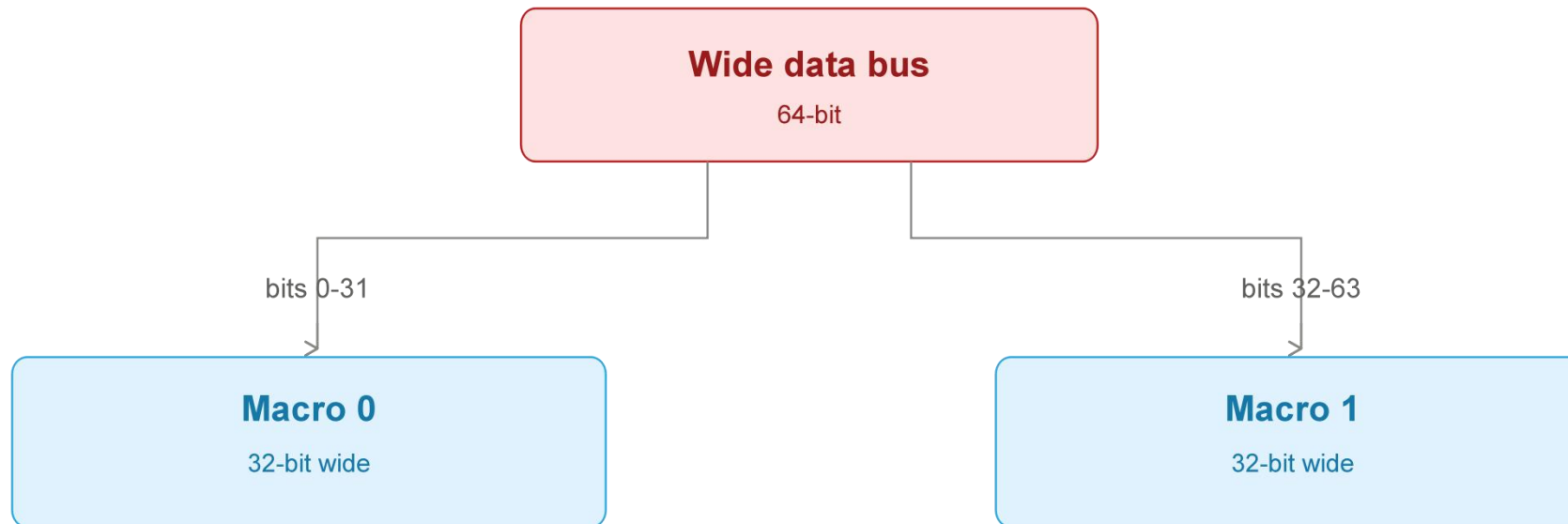
We convert the TDP macro interface into an SDP interface by statically pinning the ports



## Techniques used:

### Technique B: Width Slicing (Bit-Width Expansion)

When the required memory width exceeds the maximum width of a single SAED32 macro (32), the bus is systematically sliced across multiple parallel macros



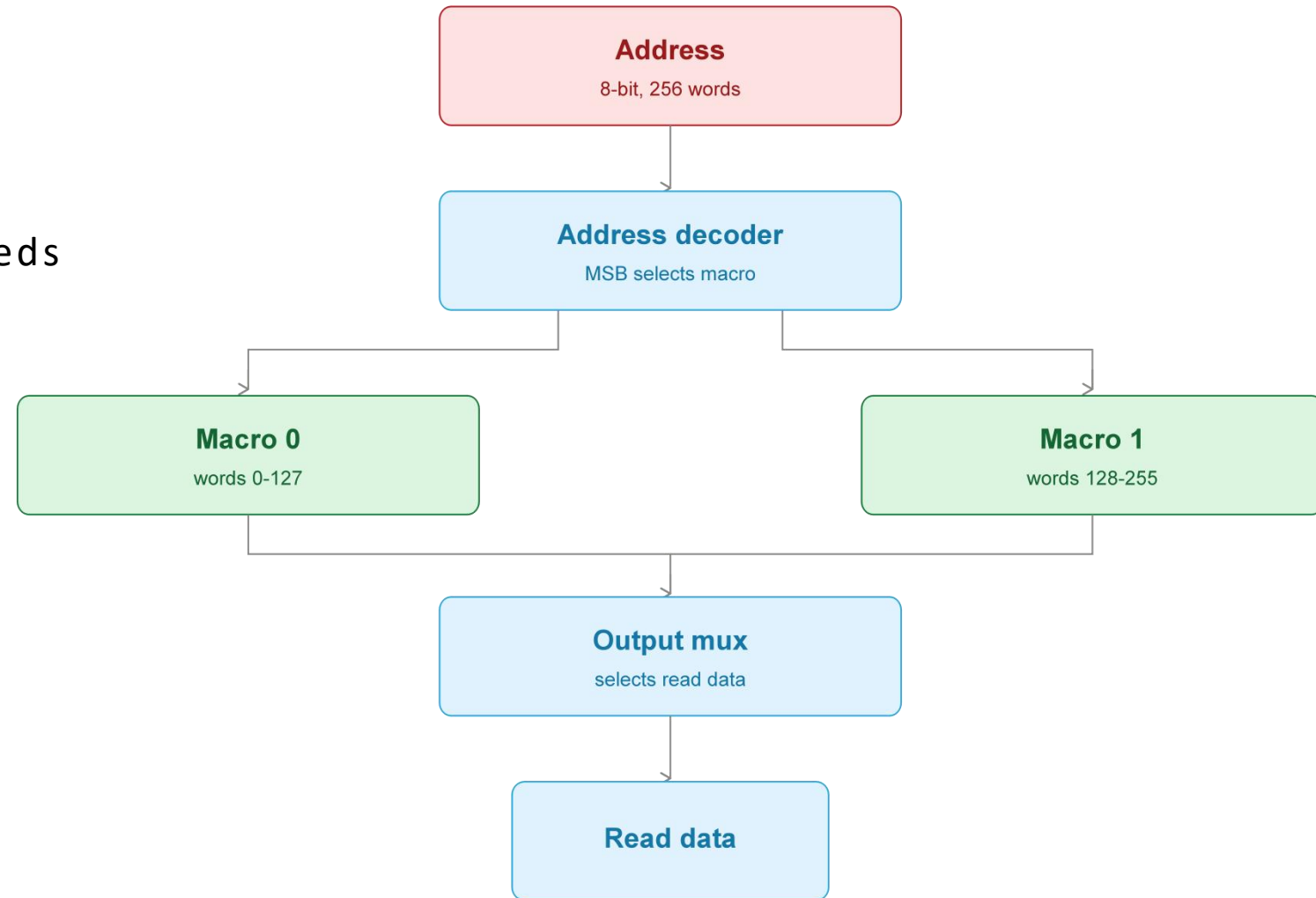


## Techniques used:

### Technique C: Depth Expansion & MSB

#### Address Decoding

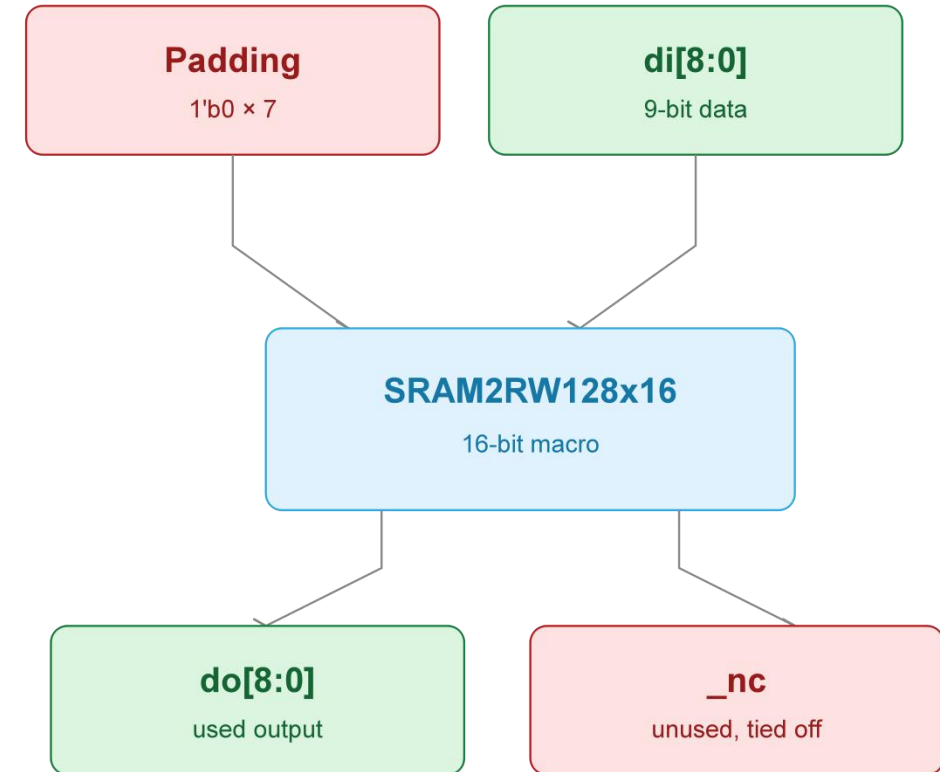
When the required memory depth exceeds standard macro depth (e.g., 256 words deep mapped using 128-word deep macros):



## Techniques used:

### Technique D: Data Padding & Unused Bit Tie-Offs

- **Input Padding:** If memory width is not a multiple of available macro width (e.g., 9-bit width into a 16-bit macro), upper input bits are padded with 1'b0
- **Output Tie-offs:** Unconnected output bits from the macro are tied to dummy \_nc (no-connect) wires to suppress synthesis lint warnings.

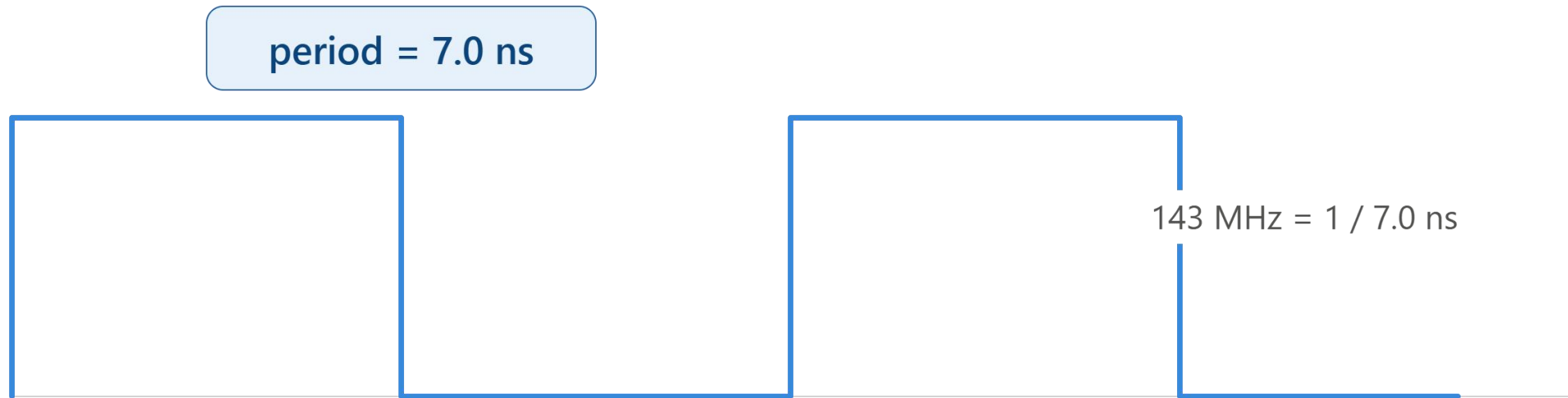


Zero-pad the extra input bits, tie off the extra output bits

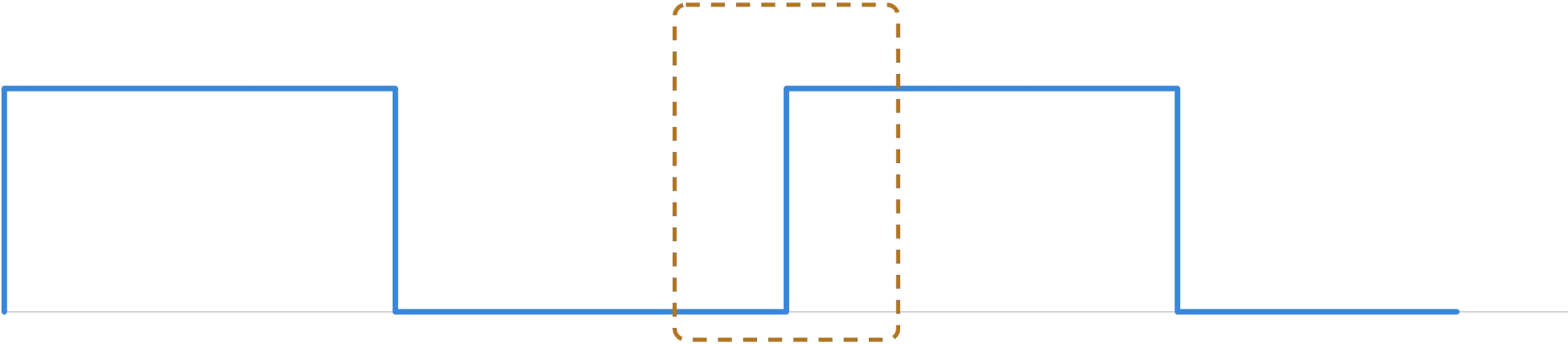
| #  | Behavioral RTL Module      | Depth × Width | Physical SAED32 SRAM Macro Instantiations |
|----|----------------------------|---------------|-------------------------------------------|
| 1  | nv_ram_rws_128x18_logic.v  | 128 × 18      | 1x SRAM2RW128x32                          |
| 2  | nv_ram_rws_16x256_logic.v  | 16 × 256      | 8x SRAM2RW16x32                           |
| 3  | nv_ram_rws_16x272_logic.v  | 16 × 272      | 8x SRAM2RW16x32 + 1x SRAM2RW16x16         |
| 4  | nv_ram_rws_16x64_logic.v   | 16 × 64       | 2x SRAM2RW16x32                           |
| 5  | nv_ram_rws_256x3_logic.v   | 256 × 3       | 2x SRAM2RW128x8                           |
| 6  | nv_ram_rws_256x64_logic.v  | 256 × 64      | 4x SRAM2RW128x32                          |
| 7  | nv_ram_rws_256x7_logic.v   | 256 × 7       | 2x SRAM2RW128x8                           |
| 8  | nv_ram_rwsp_128x11_logic.v | 128 × 11      | 1x SRAM2RW128x16                          |
| 9  | nv_ram_rwsp_128x6_logic.v  | 128 × 6       | 1x SRAM2RW128x8                           |
| 10 | nv_ram_rwsp_160x16_logic.v | 160 × 16      | 2x SRAM2RW128x16                          |
| 11 | nv_ram_rwsp_160x65_logic.v | 160 × 65      | 4x SRAM2RW128x32 + 2x SRAM2RW128x4        |
| 12 | nv_ram_rwsp_61x65_logic.v  | 61 × 65       | 2x SRAM2RW64x32 + 1x SRAM2RW64x4          |
| 13 | nv_ram_rwsp_80x14_logic.v  | 80 × 14       | 1x SRAM2RW128x16                          |
| 14 | nv_ram_rwsp_80x65_logic.v  | 80 × 65       | 2x SRAM2RW128x32 + 1x SRAM2RW128x4        |
| 15 | nv_ram_rwsp_8x65_logic.v   | 8 × 65        | 2x SRAM2RW16x32 + 1x SRAM2RW16x4          |
| 16 | nv_ram_rwst_256x8_logic.v  | 256 × 8       | 2x SRAM2RW128x8                           |
| 17 | nv_ram_rwthp_60x21_logic.v | 60 × 21       | 1x SRAM2RW64x32                           |
| 18 | nv_ram_rwthp_80x15_logic.v | 80 × 15       | 1x SRAM2RW128x16                          |
| 19 | nv_ram_rwthp_80x9_logic.v  | 80 × 9        | 1x SRAM2RW128x16                          |

- Libs
- Memory mapping
- **Constraints**
- Formality
- MMMC and library creation

| SDC Constraint | Generic / Raw NVDLA SDC | Our Enhanced SAED32 SDC |
|----------------|-------------------------|-------------------------|
| Clock Period   | Unconstrained           | <b>7.0 ns (143 MHz)</b> |

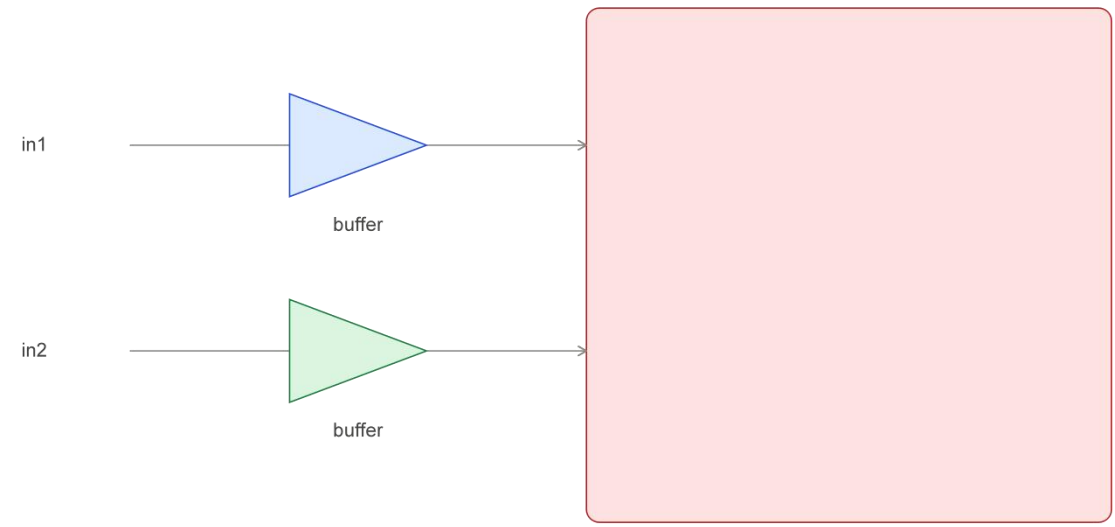


| SDC Constraint    | Generic / Raw NVDLA SDC     | Our Enhanced SAED32 SDC    |
|-------------------|-----------------------------|----------------------------|
| Clock Period      | Unconstrained / 16nm target | 7.0 ns (143 MHz)           |
| Clock Uncertainty | 0.0 ns                      | Setup: 0.3ns / Hold: 0.1ns |

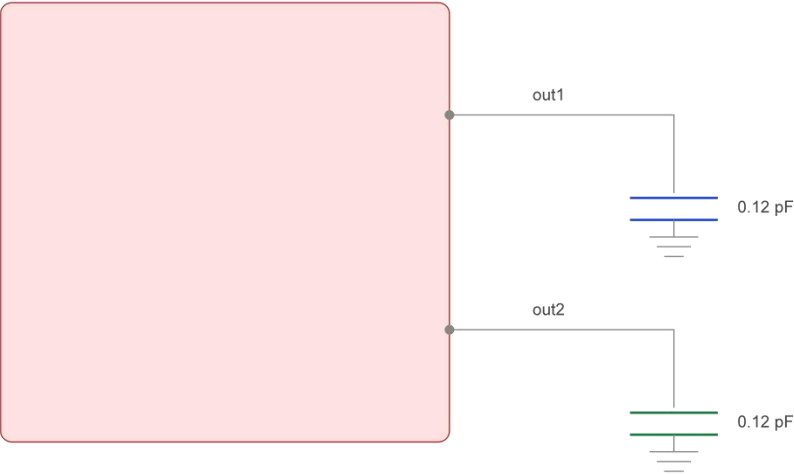


setup 0.3 ns ← —→ hold 0.1 ns

| SDC Constraint    | Generic / Raw NVDLA SDC     | Our Enhanced SAED32 SDC    |
|-------------------|-----------------------------|----------------------------|
| Clock Period      | Unconstrained / 16nm target | 7.0 ns (143 MHz)           |
| Clock Uncertainty | 0.0 ns                      | Setup: 0.3ns / Hold: 0.1ns |
| Input Drive       | Infinite / Ideal            | IBUFX4_RVT Buffer          |



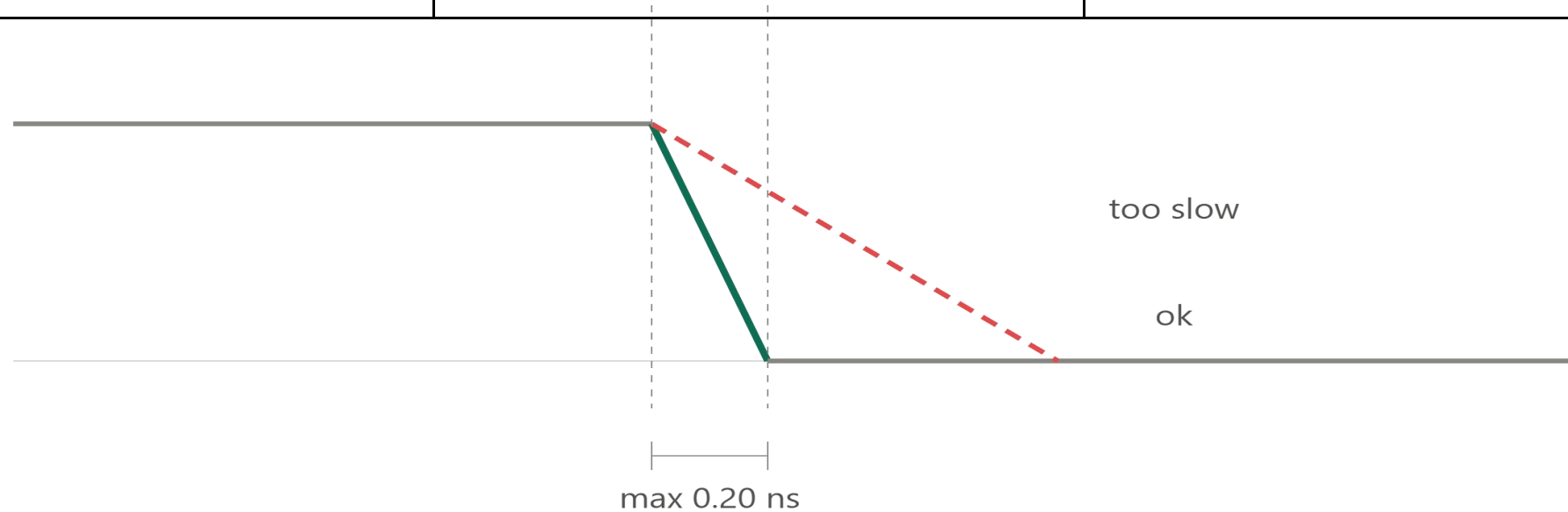
| SDC Constraint    | Generic / Raw NVDLA SDC     | Our Enhanced SAED32 SDC    |
|-------------------|-----------------------------|----------------------------|
| Clock Period      | Unconstrained / 16nm target | 7.0 ns (143 MHz)           |
| Clock Uncertainty | 0.0 ns                      | Setup: 0.3ns / Hold: 0.1ns |
| Input Drive       | Infinite / Ideal            | IBUFFX4_RVT Buffer         |
| Output Load       | 0.0 pF                      | 0.12 pF (120 fF)           |



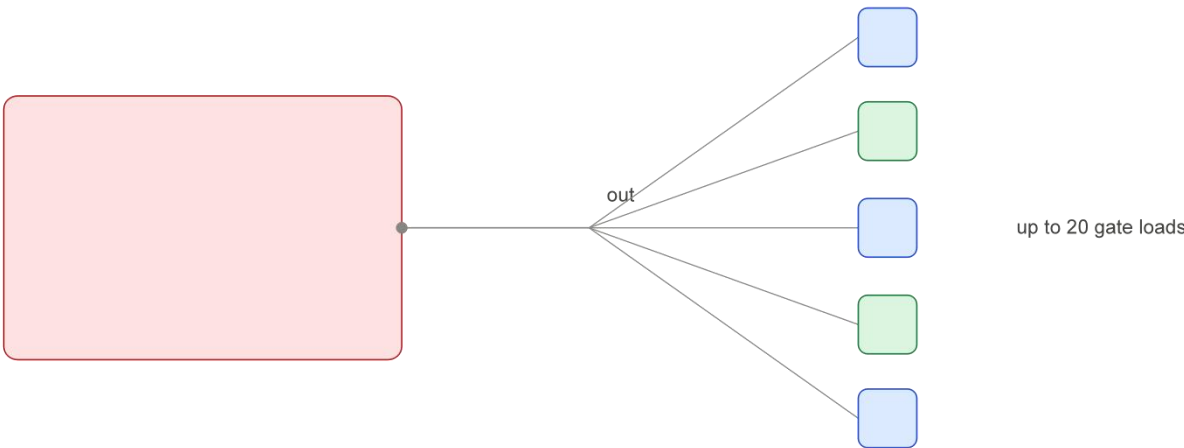


# Transitioning from Generic SDC to Realistic SAED32 Constraints

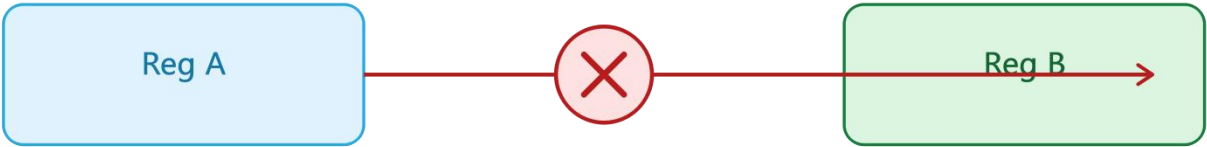
| SDC Constraint    | Generic / Raw NVDLA SDC     | Our Enhanced SAED32 SDC    |
|-------------------|-----------------------------|----------------------------|
| Clock Period      | Unconstrained / 16nm target | 7.0 ns (143 MHz)           |
| Clock Uncertainty | 0.0 ns                      | Setup: 0.3ns / Hold: 0.1ns |
| Input Drive       | Infinite / Ideal            | IBUFX4_RVT Buffer          |
| Output Load       | 0.0 pF                      | 0.12 pF (120 fF)           |
| Max Transition    | None                        | 0.20 ns (200ps)            |



| SDC Constraint    | Generic / Raw NVDLA SDC     | Our Enhanced SAED32 SDC    |
|-------------------|-----------------------------|----------------------------|
| Clock Period      | Unconstrained / 16nm target | 7.0 ns (143 MHz)           |
| Clock Uncertainty | 0.0 ns                      | Setup: 0.3ns / Hold: 0.1ns |
| Input Drive       | Infinite / Ideal            | IBUFX4_RVT Buffer          |
| Output Load       | 0.0 pF                      | 0.12 pF (120 fF)           |
| Max Transition    | None                        | 0.20 ns (200ps)            |
| Max Fanout        | None                        | 20                         |



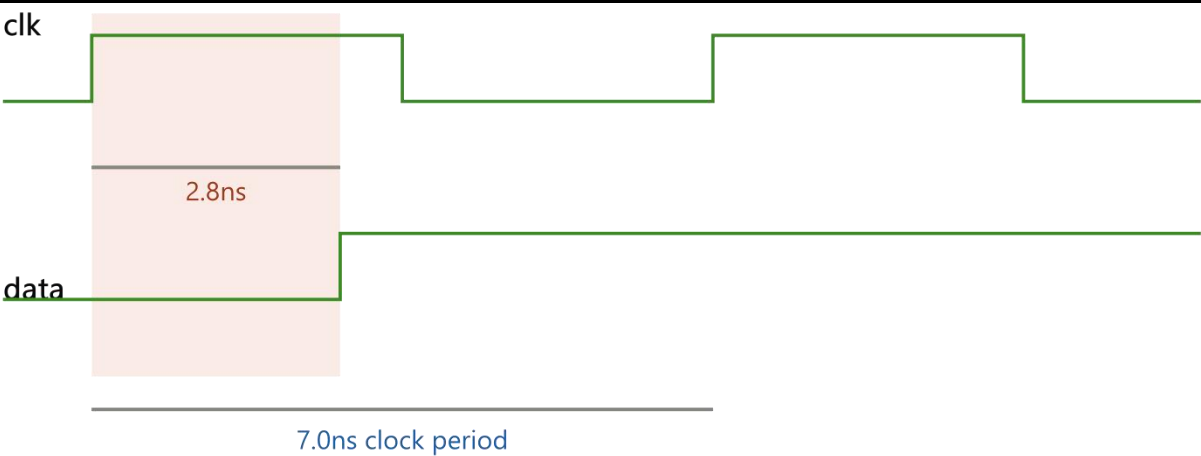
| SDC Constraint    | Generic / Raw NVDLA SDC     | Our Enhanced SAED32 SDC    |
|-------------------|-----------------------------|----------------------------|
| Clock Period      | Unconstrained / 16nm target | 7.0 ns (143 MHz)           |
| Clock Uncertainty | 0.0 ns                      | Setup: 0.3ns / Hold: 0.1ns |
| Input Drive       | Infinite / Ideal            | IBUFX4_RVT Buffer          |
| Output Load       | 0.0 pF                      | 0.12 pF (120 fF)           |
| Max Transition    | None                        | 0.20 ns (200ps)            |
| Max Fanout        | None                        | 20                         |
| RAM Power Paths   | Unconstrained               | set_false_path             |



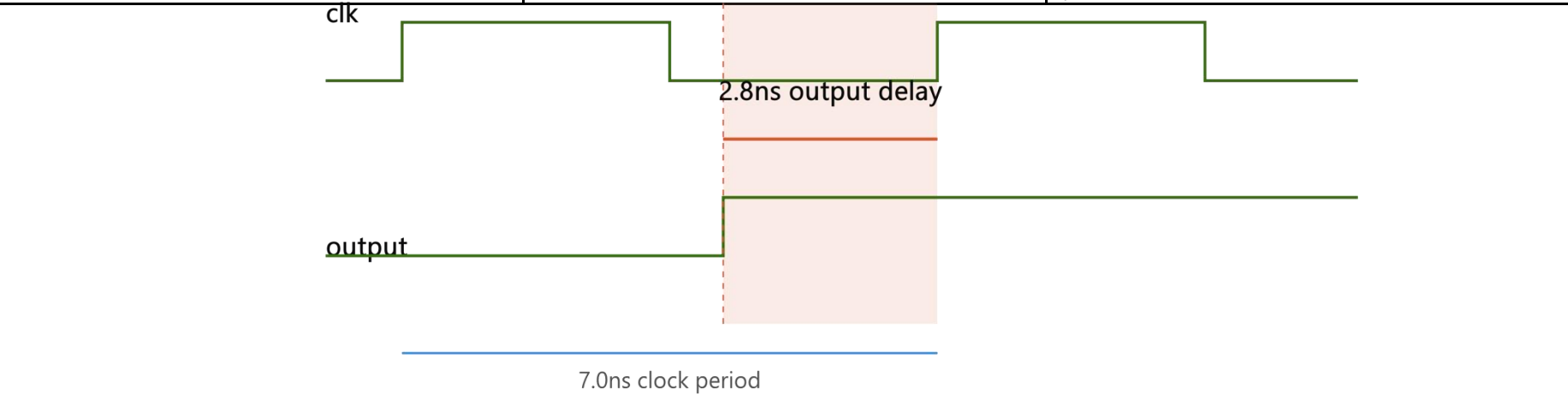
false path — excluded from timing checks

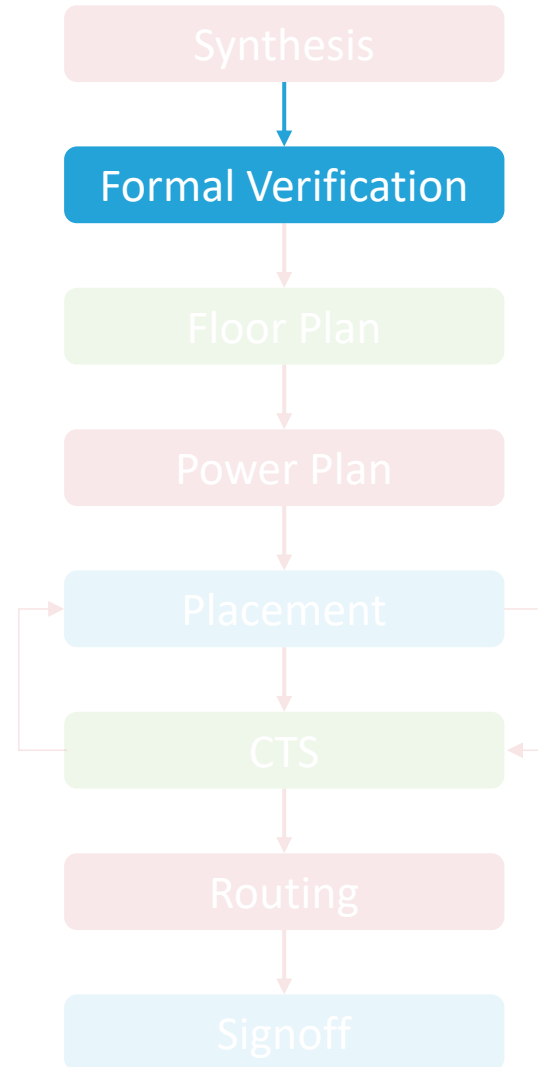
# Transitioning from Generic SDC to Realistic SAED32 Constraints

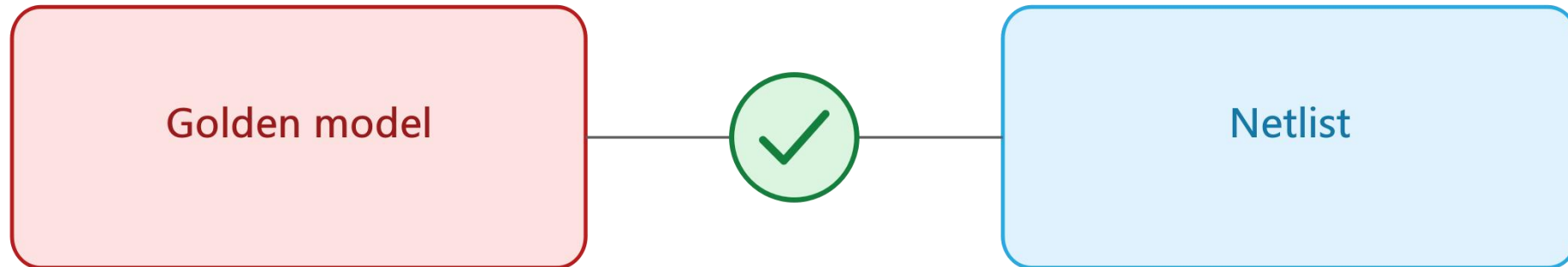
| SDC Constraint     | Generic / Raw NVDLA SDC     | Our Enhanced SAED32 SDC             |
|--------------------|-----------------------------|-------------------------------------|
| Clock Period       | Unconstrained / 16nm target | 7.0 ns (143 MHz)                    |
| Clock Uncertainty  | 0.0 ns                      | Setup: 0.3ns / Hold: 0.1ns          |
| Input Drive        | Infinite / Ideal            | IBUFFX4_RVT Buffer                  |
| Output Load        | 0.0 pF                      | 0.12 pF (120 fF)                    |
| Max Transition     | None                        | 0.20 ns (200ps)                     |
| Max Fanout         | None                        | 20                                  |
| RAM Power Paths    | Unconstrained               | set_false_path                      |
| <i>Input Delay</i> | <i>None</i>                 | <i>2.8 ns (40% of clock period)</i> |



| SDC Constraint    | Generic / Raw NVDLA SDC     | Our Enhanced SAED32 SDC      |
|-------------------|-----------------------------|------------------------------|
| Clock Period      | Unconstrained / 16nm target | 7.0 ns (143 MHz)             |
| Clock Uncertainty | 0.0 ns                      | Setup: 0.3ns / Hold: 0.1ns   |
| Input Drive       | Infinite / Ideal            | IBUFX4_RVT Buffer            |
| Output Load       | 0.0 pF                      | 0.12 pF (120 fF)             |
| Max Transition    | None                        | 0.20 ns (200ps)              |
| Max Fanout        | None                        | 20                           |
| RAM Power Paths   | Unconstrained               | set_false_path               |
| Input Delay       | None                        | 2.8 ns (40% of clock period) |
| Output Delay      | None                        | 2.8 ns (40% of clock period) |

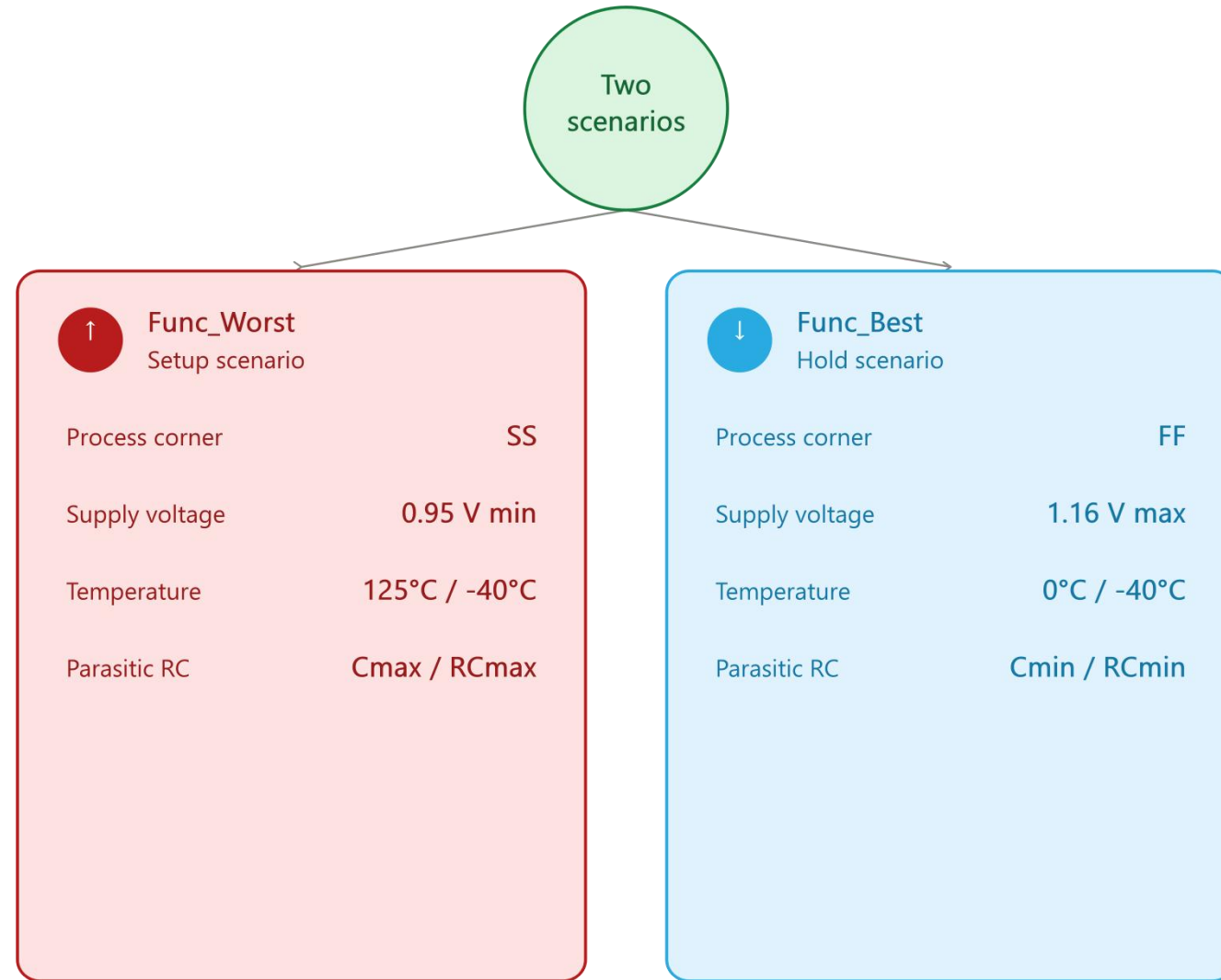


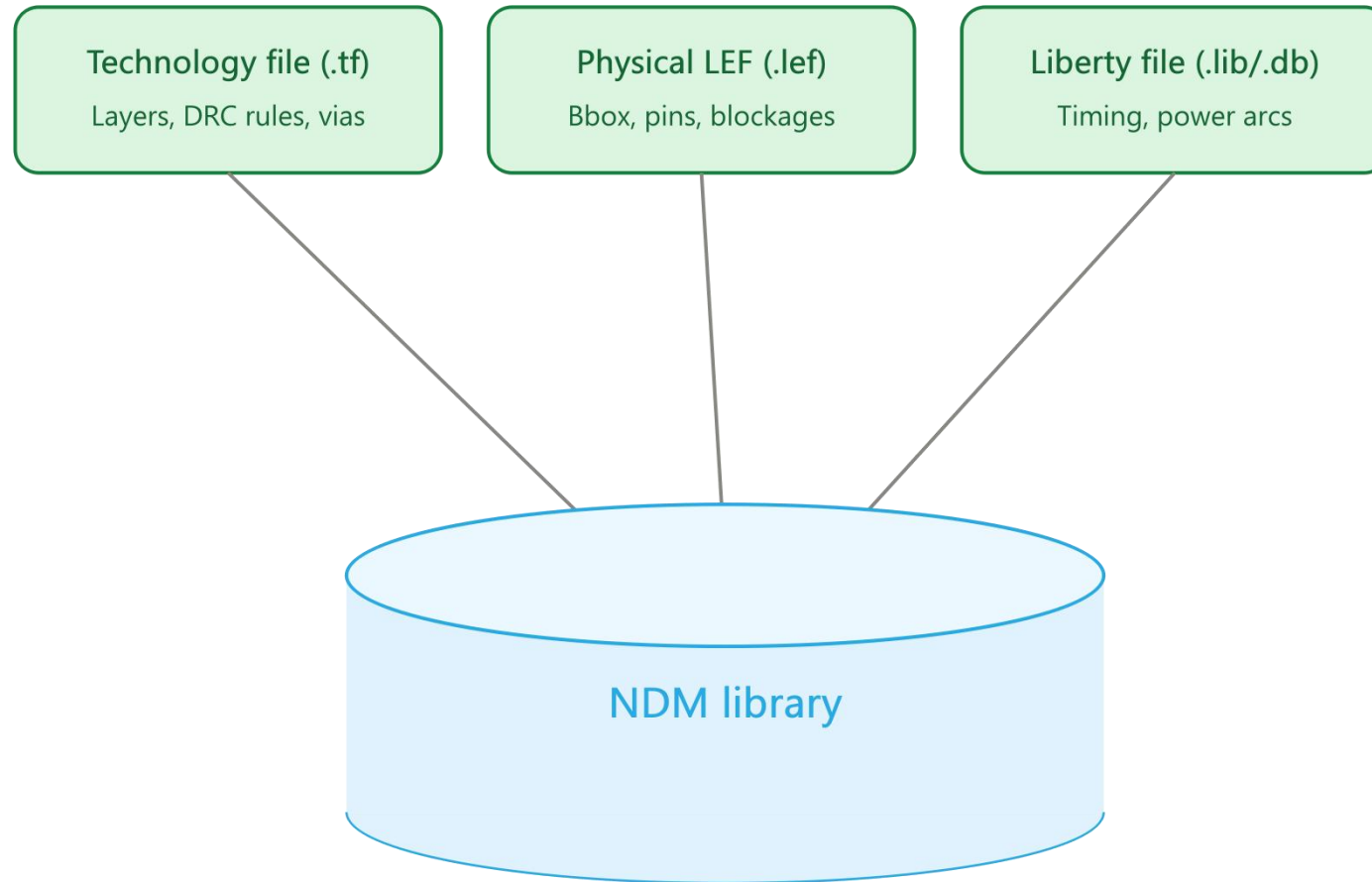


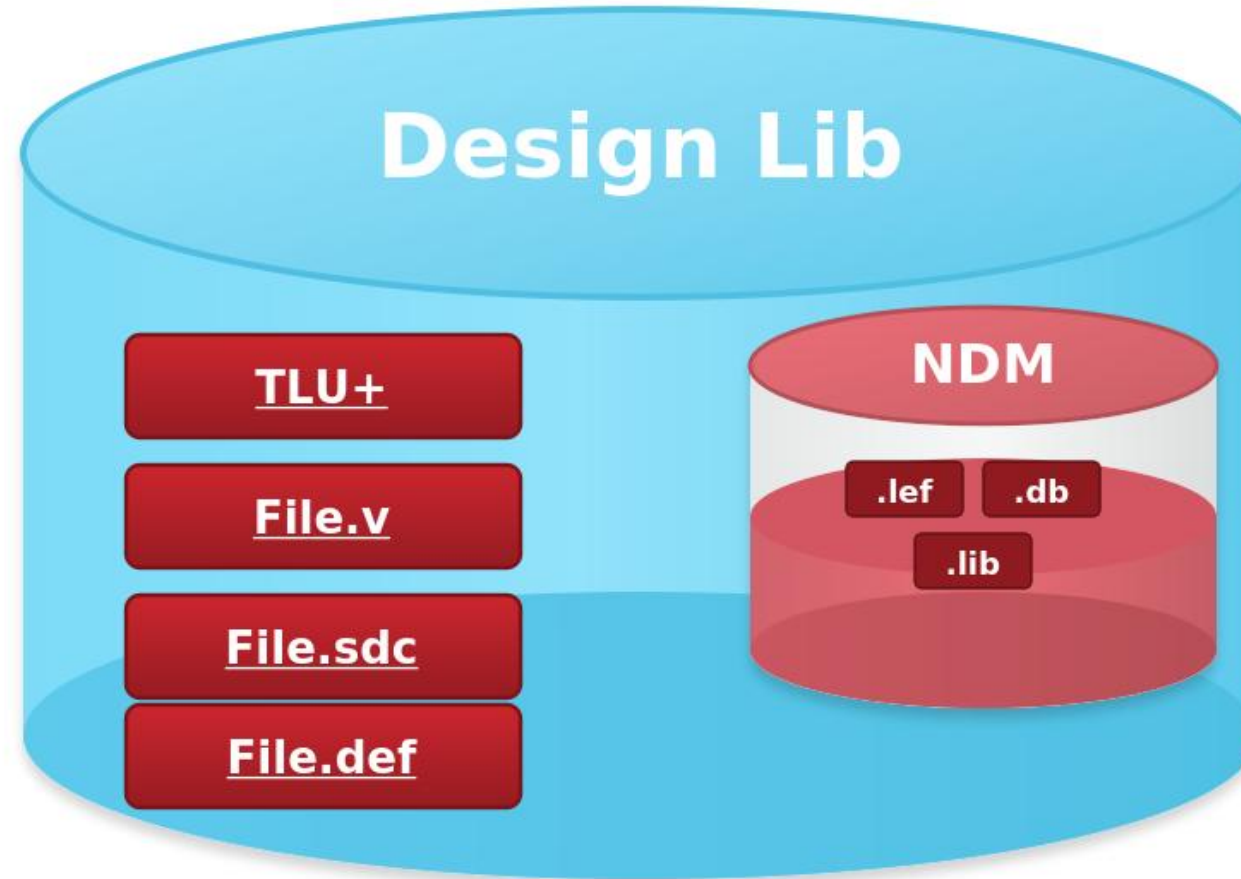


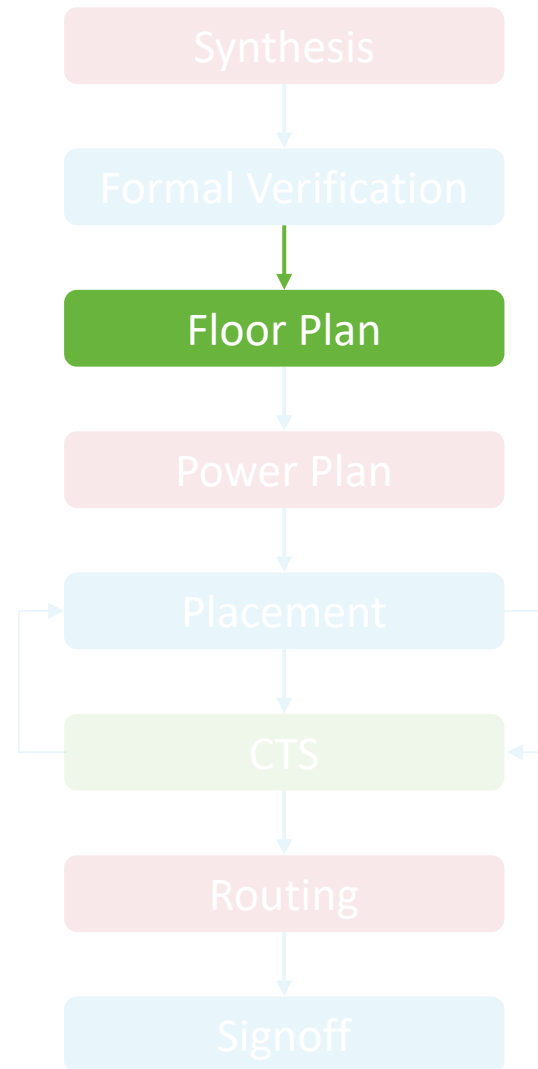
- Libs
- Memory mapping
- Constraints
- Formality
- **MMMC and library creation**











1. Initialize Floor Planning
2. I/O Ports Placed
3. Macro Placed
4. Add Blockage
5. Optimized Floorplan

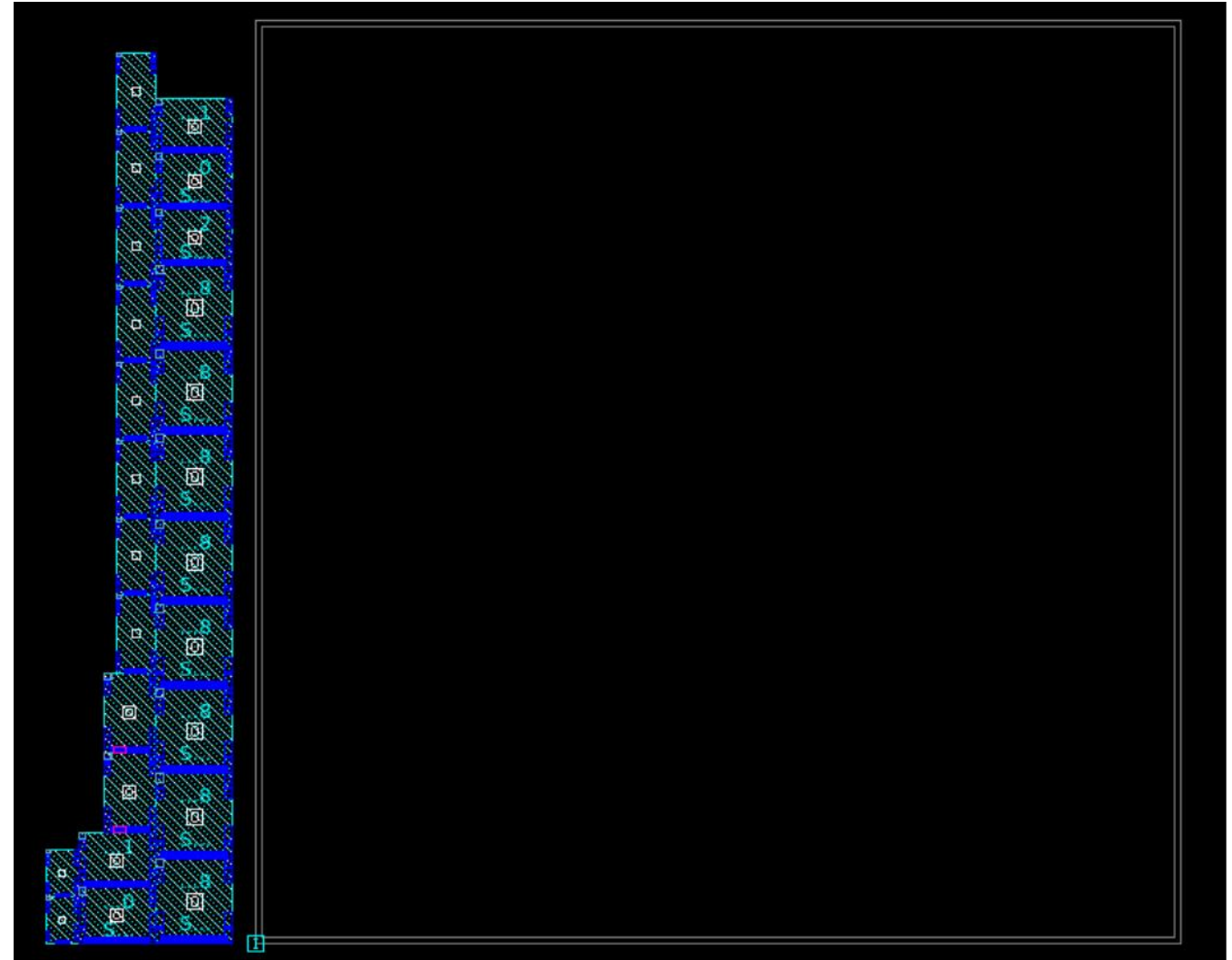
- Define the `core_utilization` = 0.35

Area

```
-----  
Combinational Area:      277126.2447  
Noncombinational Area:  255764.1743  
Buf/Inv Area:           25268.5217  
Total Buffer Area:       6092.0861  
Total Inverter Area:     19176.4355  
Macro/Black Box Area:   316260.4834  
Net Area:               337046.1221  
-----
```

```
-----  
Cell Area:               849150.9024  
Design Area:            1186197.0245  
-----
```

- Utilization = Cell Area / core\_area
- core\_area = Cell Area / Utilization
- core\_area = 2.426 (mm<sup>2</sup>)



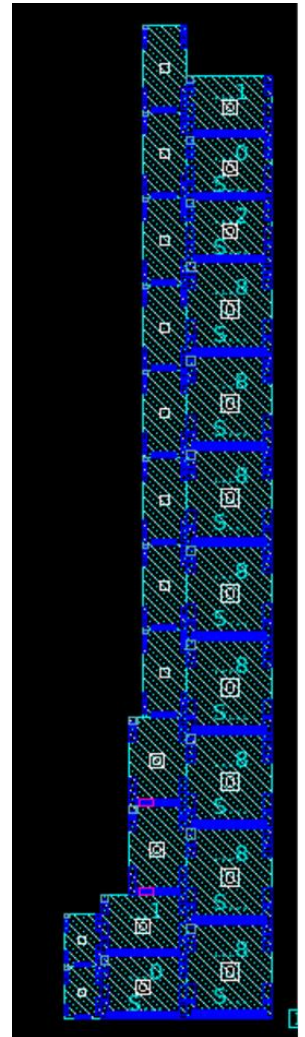
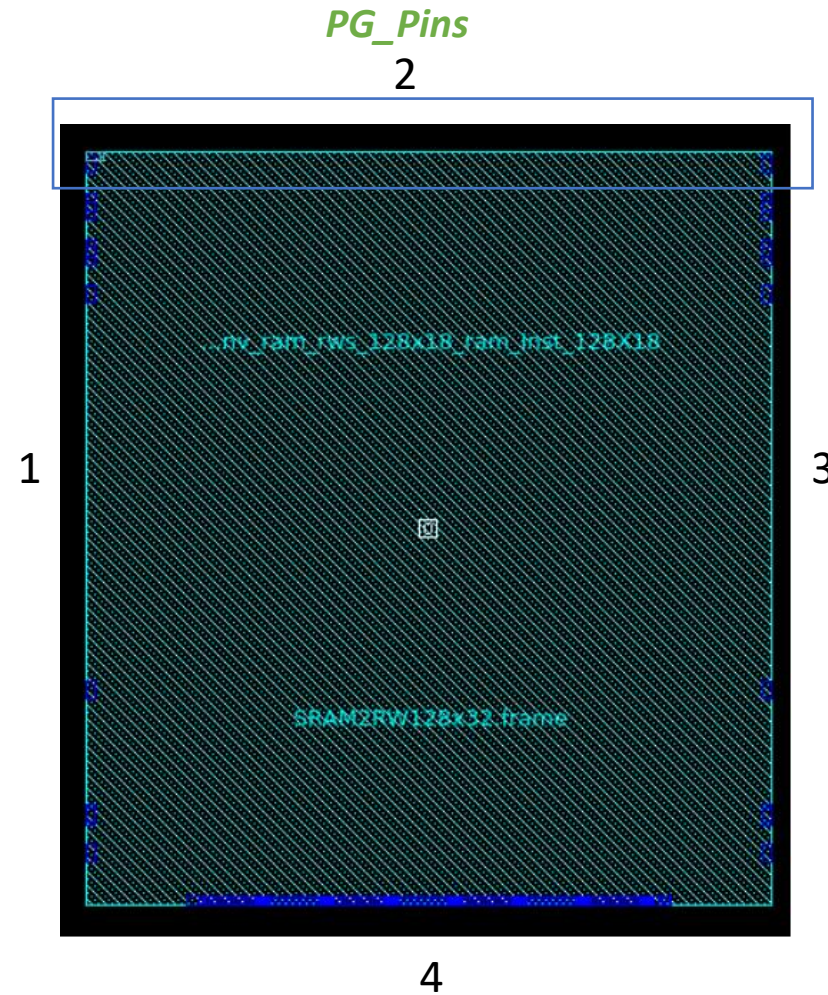
- Auto Placement Ports (M2 | M3)

```
icc2_shell>  
icc2_shell> place_pins -ports [get_ports *]  
  
Min routing layer: M1  
Max routing layer: M9  
  
Number of block ports: 1750  
Number of block pin locations assigned from router: 0  
Number of PG ports on blocks: 0  
Number of pins created: 1750  
CPU Time for Pin Placement: 00:00:05.08u 00:00:00.00s 00:00:05.09e:  
Total Pin Placement CPU Time: 00:00:05.29u 00:00:00.00s 00:00:05.30e:  
1
```





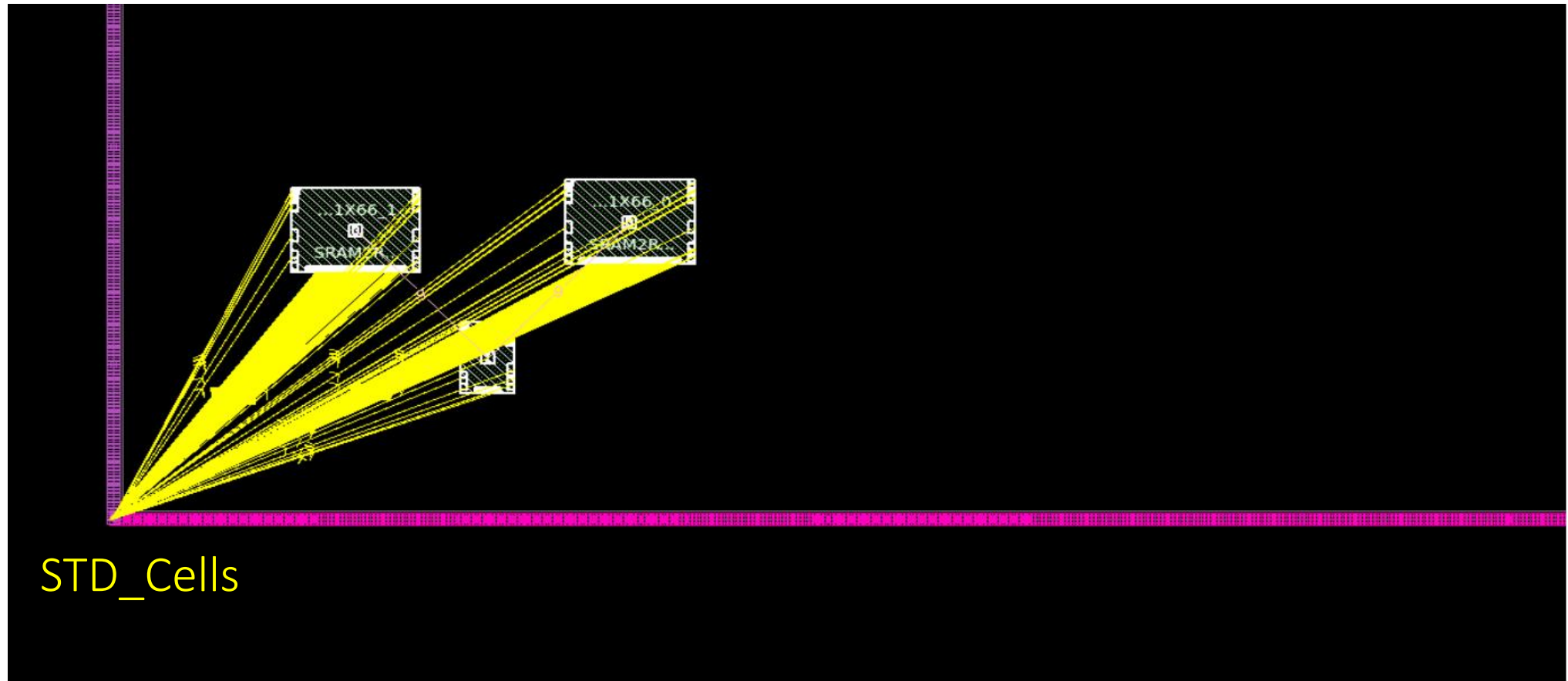
```
MACRO SRAM2RW128x16  
  CLASS BLOCK ;  
  SOURCE USER ;  
  ORIGIN 0 0 ;  
  SIZE 87.63 BY 135.673 ;  
  SYMMETRY X Y R90 ;
```



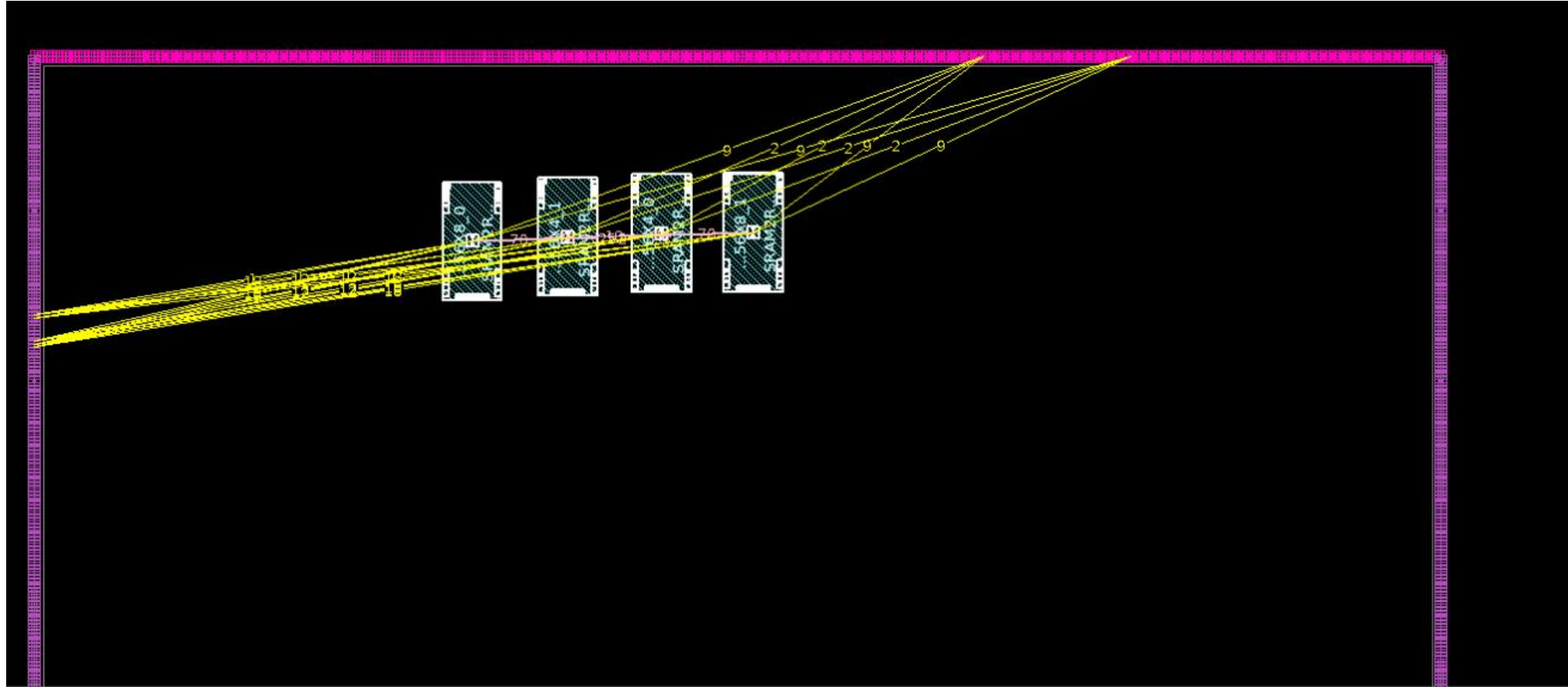


- We started by analyzing the connectivity between the macros and identifying their relationships with the I/O ports using **flylines**.
- Net\_Connections.
- Data\_Flow.
- Register\_Tracing.

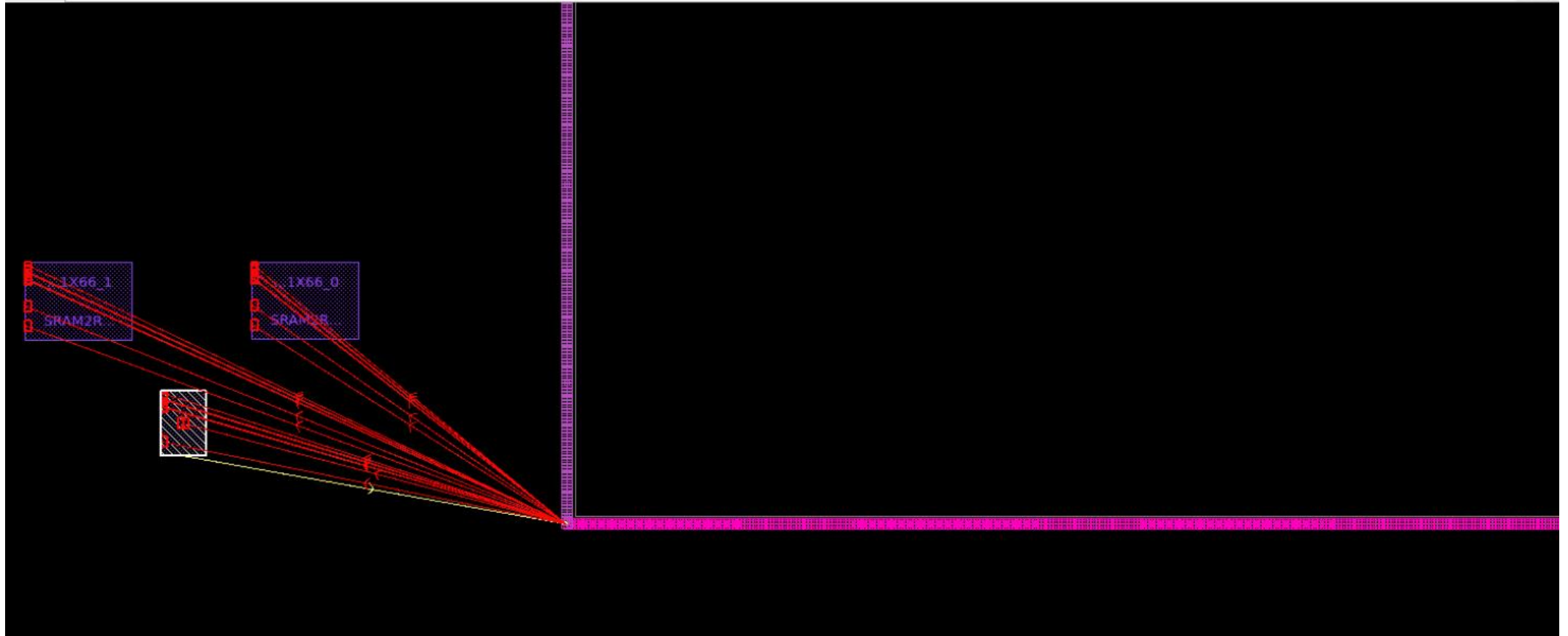
- **Net\_Connections:** Displays **direct electrical connections** from macro pins to the first standard cell, other macros, or I/O ports to optimize orientation and channel spacing.



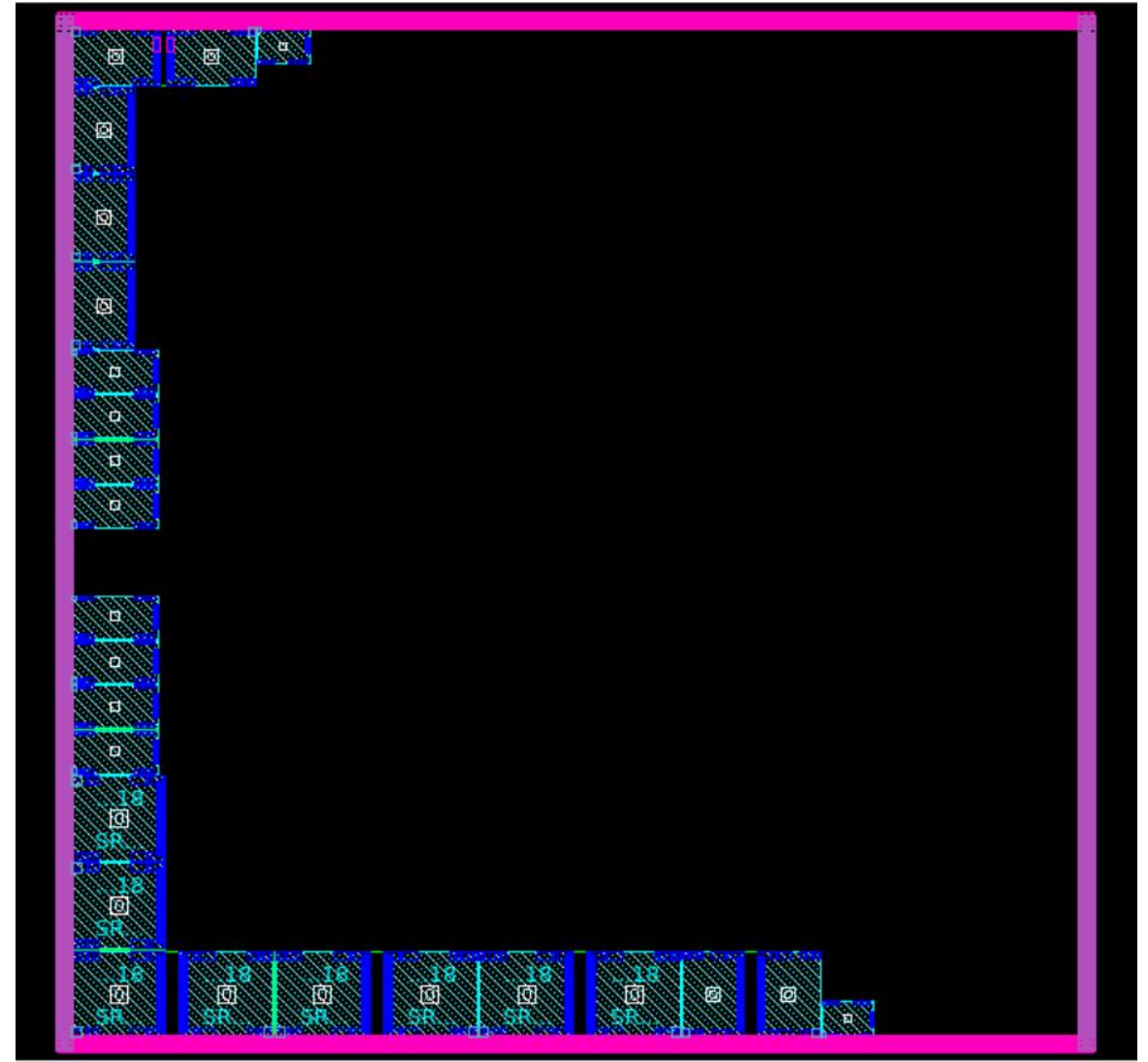
- **Data\_Flow:** Visualizes the high-level architectural communication between macro groups and I/O ports to guide the overall floorplan topology and minimize total wirelength.



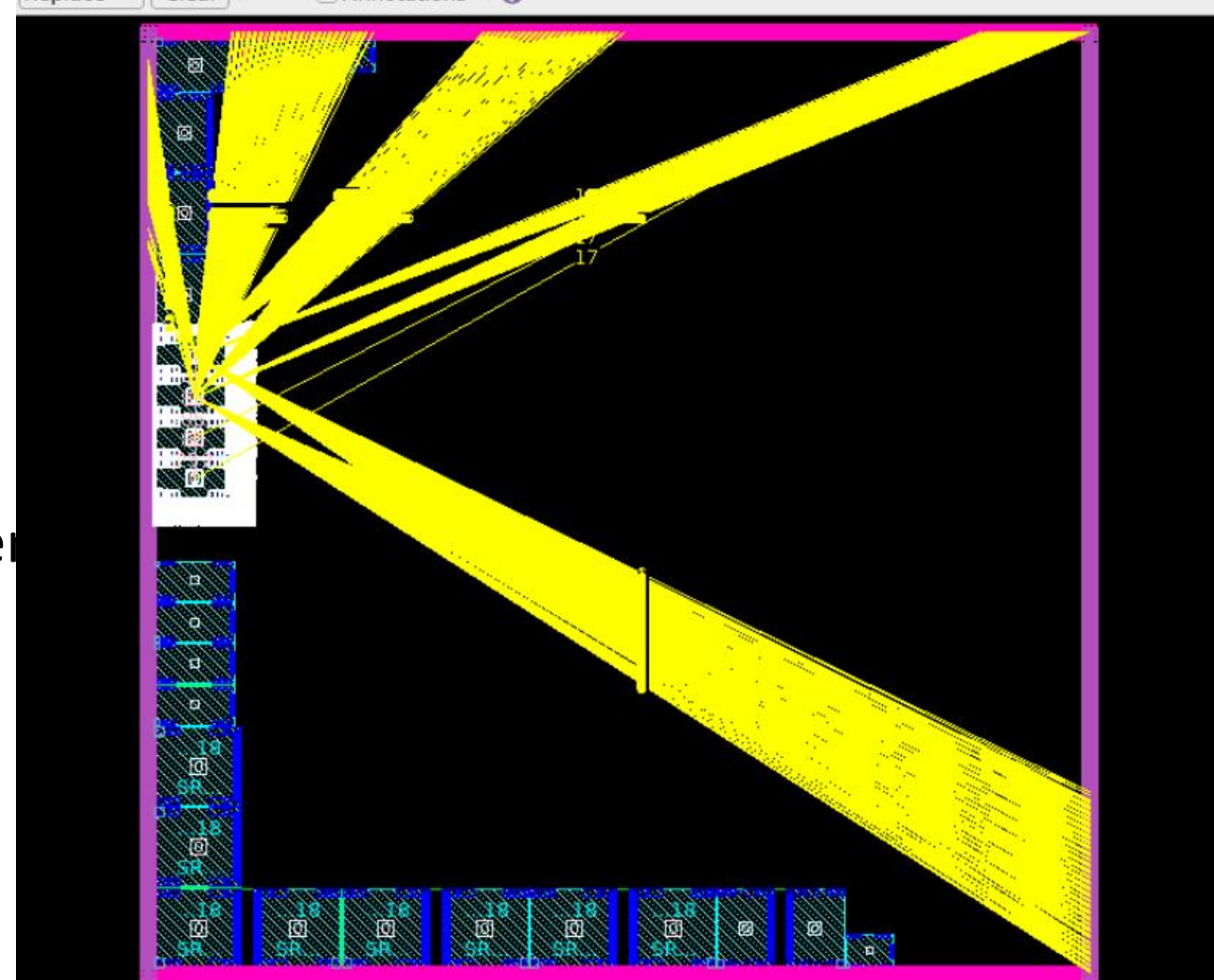
- **Register\_Tracing**: Tracks the number of intermediate flip-flops (pipeline stages) between macros to reveal **timing criticality** and dictate their required physical proximity.



- However, the automatically generated macro placement was inefficient and did not follow recommended macro placement techniques, resulting in poor floorplan quality.

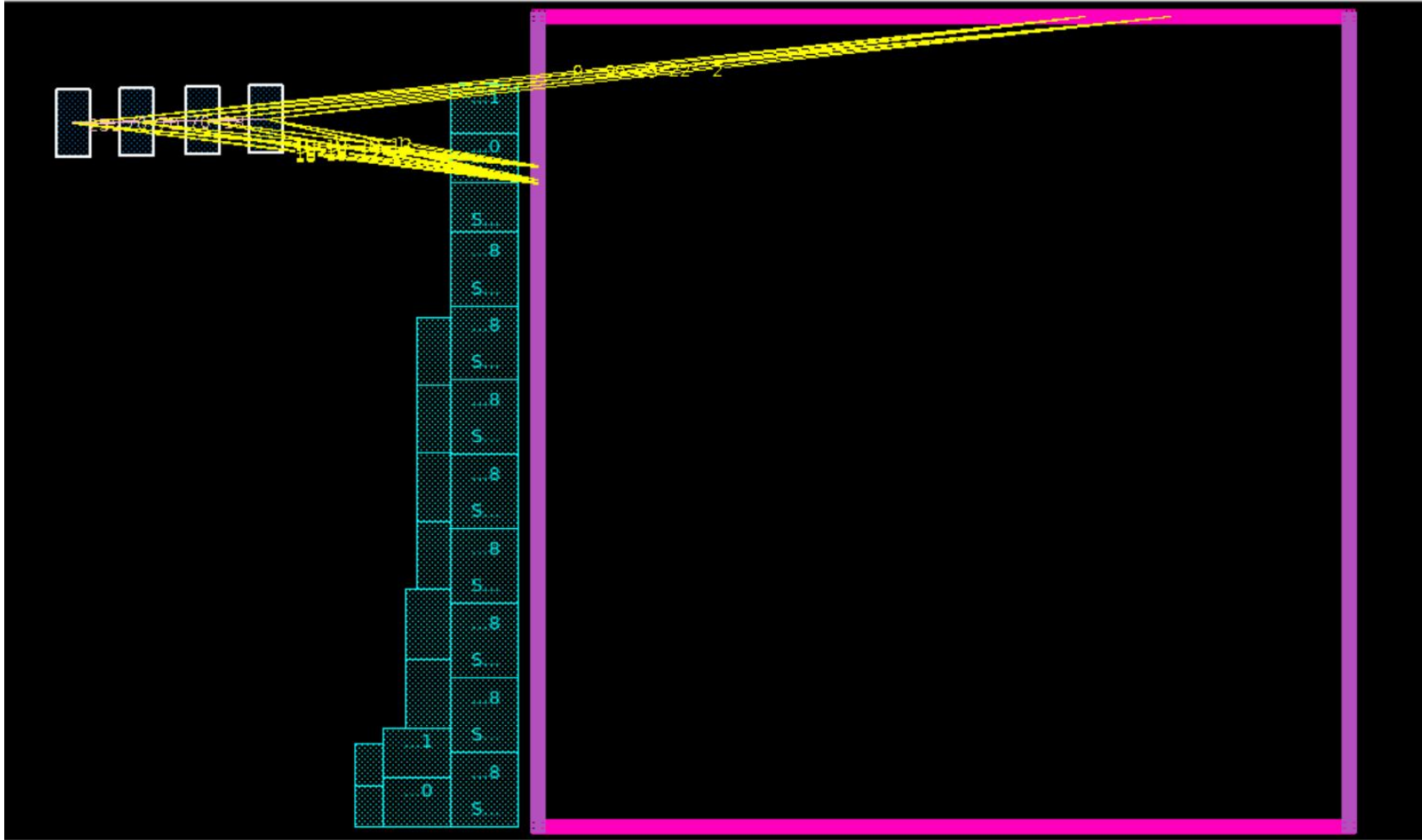


- The I/O ports were not aligned with the macro pins.
- We decided to relocate the macros following the recommended macro placement guidelines. In addition, we optimized the I/O port locations to better align with the macro pins, resulting in shorter interconnects and improved routability.



# Dividing the macros into different groups

Group\_1 communicates with each other and I/O Ports





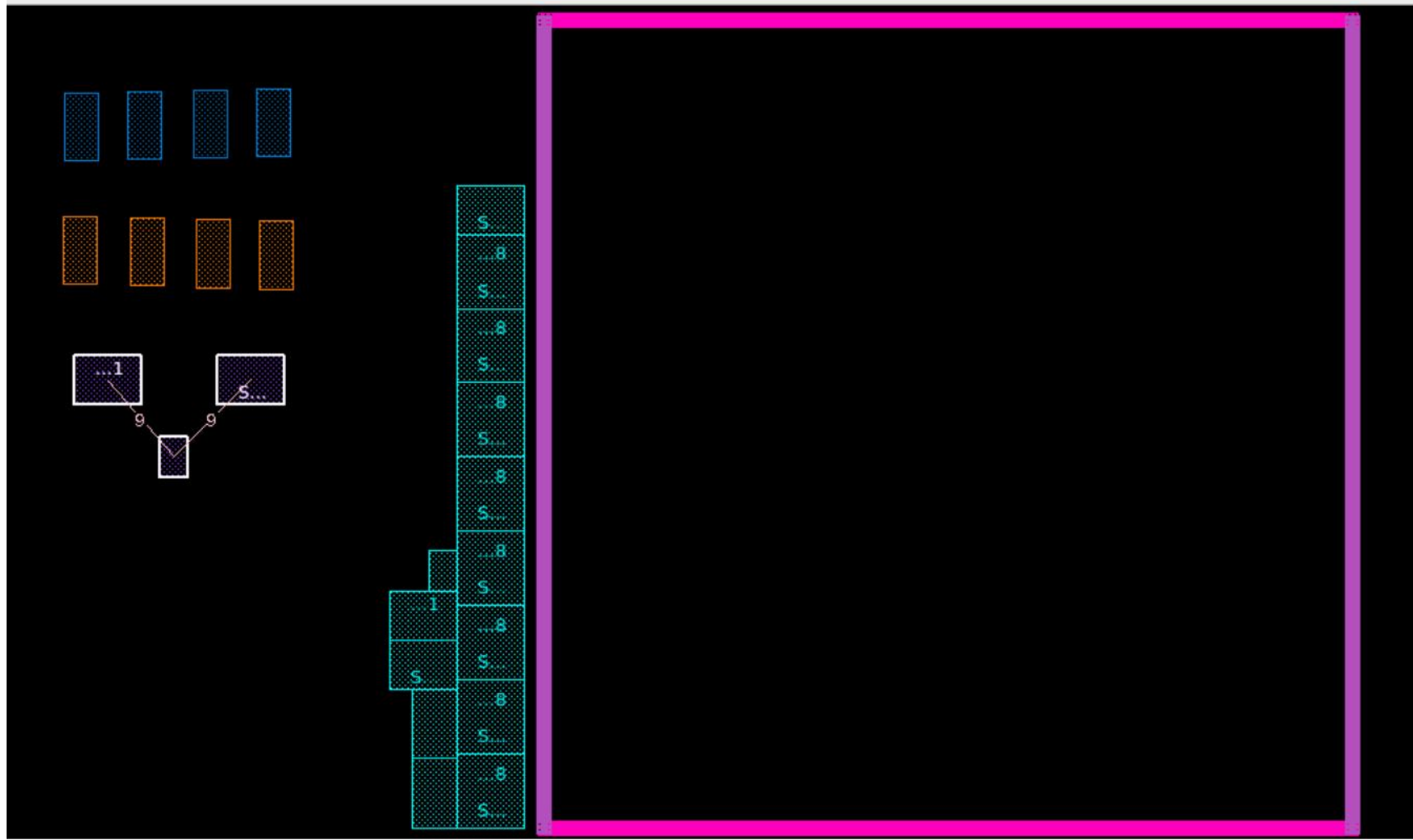


The diagram illustrates the mapping of a 2D grid to a 1D array for a 2D convolution operation. On the left, a 2D grid of size 4x4 is shown, with the bottom row highlighted in red. This grid is mapped to a 1D array of size 16, where the bottom row of the 2D grid is mapped to the first 4 elements of the 1D array. The 1D array is then mapped to a 2D grid of size 4x4, where the bottom row of the 2D grid is mapped to the first 4 elements of the 1D array. The diagram shows the mapping of the bottom row of the 2D grid to the first 4 elements of the 1D array, and the mapping of the first 4 elements of the 1D array to the bottom row of the 2D grid.



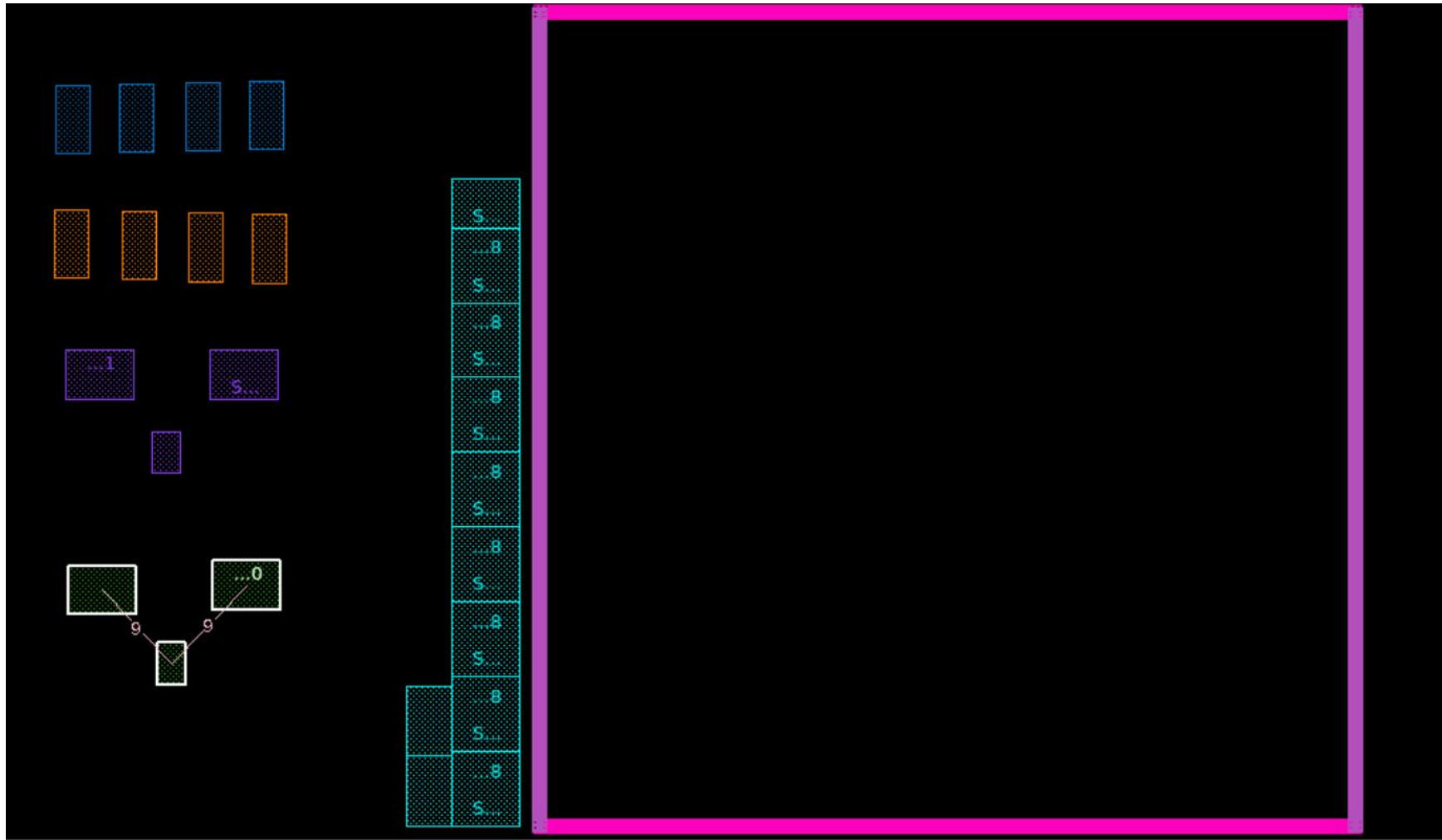
# Dividing the macros into different groups

Group\_3 communicates with each other and STD cells



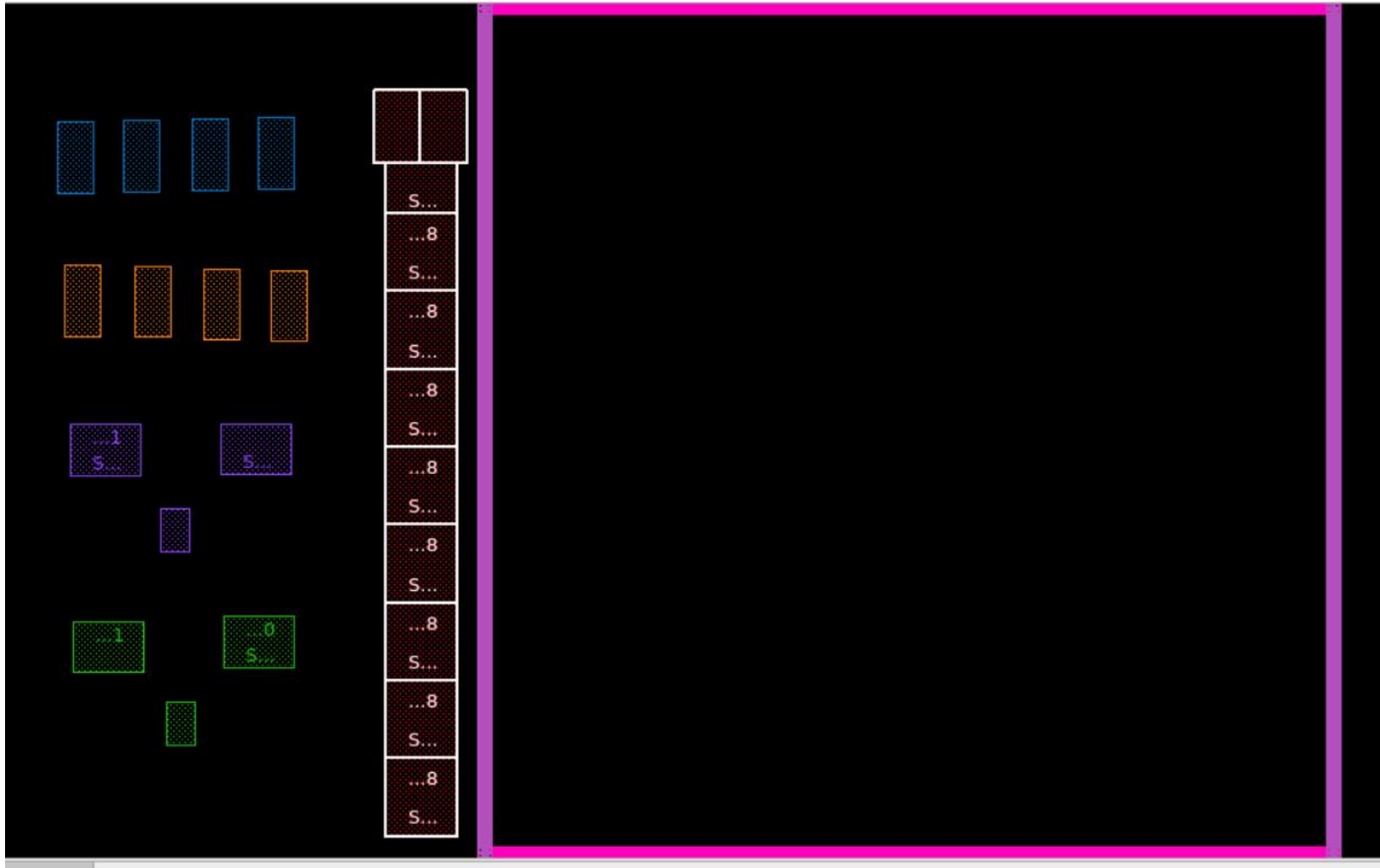
# Dividing the macros into different groups

Group\_4 communicates with each other and STD cells



# Dividing the macros into different groups

Group\_5 communicates with STD cells only.



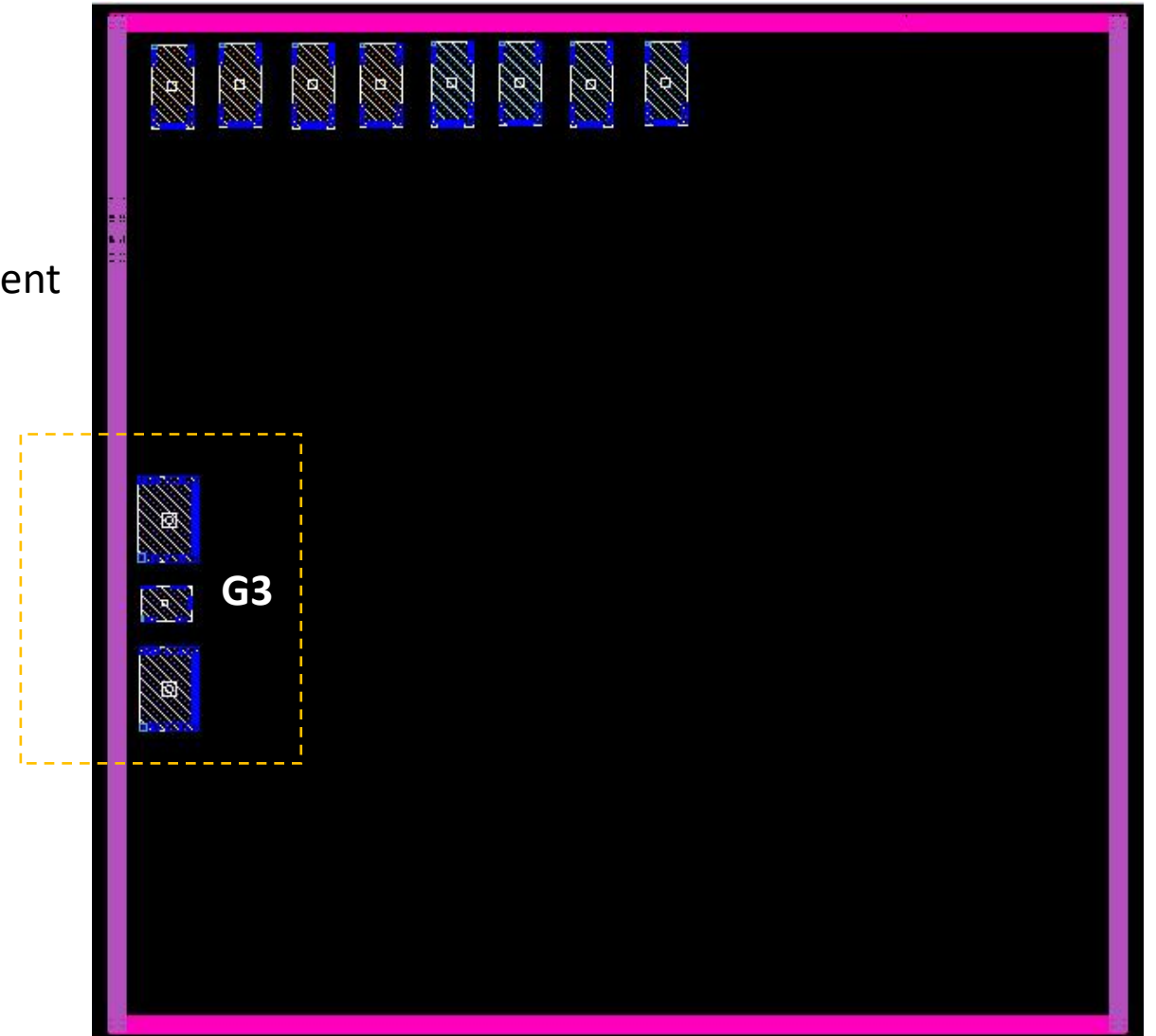


-

- We started by placing each macro group in its appropriate location based on the fly lines.
- Placed **Group\_1** and **Group\_2** in the Core Area



- Placed **Group\_3** in the Core Area
- Rotated macros in **Group\_3** by 90° **counterclockwise** to orient their top sides toward the core boundary.



- Placed **Group\_4** in the Core Area
- Rotated macros in **Group\_4** by 90° **counterclockwise** to orient their top sides toward the core boundary.



- Placed **Group\_5** in the Core Area
- Rotated some macros in **Group\_5** by 90° **clockwise** to orient their top sides toward the core boundary.

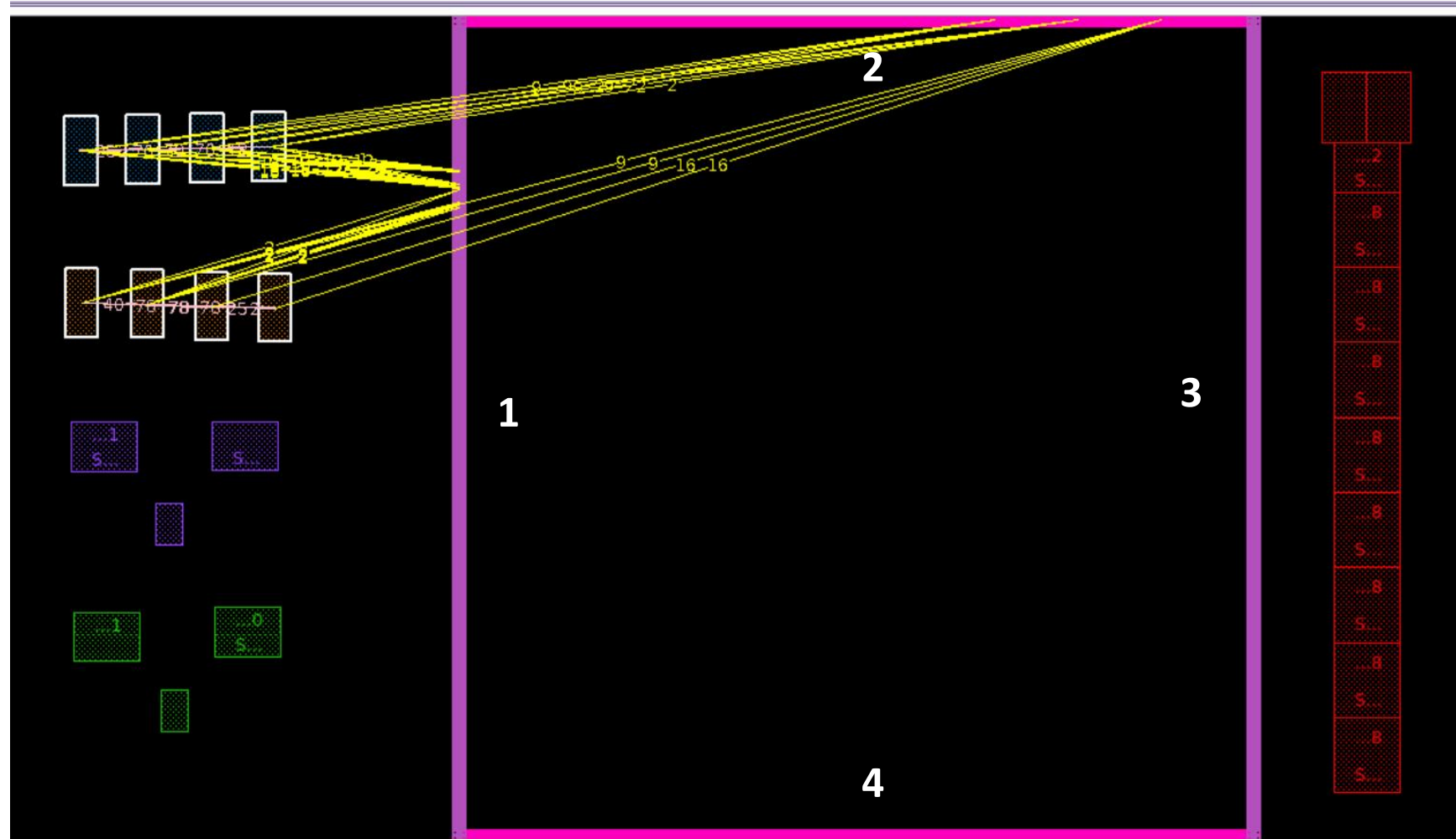




- After placing all the macro groups

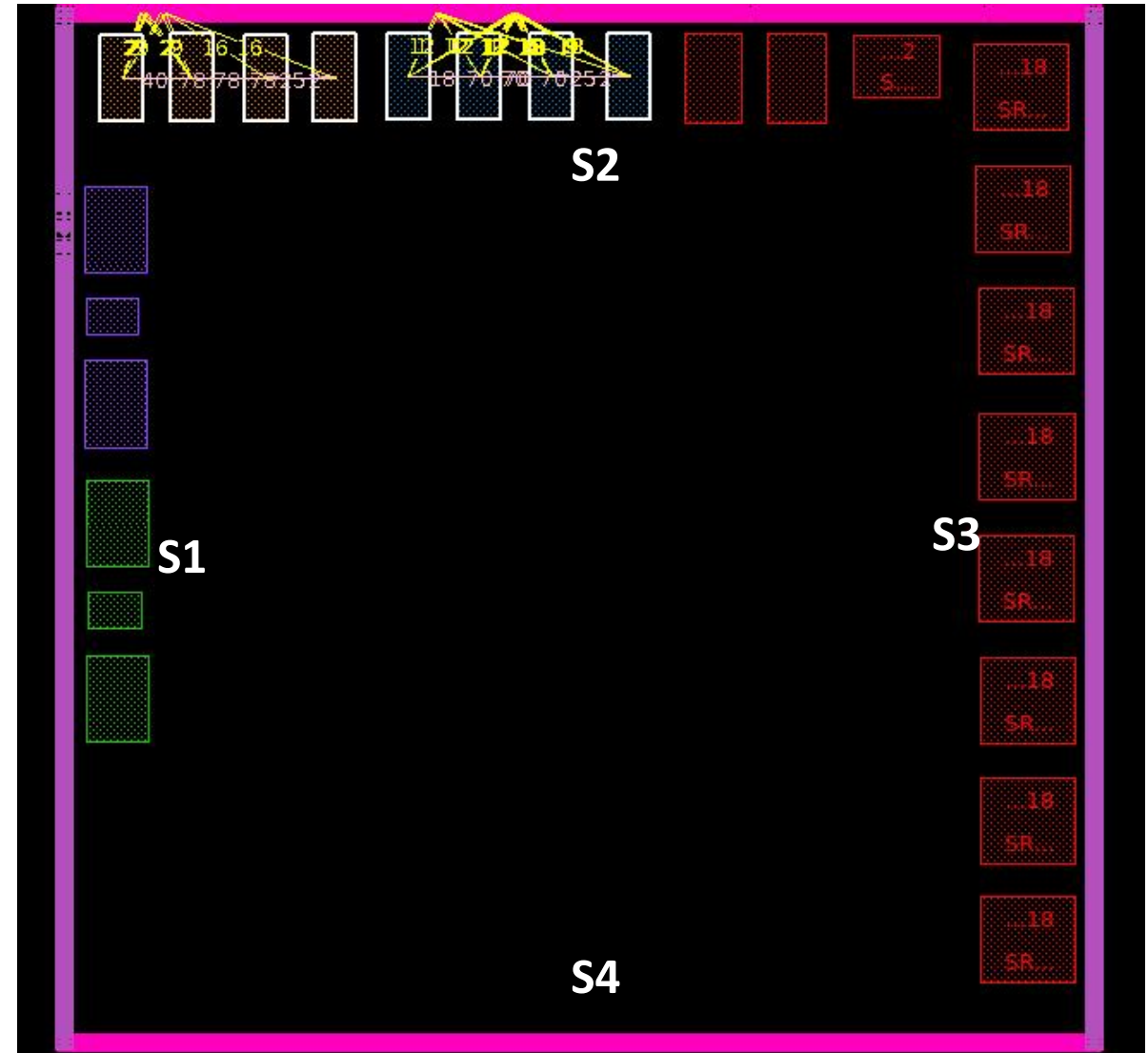


- Adjust the I/O port locations to align with each macro group.

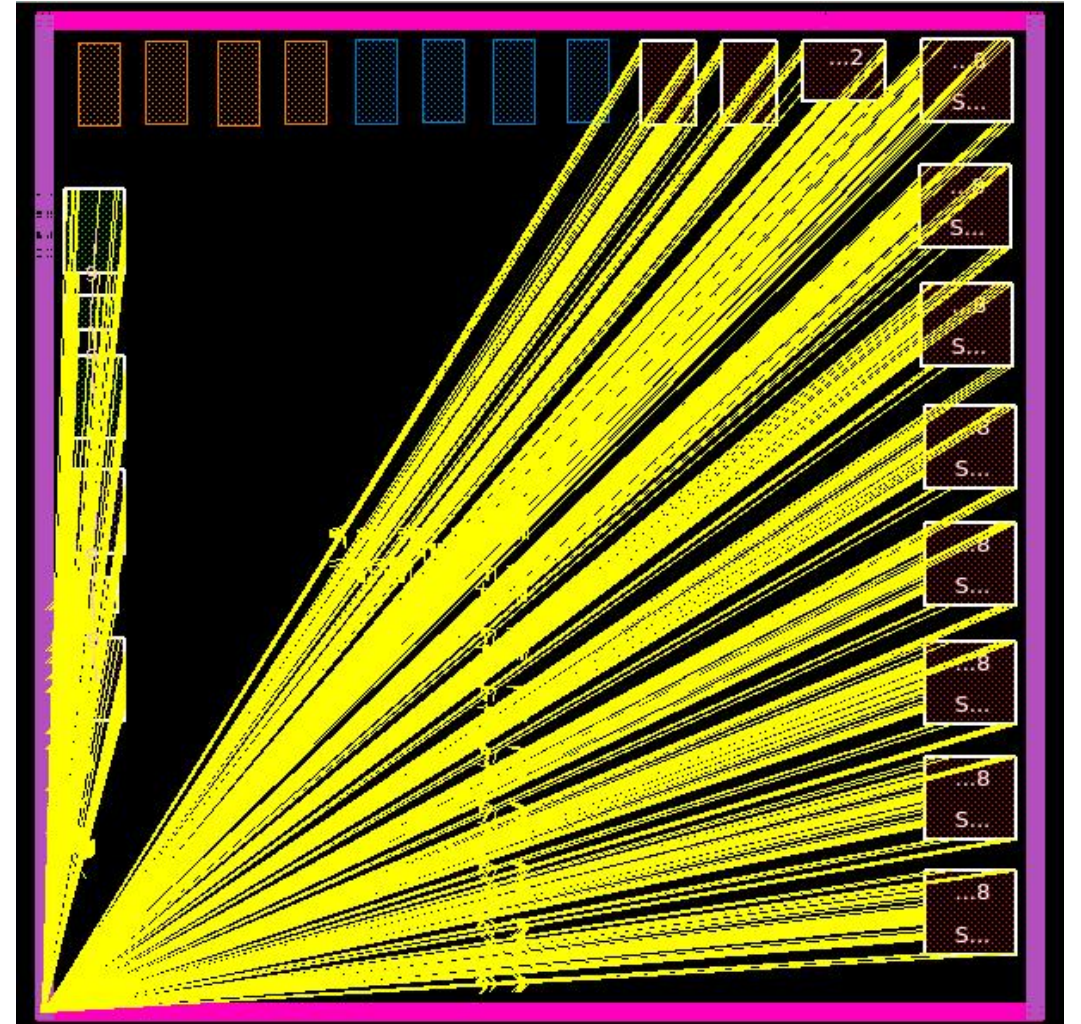


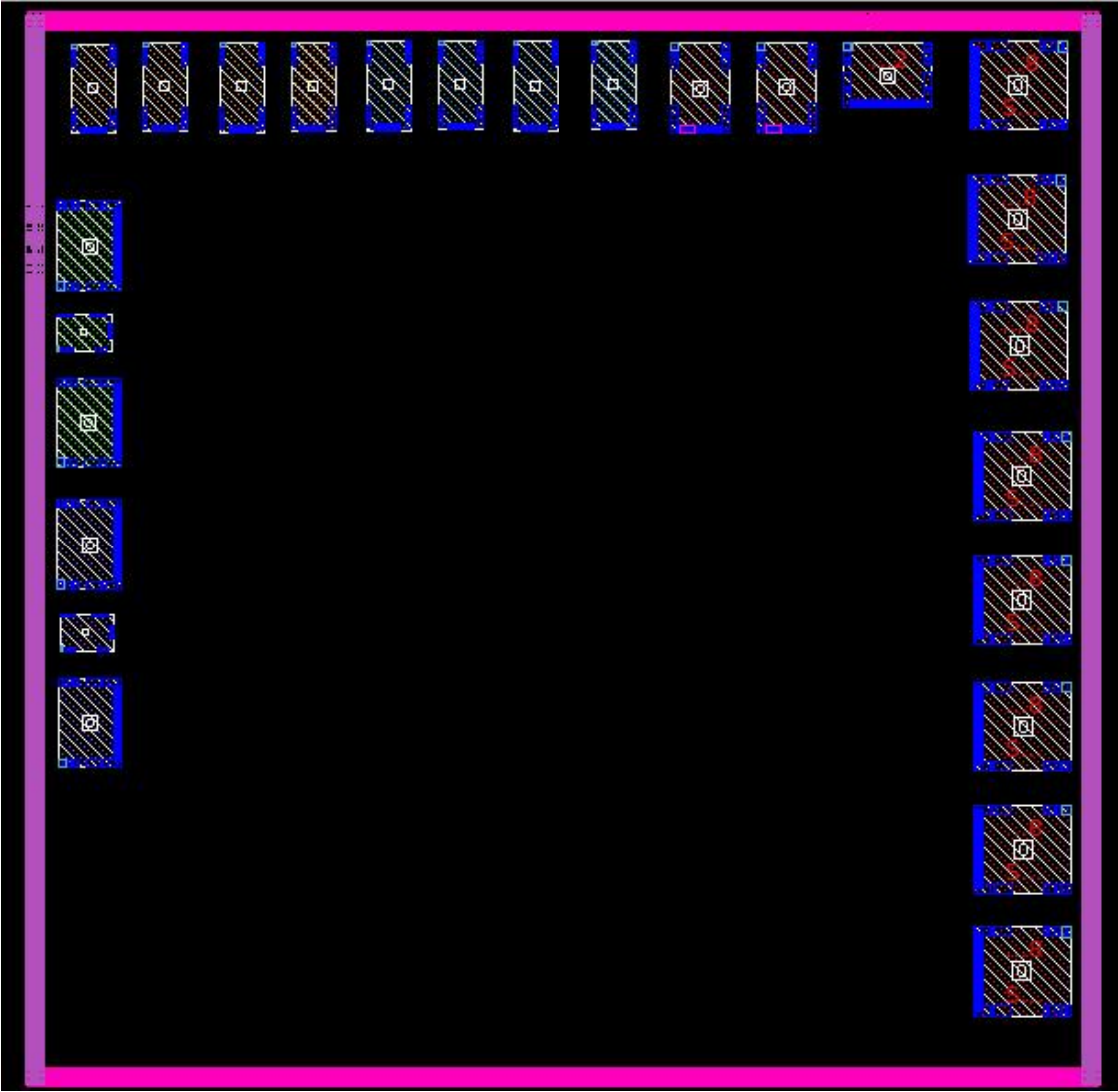
# Adjusted the I/O Port Locations

- Moved individual I/O ports from Side 1 to Side 2 to align them with the macro pins.



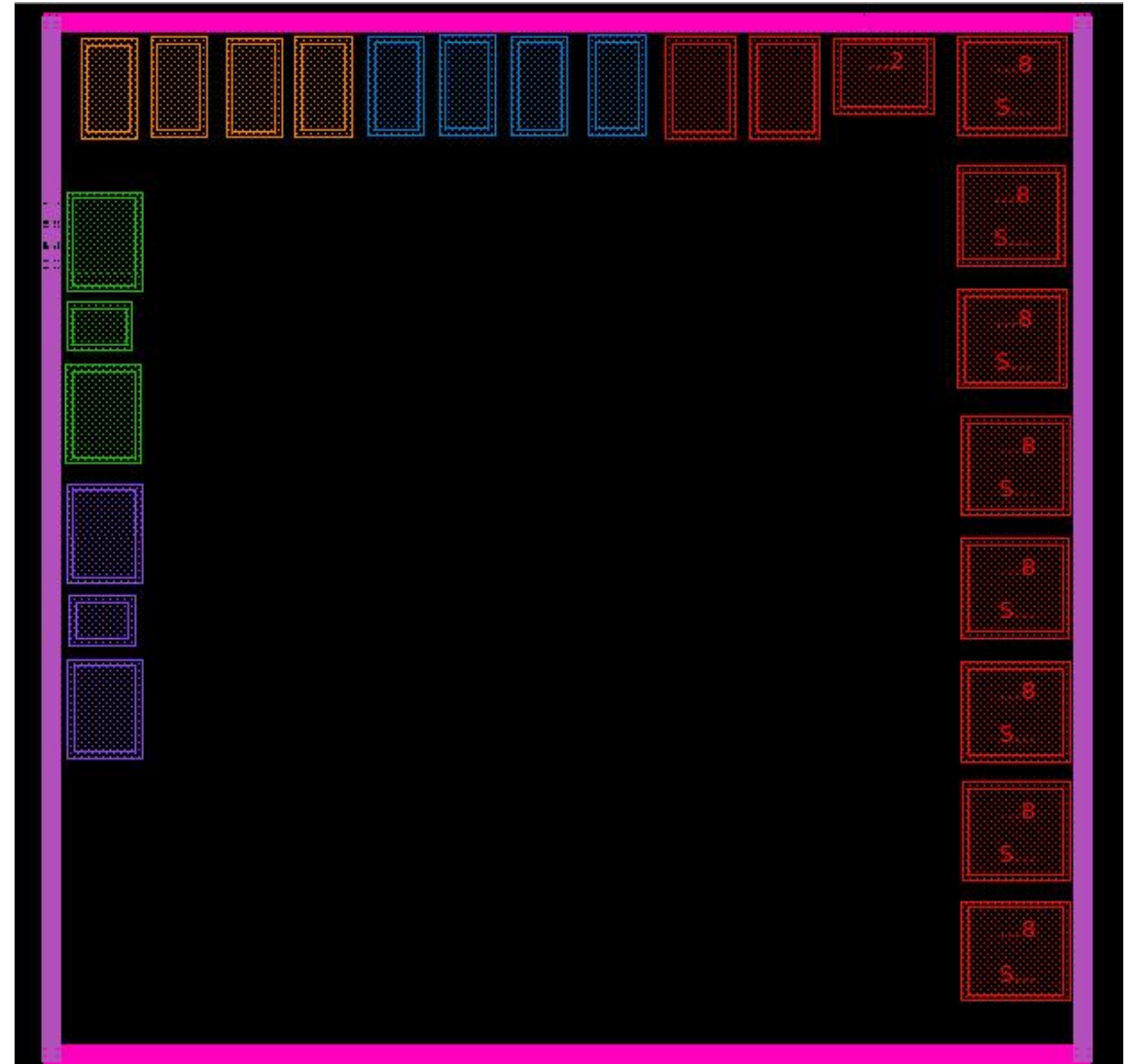
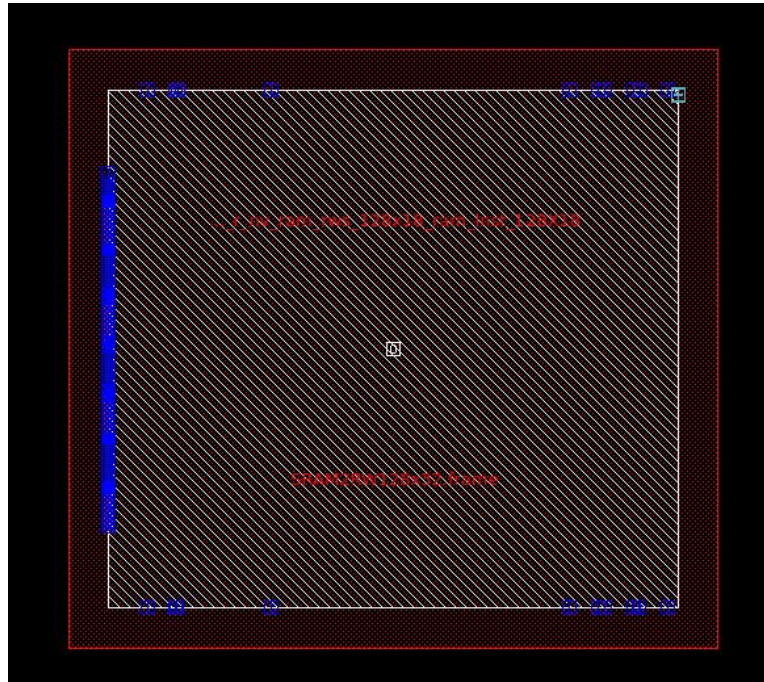








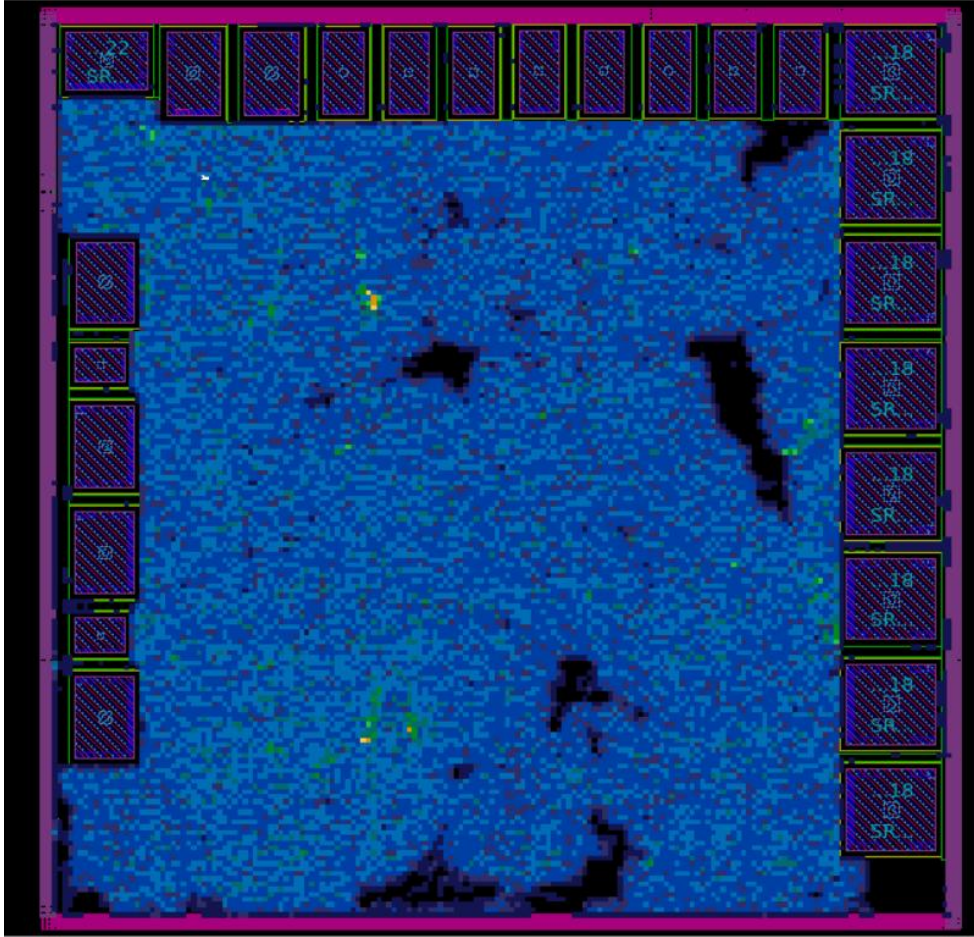
- Keepout margins (Halo): Prevents standard cells from being placed too close to macro boundaries to preserve dedicated channel space for pin access and avoid placement congestion.



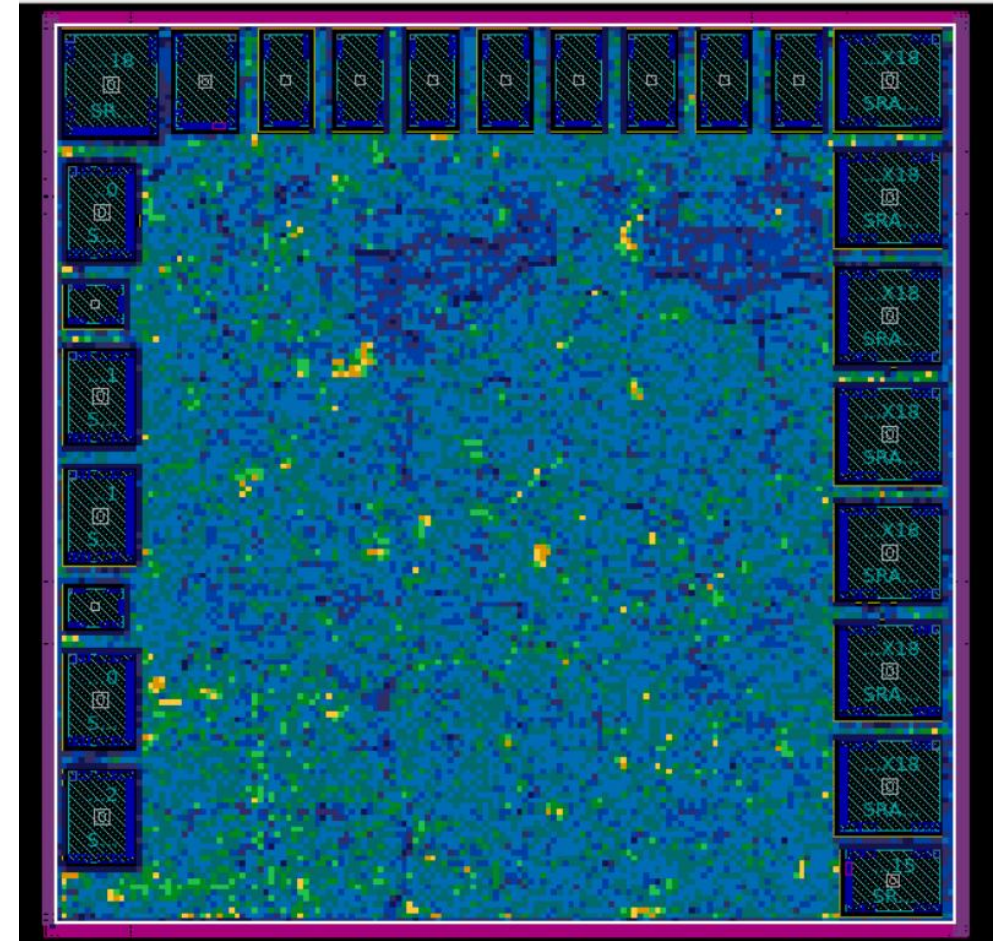
- After many iterations of the power planning and placement stages, we continuously refined the floorplan to resolve various implementation issues.
- Reducing the core area (U from 0.35 to 0.5)
- Added M1 routing blockages around the macros to prevent shorts and minimum-spacing DRC violations between the M1 power rails and the macro pins.



- Reducing the core area (Utilization from 0.35 to 0.5)



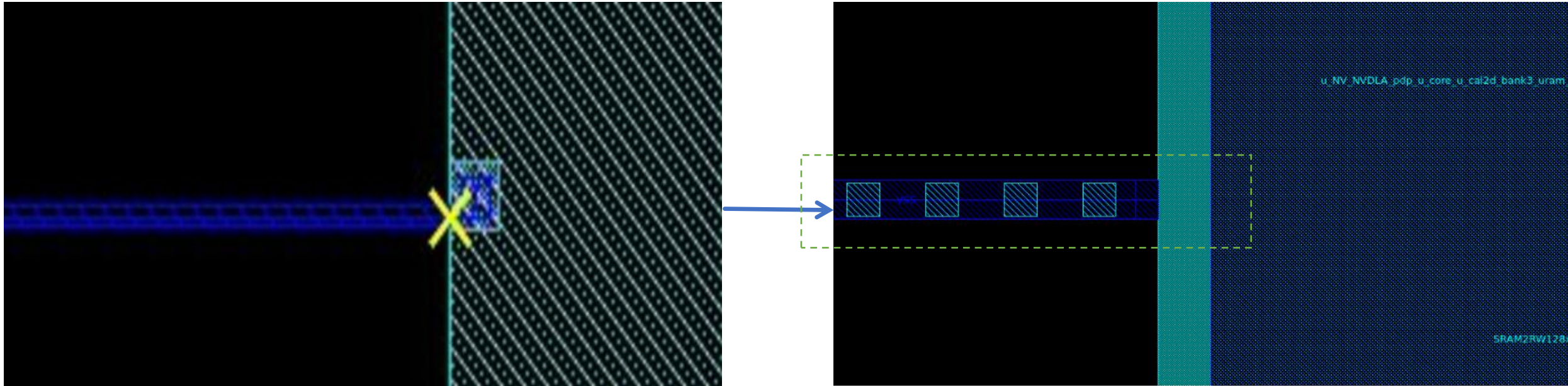
- Utilization = 0.35



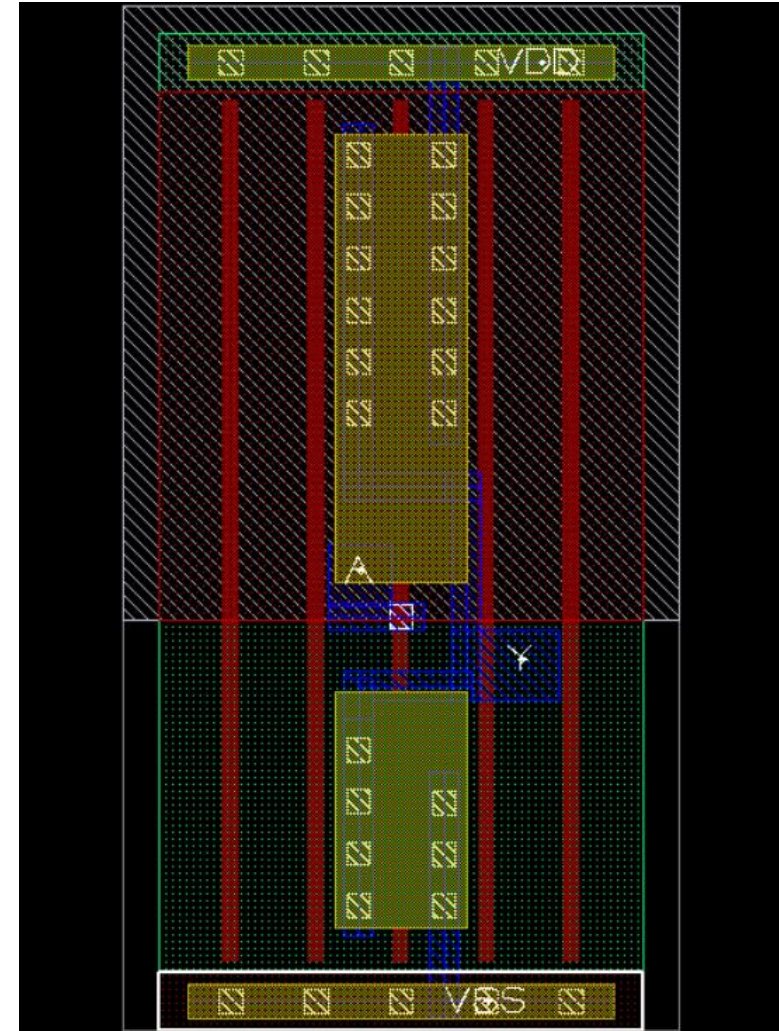
- Utilization = 0.5

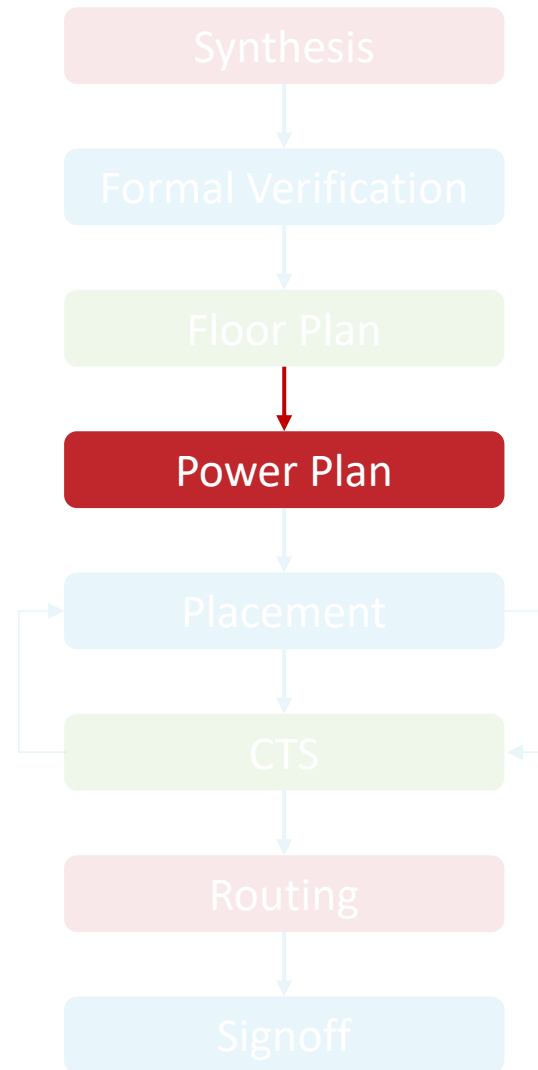


- Added **M1 routing blockages** around the macros to prevent shorts and minimum-spacing DRC violations between the M1 power rails and the macro pins.



- No well tap cells were added because the PDK contains traditional standard cells, not tapless cells.
- Endcap cells were inserted during the pre-placement stage, both around each macro and at the ends of all sites rows.



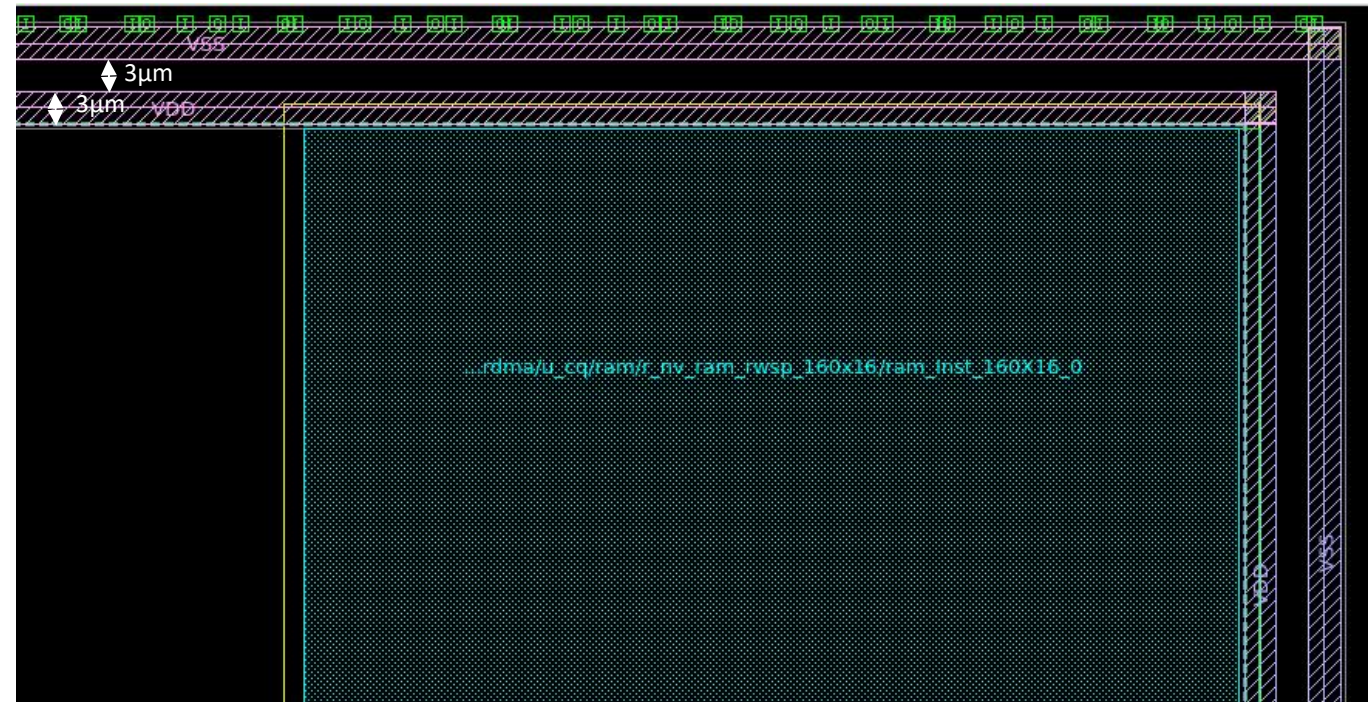


1. Core Power Ring
2. Power Mesh
3. Standard Cell M1 Power Rails
4. DRCs
5. Macro PG Pins Connectivity Strategies



- Constructed VDD and VSS rings around the core boundary using **top thick metal layers (M8 and M9)** to evenly distribute current from rings into the internal floorplan.

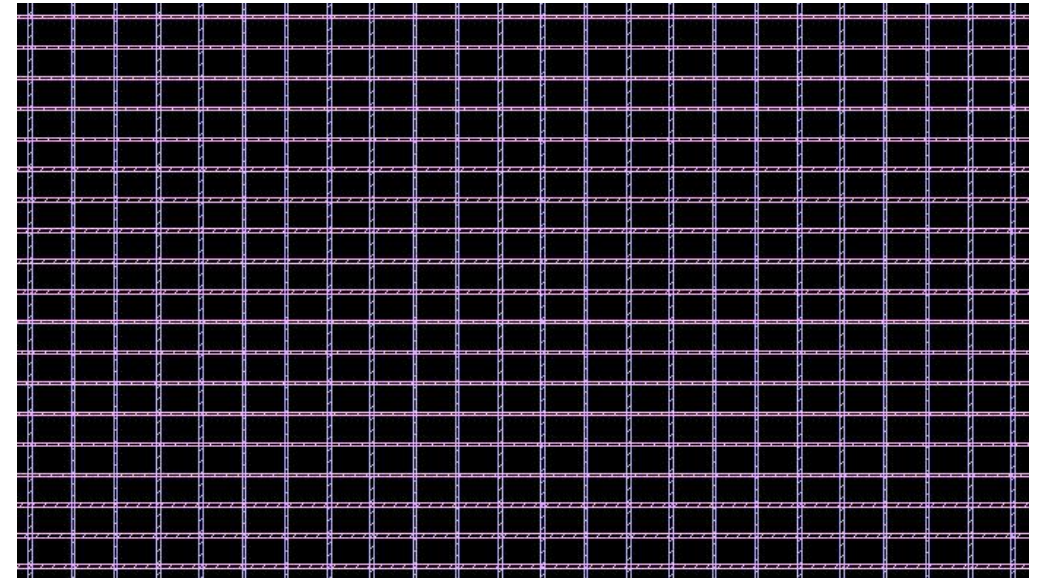
| Variable               | Value       |
|------------------------|-------------|
| Ring Offset            | 0.5 $\mu$ m |
| Ring Width             | 3 $\mu$ m   |
| Ring Spacing           | 3 $\mu$ m   |
| Horizontal Metal Layer | M9          |
| Vertical Metal Layer   | M8          |



- Implementing the internal Power/Ground (PG) mesh inside the core area
- It creates interleaved VDD/VSS straps on **top metal layers (M8 & M9)**, extending them to the design boundary to generate pins automatically.

| Vertical Layer | M8           |
|----------------|--------------|
| Strap Offset   | 7μm          |
| Strap Width    | 3μm          |
| Strap Spacing  | Interleaving |
| Strap Pitch    | 70μm         |
| Trim           | True         |

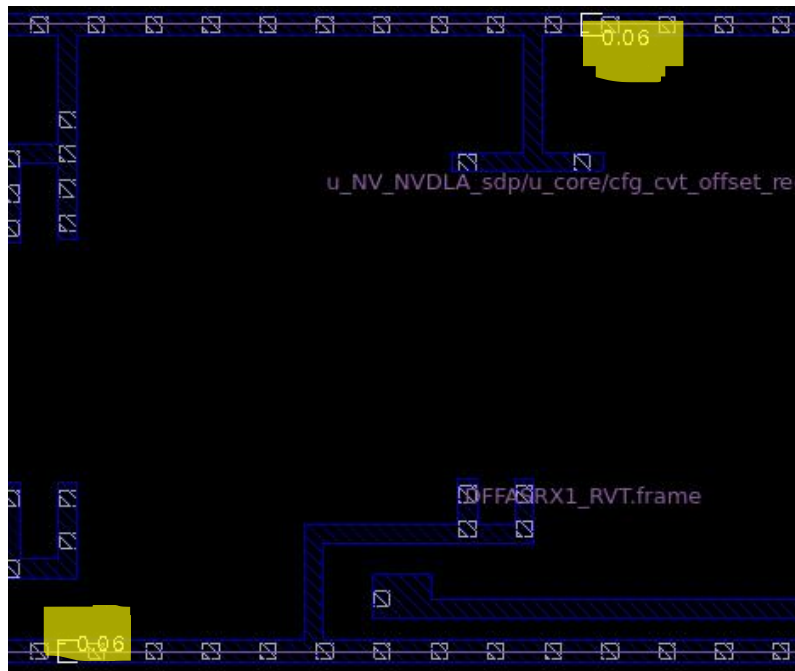
| Horizontal Layer | M9           |
|------------------|--------------|
| Strap Offset     | 7μm          |
| Strap Width      | 3μm          |
| Strap Spacing    | Interleaving |
| Strap Pitch      | 50μm         |
| Trim             | True         |



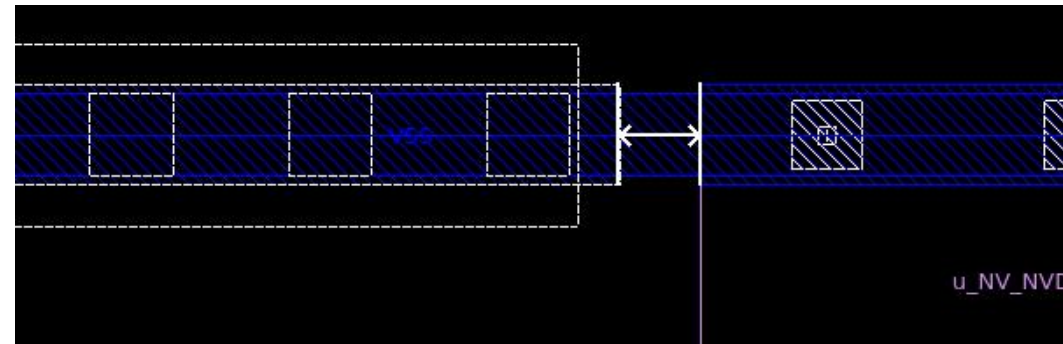
- The **Tech File** specifies an M1 **minWidth** of **0.05  $\mu\text{m}$**

```
Layer      "M1" {  
            /* layout attributes */  
            layerNumber      = 11  
            isDefaultLayer    = 1  
            maskName          = "metall1"  
            pitch             = 0.152  
            /* display attributes */  
            visible           = 1  
            selectable         = 1  
            blink              = 0  
            color              = "blue"  
            lineStyle          = "solid"  
            pattern            = "impdot"  
            /* design rule attributes */  
            minWidth           = 0.05  
            maxWidth           = 5  
            defaultWidth       = 0.05  
            minSpacing         = 0.05
```

- **Root Cause:** When the 0.05  $\mu\text{m}$  PG rail meets the 0.06  $\mu\text{m}$  cell rail at the boundary, it produces a horizontal step/notch. The DRC engine treats these misaligned edges as separate metal geometries on the same net



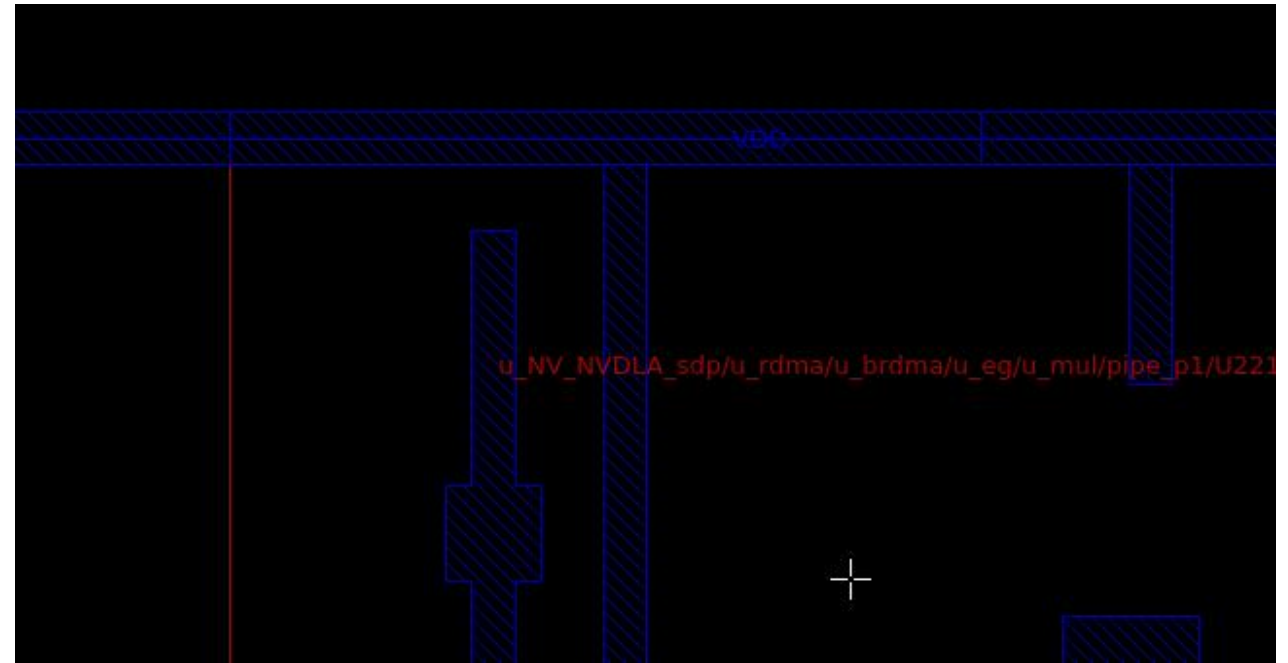
```
icc2_shell> check_pg_drc
Command check_pg_drc started at Thu Aug 6 19:21:35 2026
Command check_pg_drc finished at Thu Aug 6 19:21:53 2026
CPU usage for check_pg_drc: 133.07 seconds ( 0.04 hours)
Elapsed time for check_pg_drc: 17.74 seconds ( 0.00 hours)
Total number of errors found: 117
117 insufficient spacings on M1
```





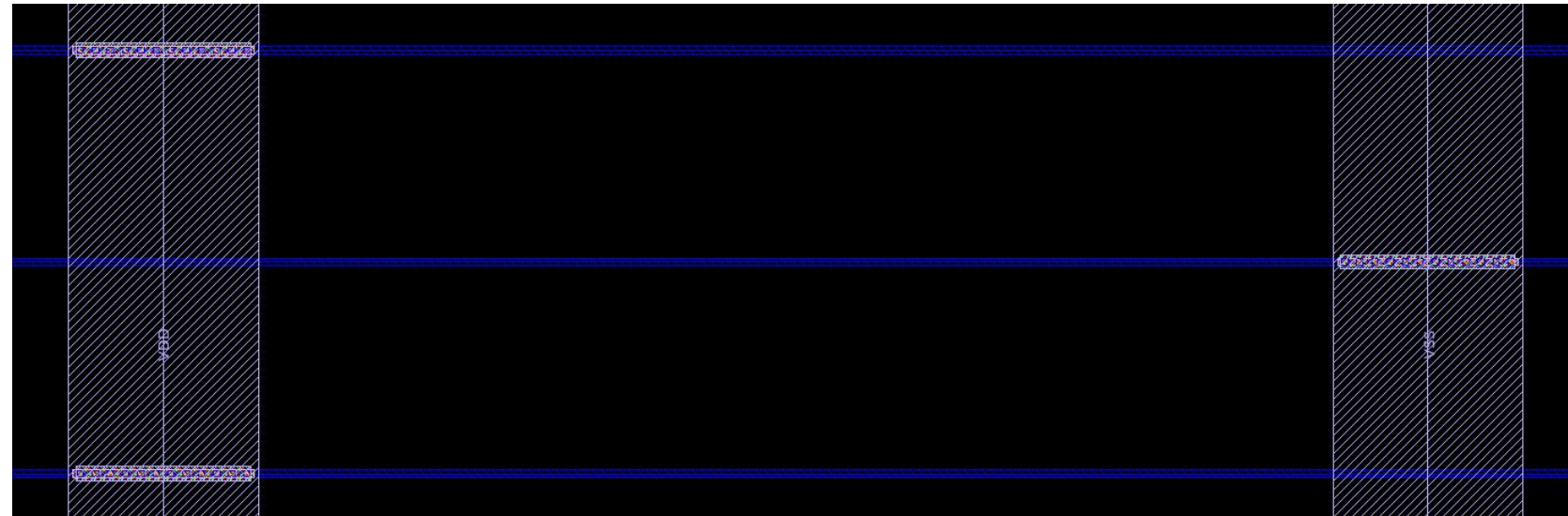
- **Solution:** Change **rail\_width** to **0.06  $\mu\text{m}$**  to match the standard cell layout

```
icc2_shell> check_pg_drc  
Command check_pg_drc started at Sun Aug 9 03:20:59 2026  
Command check_pg_drc finished at Sun Aug 9 03:22:24 2026  
CPU usage for check_pg_drc: 85.73 seconds ( 0.02 hours)  
Elapsed time for check_pg_drc: 85.74 seconds ( 0.02 hours)  
No errors found.
```



- Constructed VDD and VSS rails on layer **M1** and the via transition **from layer M8 to layer**
- Flipped alternating standard cell rows to enable direct VDD and VSS M1 rail sharing through abutment
- Maximized core cell density and saved horizontal routing tracks by merging adjacent power supply rails

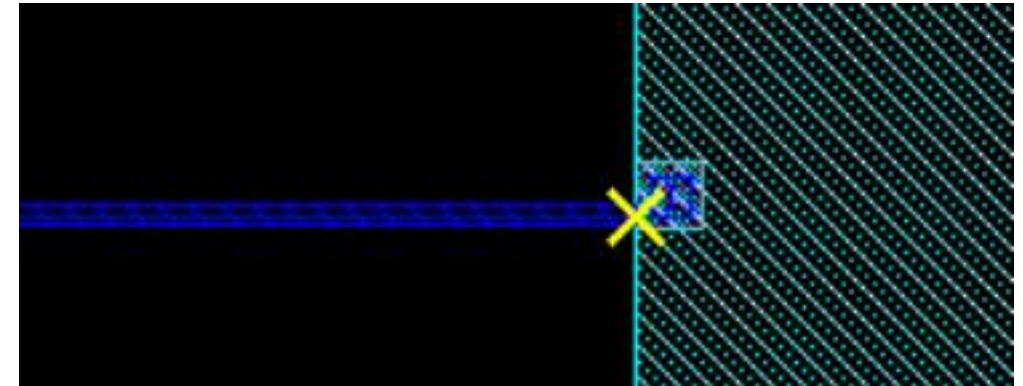
| Variable          | Value  |
|-------------------|--------|
| Rails Metal Layer | M1     |
| Rails Width       | 0.06μm |



- `compile_pg -strategies $rail_startegie -ignore_drc`

```
Committed 1541 wires.  
Committing wires takes 0.00 seconds.  
Committed 39697 vias.
```

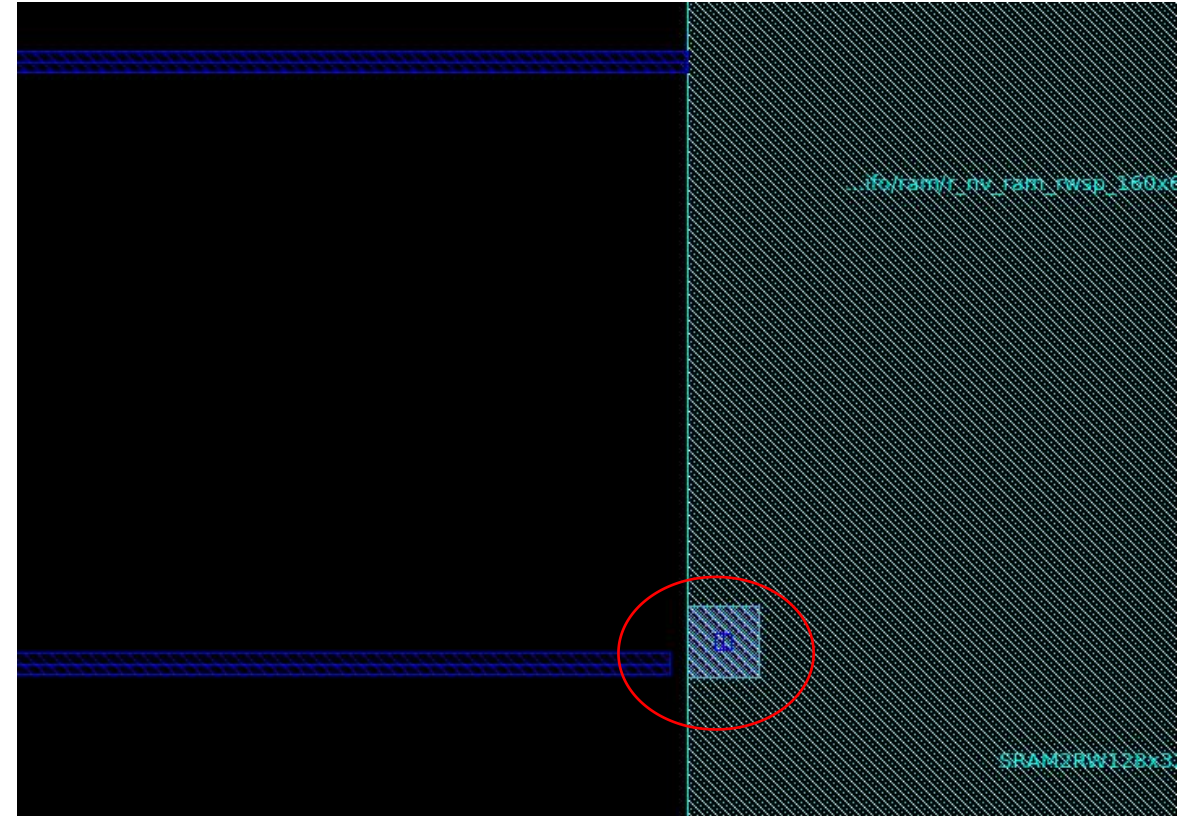
```
icc2_shell> check_pg_drc  
Command check_pg_drc started at S  
Command check_pg_drc finished at S  
CPU usage for check_pg_drc: 6.97 s  
Elapsed time for check_pg_drc: 6.9  
Total number of errors found: 504  
58 shorts on M1
```



- `compile_pg -strategies $rail_startegie`

```
Committed 1542 wires.  
Committing wires takes 0.00 seconds.  
Committed 0 vias.
```

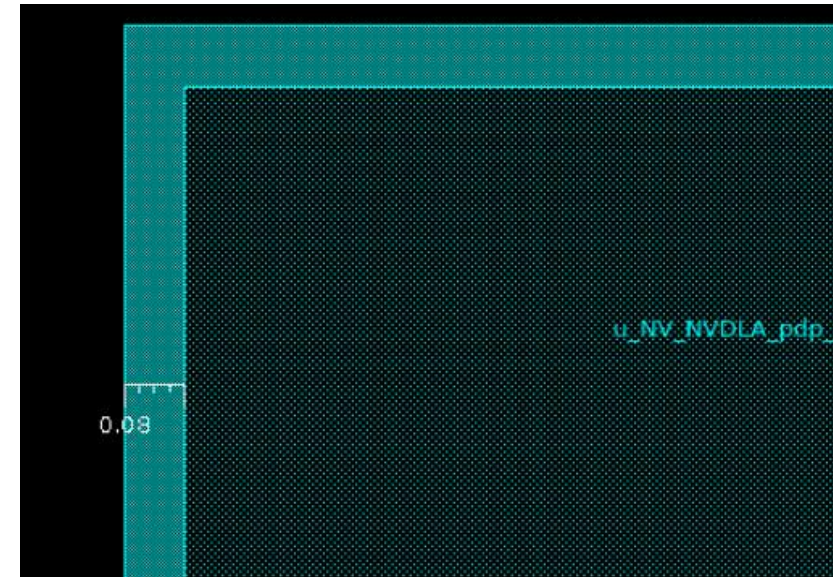
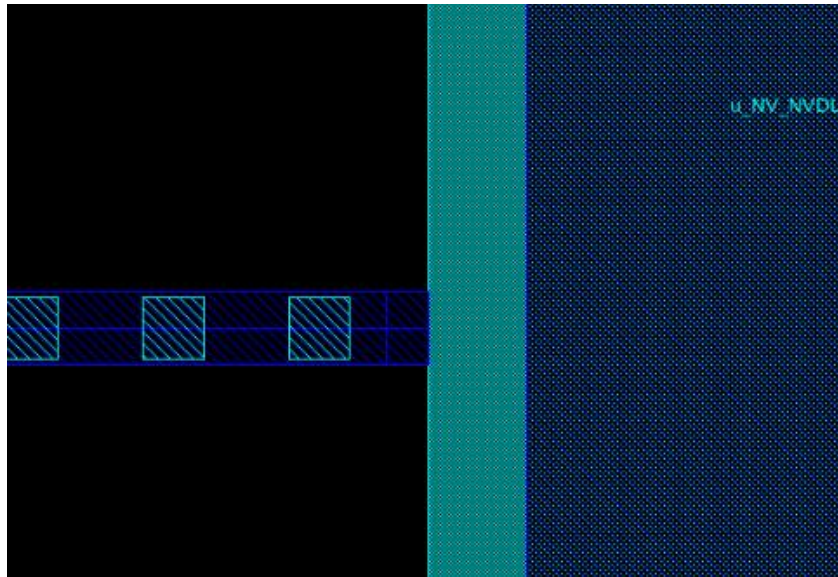
- `create_pg_vias -from_layers M8 -to_layers M1 \`  
  `-nets {VDD VSS} -drc no_check`

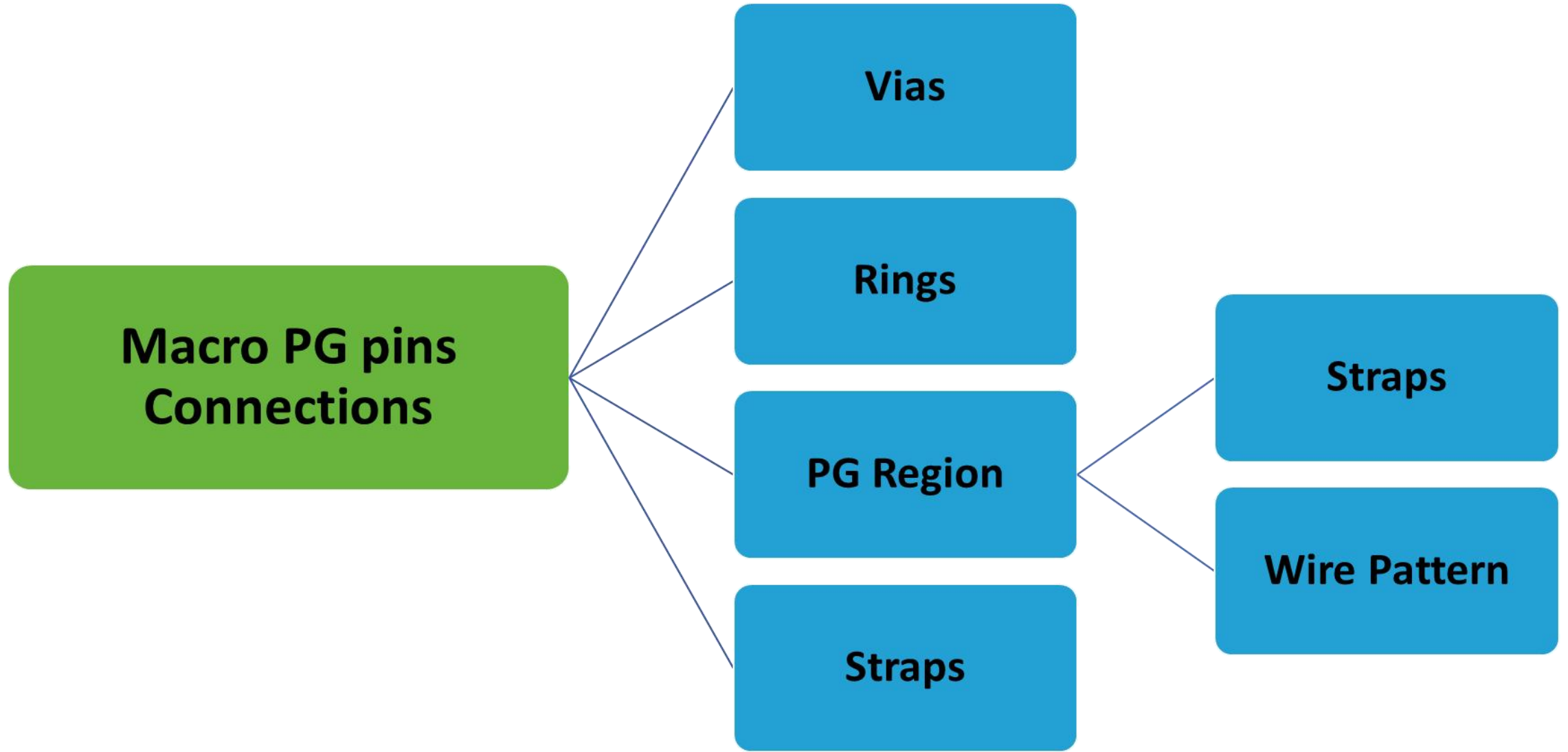




- Adding **M1 routing blockages** around the macros to prevent shorts and minimum-spacing DRC violations between the **M1 power rails** and the **macro pins**

```
set my_macros [get_cells -hierarchical -filter "is_hard_macro==true || design_type==macro"]  
  
create_keepout_margin -type routing_blockage -outer {0.08 0.08 0.08 0.08} -layers {M1} $my_macros
```

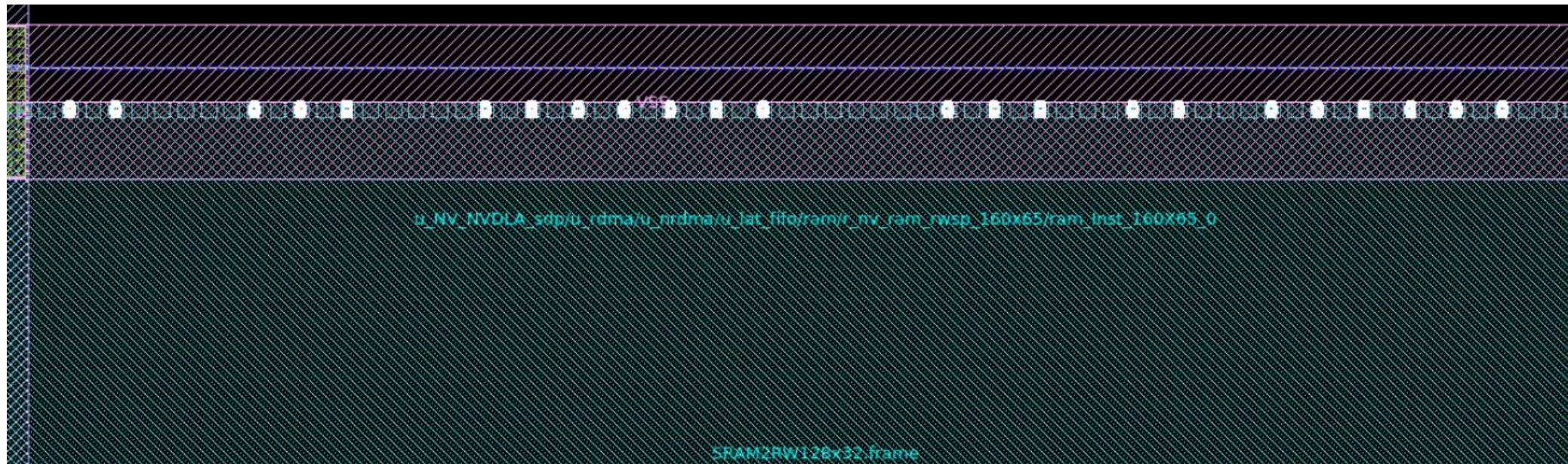




| Strategies                | Connection Rate<br>For Macro Pins | Routing<br>Resources |
|---------------------------|-----------------------------------|----------------------|
| Vias                      | 7%                                | Reduced              |
| PG Region Straps          | 15%                               | 19.8%                |
| PG Region Wire<br>Pattern | 15%                               | 16.6%                |
| Rings                     | 75%                               | 3.2%                 |
| Straps                    | 95%                               | 1%                   |



- Only **7%** of the macro PG pins are connected to the power mesh
- **Saves routing resources** and mitigates congestion on upper layers

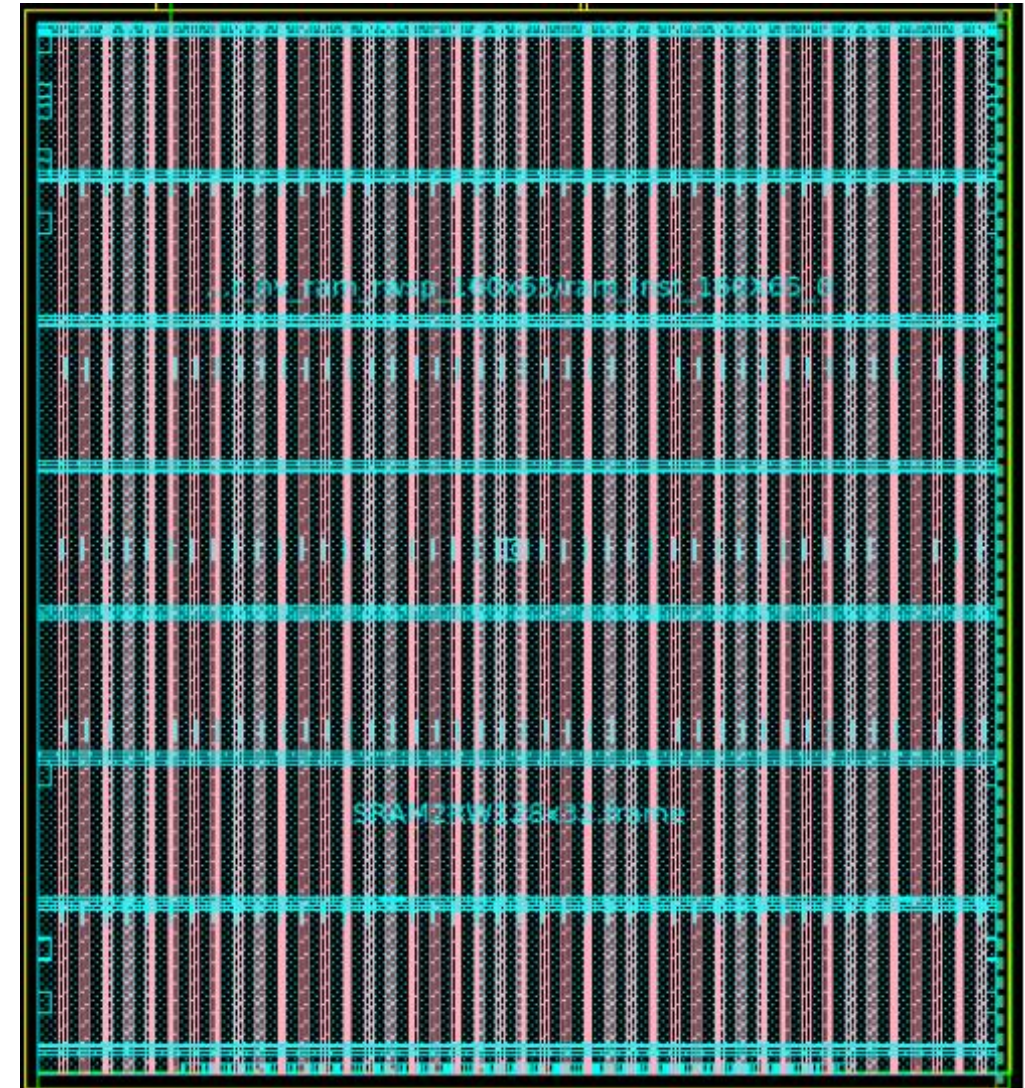




# Connecting Macro PG Pins via PG Region Straps

- Dropping M6 & M7 straps onto the macro pins achieves **15%** connection rate for macro PG
- Power grid allocation consumes **~19.9** routing area, consuming high routing resources.

| layer | Shape   | Spacing | Total   |
|-------|---------|---------|---------|
| P0    | 0.000%  | 0.000%  | 0.000%  |
| M1    | 0.000%  | 0.000%  | 0.000%  |
| M2    | 0.000%  | 0.000%  | 0.000%  |
| M3    | 0.000%  | 0.000%  | 0.000%  |
| M4    | 0.000%  | 0.000%  | 0.000%  |
| M5    | 0.000%  | 0.000%  | 0.000%  |
| M6    | 10.294% | 6.331%  | 16.625% |
| M7    | 1.965%  | 1.315%  | 3.279%  |

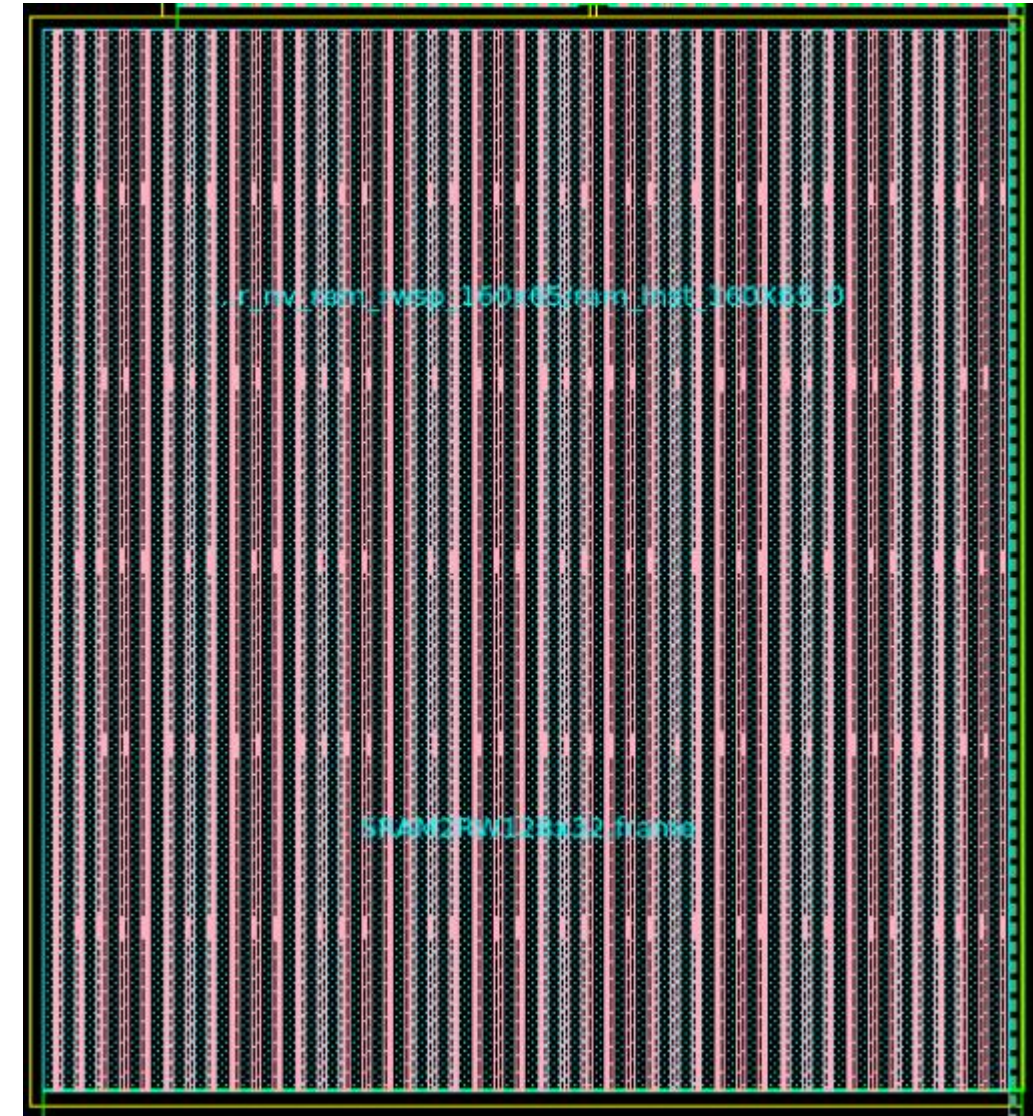




# Connecting Macro PG Pins via PG Region Wire Pattern

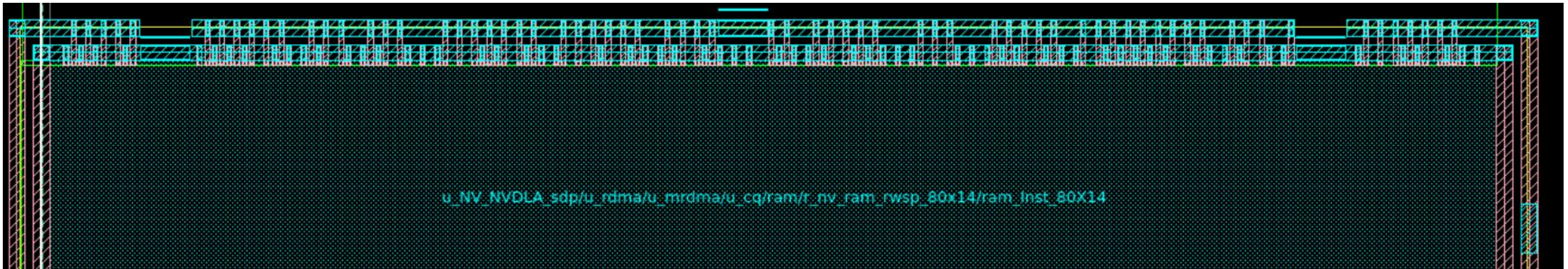
- Dropping M6 & M7 straps onto the macro pins achieves **15%** connection rate for macro PG
- Power grid allocation consumes **~16.6** routing area, consuming high routing resources.

| layer | Shape   | Spacing | Total   |
|-------|---------|---------|---------|
| P0    | 0.000%  | 0.000%  | 0.000%  |
| M1    | 0.000%  | 0.000%  | 0.000%  |
| M2    | 0.000%  | 0.000%  | 0.000%  |
| M3    | 0.000%  | 0.000%  | 0.000%  |
| M4    | 0.000%  | 0.000%  | 0.000%  |
| M5    | 0.000%  | 0.000%  | 0.000%  |
| M6    | 10.330% | 6.269%  | 16.598% |





- Power Rings on M6 & M7 onto the macro pins achieves **75%** connection rate for macro power delivery
- Power grid allocation consumes only **~3.2%** of total M6 & M7 routing area, keeping optimal routing resources available for signal nets

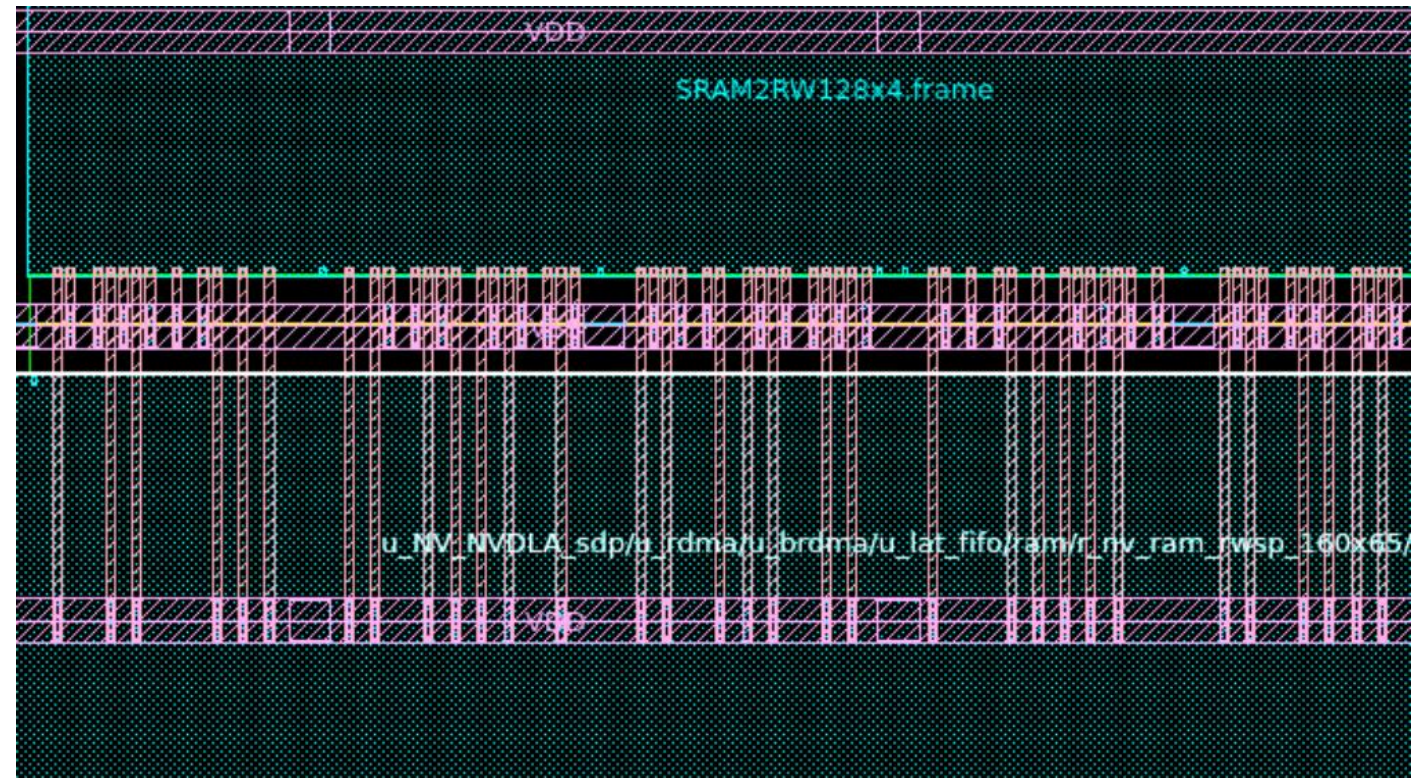


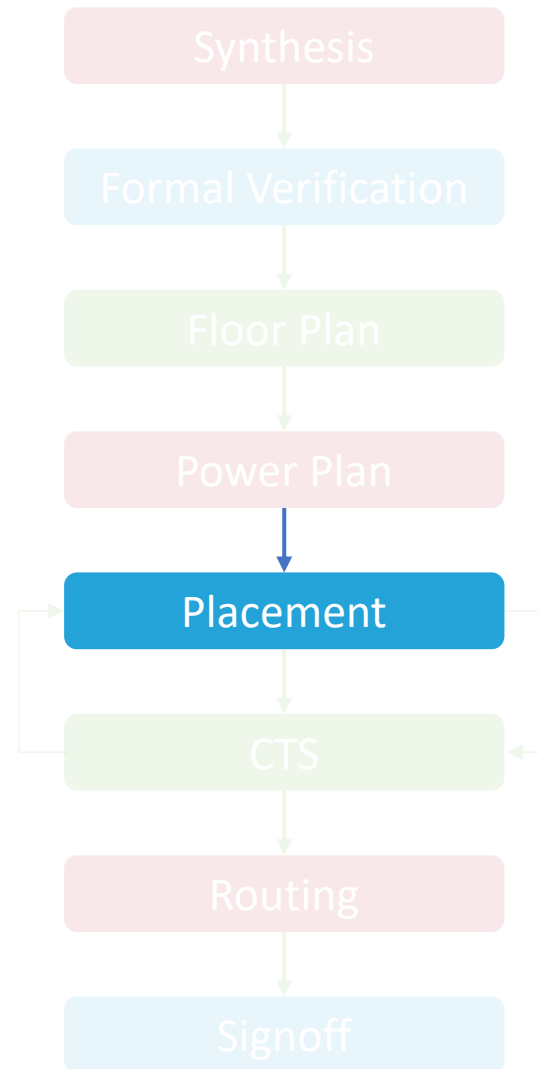
|    |  |        |  |        |  |        |
|----|--|--------|--|--------|--|--------|
| M6 |  | 1.242% |  | 0.716% |  | 1.958% |
| M7 |  | 1.008% |  | 0.317% |  | 1.325% |



- Dropping straps on M6 onto the macro pins achieves **95%** connection rate for macro power delivery
- Power grid allocation consumes only **~1% (~1.061%)** of total M6 routing area, keeping **optimal routing resources** available for signal nets

| layer | Shape  | Spacing | Total  |
|-------|--------|---------|--------|
| P0    | 0.000% | 0.000%  | 0.000% |
| M1    | 6.758% | 0.000%  | 6.758% |
| M2    | 0.181% | 0.505%  | 0.686% |
| M3    | 0.456% | 0.000%  | 0.456% |
| M4    | 0.182% | 0.508%  | 0.690% |
| M5    | 0.465% | 0.010%  | 0.476% |
| M6    | 0.401% | 0.660%  | 1.061% |





1. Pre-placement Checks
2. Scenario Filtering
3. Boundary Cells (Decaps)
4. Application Options
5. High-Fanout Synthesis (HFS)
6. Tie Cells and Spare Cells

**Placement** : is the process of placing the standard cells inside the core in an **optimal location**

- Ensure the PG network connectivity is good (**check\_pg\_connectivity**)
- Catch PG shorts/spacing issues early before placement (**check\_pg\_drc**)
- Ensure all required stack-ups are present for PG (**check\_pg\_missing\_vias**)
- Legality check before placement for Macros (**check\_legality**)
- check for issues that could block placement (**check\_design**)



## \*\*\*\*\*Verify net VDD connectivity\*\*\*

Number of floating wires: 0  
Number of floating vias: 0  
Number of floating std cells: 0  
Number of floating hard macros: 0  
Number of floating I/O pads: 0  
Number of floating terminals: 0  
Number of floating hierarchical blocks: 0

\*\*\*\*\*

## \*\*\*\*\*Verify net VSS connectivity\*\*\*

Number of floating wires: 0  
Number of floating vias: 0  
Number of floating std cells: 0  
Number of floating hard macros: 0  
Number of floating I/O pads: 0  
Number of floating terminals: 0  
Number of floating hierarchical blocks: 0

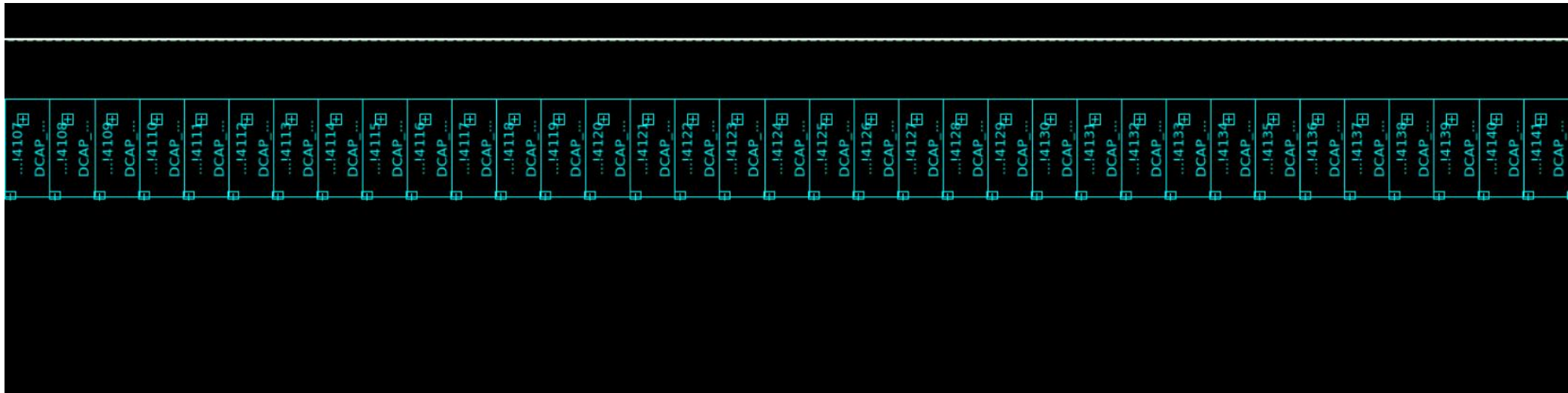
\*\*\*\*\*

```
icc2_shell> check_pg_drc -ignore_std_cells
Command check_pg_drc started at Sat Aug 8 17:07:17 2026
Command check_pg_drc finished at Sat Aug 8 17:07:31 2026
CPU usage for check_pg_drc: 14.08 seconds ( 0.00 hours)
Elapsed time for check_pg_drc: 14.08 seconds ( 0.00 hours)
No errors found.
```

```
icc2_shell> check_pg_missing_vias
Check net VDD vias...
Number of missing vias: 0
Checking net VDD vias took 1 seconds.
Check net VSS vias...
Number of missing vias: 0
Checking net VSS vias took 0 seconds.
Overall runtime: 1 seconds.
```

| Scenario                        | Active |
|---------------------------------|--------|
| func_slow_cmax (Late analysis)  | true   |
| func_fast_cmin (Early analysis) | false  |

- Using **Decaps on boundaries** compensates for **missing endcaps** while boosting power reliability
- **Boundary Protection:** shields gates from distortions
- **Dynamic IR Drop Reduction:** Functions as local charge reservoirs to supply instant current during simultaneous switching events



| App Option                                        | Value | Purpose                                                                                                                                                      |
|---------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| place.legalize.enable_advanced_legalizer          | True  | Enables advanced legalizer engine to prevent cell overlaps                                                                                                   |
| place.coarse.legalizer_driven_placement           | True  | legalization challenges during coarse placement to prevent large cell shifts later and keep timing predictable.                                              |
| place.coarse.auto_timing_control                  | True  | Dynamically balances timing improvement against congestion mitigation                                                                                        |
| opt.tie_cell.max_fanout,<br>opt.common.max_fanout | 5     | Limits the max fanout load connected to any Tie cell<br>Restricts the maximum fanout on general logic nets to 20 to control parasitic capacitance and delay. |
| opt.timing.effort                                 | high  | focus on reducing setup timing violations                                                                                                                    |
| place_opt.congestion.effort                       | high  | Enables congestion-reduction algorithms during place_opt to resolve routing bottlenecks.                                                                     |

Found 5904 buffer-tree drivers

Roi-HfsDrc SN: 3403505 (1174.419067)

Processing Buffer Trees (ROI) ...

[591] 10% ...

[1182] 20% ...

[1773] 30% ...

[2364] 40% ...

[2955] 50% ...

[3546] 60% ...

[4137] 70% ...

[4728] 80% ...

[5319] 90% ...

[5904] 100% Done

|            | Deleted | Added |
|------------|---------|-------|
| Buffers:   | 318     | 2646  |
| Inverters: | 1402    | 1498  |
| Total:     | 1720    | 4144  |

Number of Drivers Sized: 1588 [26.90%]

## Before HFNS

| NetName                                       | Fanout | Driver                                                                    |
|-----------------------------------------------|--------|---------------------------------------------------------------------------|
| -----                                         | -----  | -----                                                                     |
| csb2sdp_rdma_req_pvld                         | 33     | csb2sdp_rdma_req_pvld                                                     |
| csb2sdp_req_pvld                              | 33     | csb2sdp_req_pvld                                                          |
| nvdla_core_clk                                | 4926   | nvdla_core_clk                                                            |
| nvdla_core_rstn                               | 4816   | u_partition_p_reset/sync_reset_synced_rstn/UI_test_mode_outmux/U1/Y       |
| u_NV_NVDLA_sdp/reg2dp_interrupt_ptr           | 64     | u_NV_NVDLA_sdp/u_reg/U53/Y                                                |
| u_NV_NVDLA_sdp/reg2dp_output_dst              | 20     | u_NV_NVDLA_sdp/u_reg/U95/Y                                                |
| u_NV_NVDLA_sdp/u_rdma/u_mrdma/nvdla_gated_clk | 1833   | u_NV_NVDLA_sdp/u_rdma/u_mrdma/u_gate/nvdla_core_clk_slcg_0/p_clkgate/U3/Y |

## After HFNS

| NetName                                                                | Fanout | Driver                                                                               |
|------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------|
| -----                                                                  | -----  | -----                                                                                |
| nvdla_core_clk                                                         | 4926   | nvdla_core_clk                                                                       |
| u_NV_NVDLA_sdp/u_rdma/u_mrdma/nvdla_gated_clk                          | 1833   | u_NV_NVDLA_sdp/u_rdma/u_mrdma/u_gate/nvdla_core_clk_slcg_0/p_clkgate/U3/Y            |
| u_NV_NVDLA_sdp/u_rdma/u_mrdma/u_cq/nvdla_core_clk_mgated               | 33     | u_NV_NVDLA_sdp/u_rdma/u_mrdma/u_cq/nvdla_core_clk_mgate/p_clkgate/U3/Y               |
| u_NV_NVDLA_sdp/u_rdma/u_mrdma/u_eg/u_cmd/u_sfifo/nvdla_core_clk_mgated | 63     | u_NV_NVDLA_sdp/u_rdma/u_mrdma/u_eg/u_cmd/u_sfifo/nvdla_core_clk_mgate/p_clkgate/U3/Y |
| u_NV_NVDLA_sdp/u_rdma/u_mrdma/u_eg/u_cmd/u_dfifo/nvdla_core_clk_mgated | 67     | u_NV_NVDLA_sdp/u_rdma/u_mrdma/u_eg/u_cmd/u_dfifo/nvdla_core_clk_mgate/p_clkgate/U3/Y |



## Before HFNS

| NET FANOUT AND PIN COUNT INFORMATION |          |             |          |
|--------------------------------------|----------|-------------|----------|
| Fanout                               | Netcount | netPinCount | NetCount |
| <2                                   | 88026    | <2          | 8746     |
| 2                                    | 30141    | 2           | 79298    |
| 3                                    | 14919    | 3           | 30123    |
| 4                                    | 4131     | 4           | 14924    |
| 5                                    | 1900     | 5           | 4127     |
| 6-10                                 | 3385     | 6-10        | 4890     |
| 11-20                                | 4812     | 11-20       | 5085     |
| 21-30                                | 838      | 21-30       | 884      |
| 31-50                                | 1168     | 31-50       | 1243     |
| 51-100                               | 54       | 51-100      | 54       |
| 101-500                              | 39       | 101-500     | 39       |
| 501-1000                             | 2        | 501-1000    | 2        |
| >1000                                | 6        | >1000       | 6        |

## After HFNS

| NET FANOUT AND PIN COUNT INFORMATION |          |             |          |
|--------------------------------------|----------|-------------|----------|
| Fanout                               | Netcount | netPinCount | NetCount |
| <2                                   | 109208   | <2          | 631      |
| 2                                    | 37985    | 2           | 108577   |
| 3                                    | 26095    | 3           | 37985    |
| 4                                    | 23691    | 4           | 26095    |
| 5                                    | 21895    | 5           | 23691    |
| 6-10                                 | 7        | 6-10        | 21902    |
| 11-20                                | 3        | 11-20       | 3        |
| 21-30                                | 7        | 21-30       | 5        |
| 31-50                                | 1        | 31-50       | 3        |
| 51-100                               | 39       | 51-100      | 39       |
| 101-500                              | 38       | 101-500     | 38       |
| 501-1000                             | 2        | 501-1000    | 2        |
| >1000                                | 5        | >1000       | 5        |



```
saed32rvt_c/TIEL_RVT saed32rvt_c/TIEL_RVT boolean dont_use true
```

- Unlocks Tie-High (TIEH) and Tie-Low (TIEL) cells ( dont\_use=false) so the tool can place them
- Runs add\_tie\_cells to connect static gate inputs to dedicated tie cells instead of direct Vdd/Vss rails, protecting gates noise spikes
- Runs legalize\_placement to align newly added tie cells onto the placement grid.

## Before Tie Cell Insertion

|          |   |
|----------|---|
| Tie Low  | 0 |
| Tie High | 0 |
| Others   | 0 |

## After Tie Cell Insertion

|          |     |
|----------|-----|
| Tie Low  | 461 |
| Tie High | 24  |
| Others   | 0   |

- Instantiated 50 base gates (NAND2, AND2, INV) distributed randomly across the core while tying inputs to Tie Low to prevent floating gate noise and leakage power
- Legalized their locations without disturbing active functional logic
- Locked down physical locations and attributes of spare cells to ensure subsequent placement and routing passes preserve them for post-silicon ECOs

## Before Spare Cell Insertion

|                 |       |
|-----------------|-------|
| Spare cells     | 0     |
| ICG cells       | 0     |
| Flip-flop cells | 14218 |
| Latch cells     | 0     |
| Antenna cells   | 0     |

## After Spare Cell Insertion

|                 |       |
|-----------------|-------|
| Spare cells     | 50    |
| ICG cells       | 0     |
| Flip-flop cells | 14218 |
| Latch cells     | 0     |
| Antenna cells   | 0     |

# Problems and Solutions

# 1<sup>st</sup> Problem: No Valid scan def found

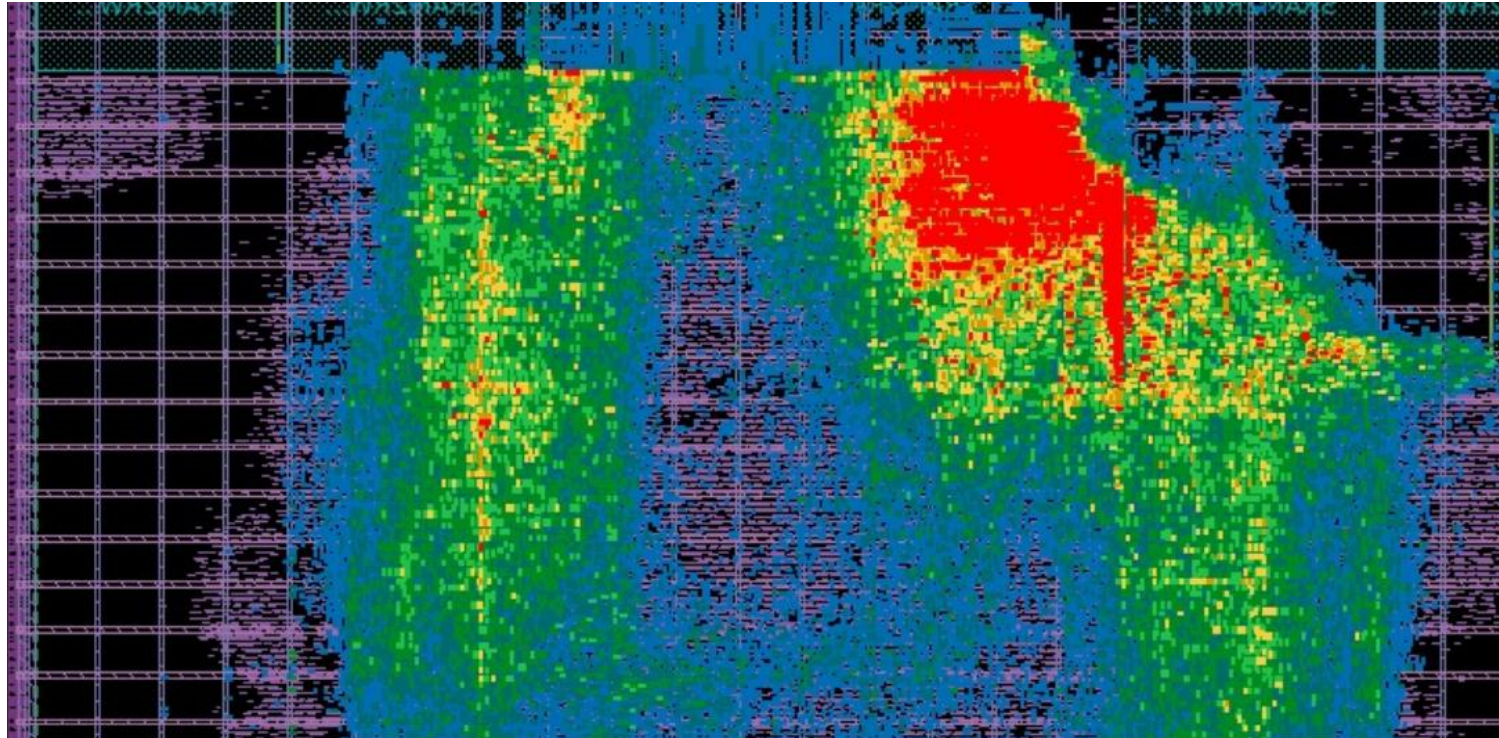
- The placer defaults to scan-chain aware to group ffs in the same scan chain close together
- Setting **continue\_on\_missing\_scandef true** allows place\_opt to complete without crashing, but scan chain ffs will be placed based on functional logic connectivity rather than scan ordering

**If you need continue without def file**

```
set_app_options -list {place.coarse.continue_on_missing_scandef {true}}
```

```
Error: No valid scan def found. (PLACE-042)  
Error: Optimization failed at create_placement  
Error: place_opt failed during initial placement.  
Error: place_opt failed during create_placement
```

- Insufficient cell spreading resulted in **high-density** regions, creating multiple placement **hotspots** in specific areas



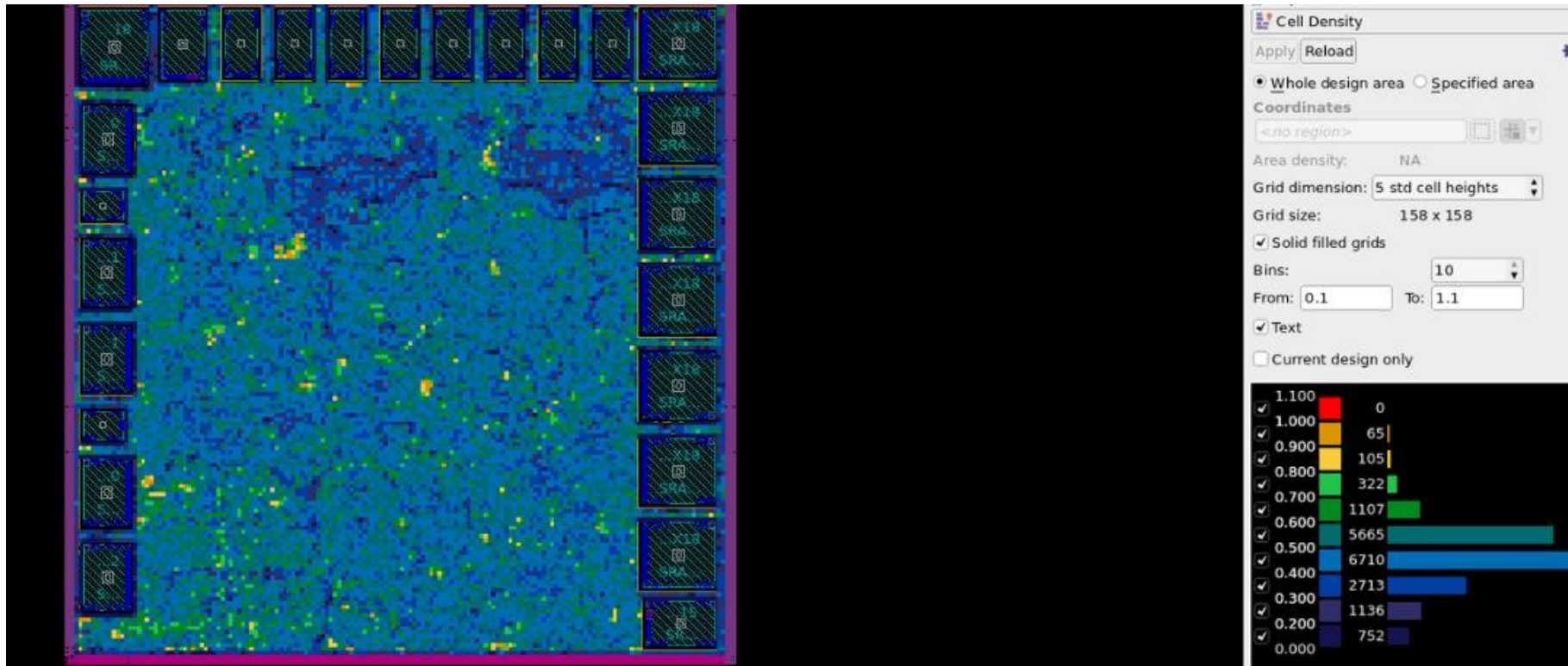


## 2<sup>nd</sup> Problem: Congestion

**We increased cell spreading to eliminate placement hotspots**

```
set_app_options -name place.coarse.max_density -value 0.3
```

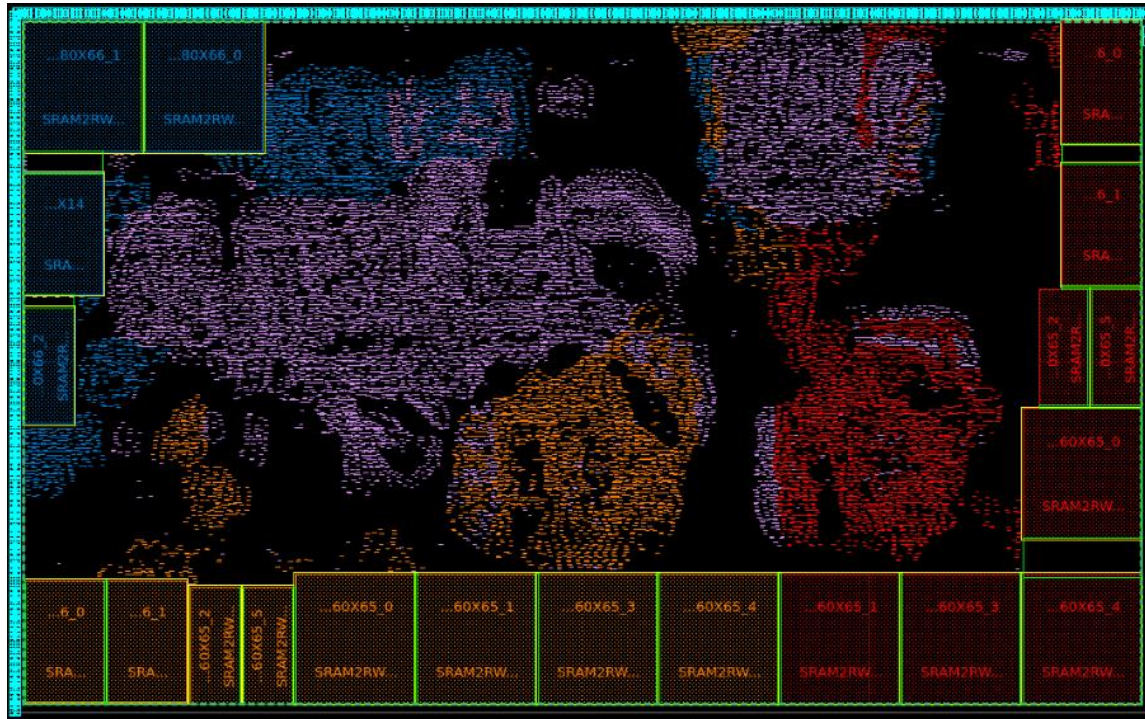
```
set_app_options -name place.coarse.congestion_driven_max_util -value 0.4
```



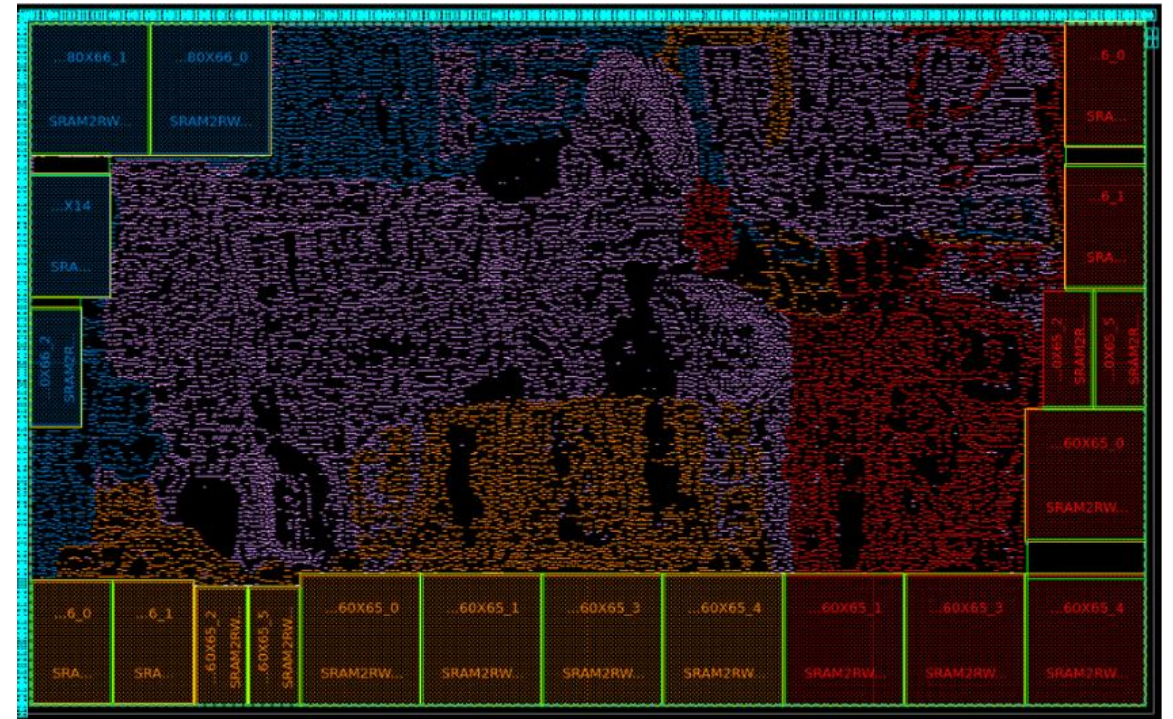


## 2<sup>nd</sup> Problem: Congestion

Before Cell Spreading



After Cell Spreading





# [3<sup>rd</sup> Problem] Max Capacitance and Max Transition Violations

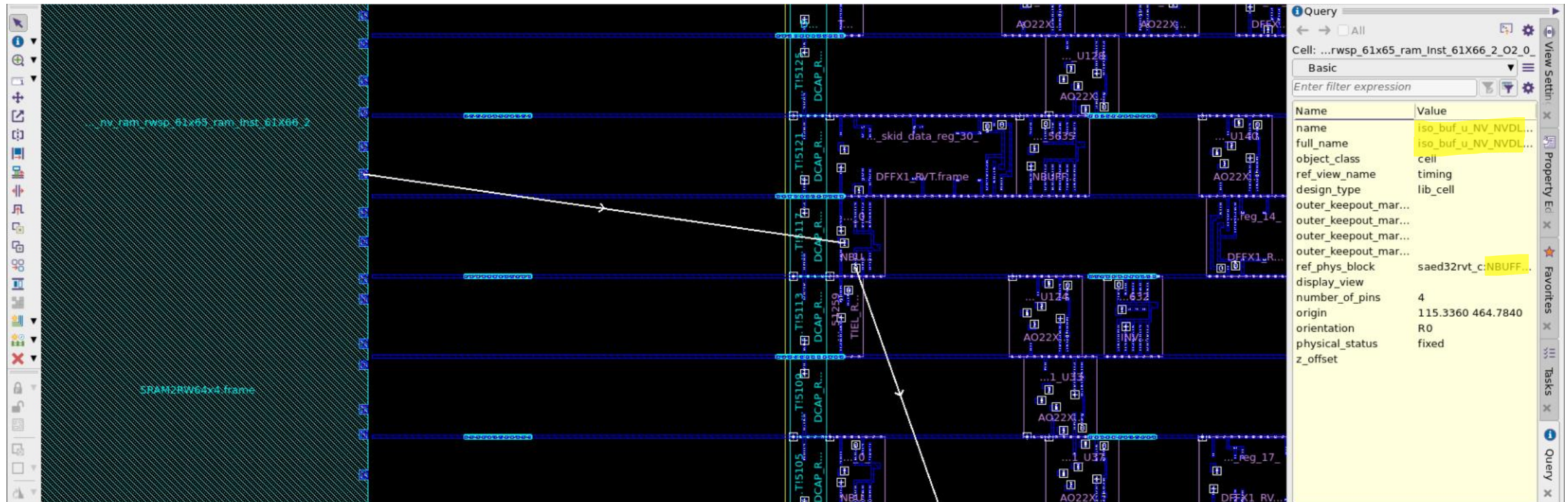
- We observed **maximum capacitance and max transition violations on the macro pins** due to excessive net lengths and the high input capacitance of the first-stage standard cells
- Consequently, the macro pins could not drive these loads effectively, resulting in max capacitance and max transition violations

| max_capacitance                                                                             | Required Capacitance | Actual Capacitance | Slack   | Violation  |
|---------------------------------------------------------------------------------------------|----------------------|--------------------|---------|------------|
| Net                                                                                         |                      |                    |         |            |
| u_NV_NVDLA_pdp_u_core_u_cal2d_mem_rdata_1[13]                                               | 25.00                | 147.16             | -122.16 | (VIOLATED) |
| PIN : u_NV_NVDLA_pdp_u_core_u_cal2d_bank1_uram_0_r_nv_ram_rws_128x18_ram_Inst_128X18/O2[13] | 25.00                | 147.16             | -122.16 | (VIOLATED) |
| u_NV_NVDLA_pdp_u_core_u_cal2d_mem_rdata_1[11]                                               | 25.00                | 147.00             | -122.00 | (VIOLATED) |
| PIN : u_NV_NVDLA_pdp_u_core_u_cal2d_bank1_uram_0_r_nv_ram_rws_128x18_ram_Inst_128X18/O2[11] | 25.00                | 147.00             | -122.00 | (VIOLATED) |
| u_NV_NVDLA_pdp_u_core_u_cal2d_mem_rdata_1[6]                                                | 25.00                | 146.28             | -121.28 | (VIOLATED) |
| PIN : u_NV_NVDLA_pdp_u_core_u_cal2d_bank1_uram_0_r_nv_ram_rws_128x18_ram_Inst_128X18/O2[6]  | 25.00                | 146.28             | -121.28 | (VIOLATED) |
| u_NV_NVDLA_pdp_u_core_u_cal2d_mem_rdata_1[5]                                                | 25.00                | 145.51             | -120.51 | (VIOLATED) |
| PIN : u_NV_NVDLA_pdp_u_core_u_cal2d_bank1_uram_0_r_nv_ram_rws_128x18_ram_Inst_128X18/O2[5]  | 25.00                | 145.51             | -120.51 | (VIOLATED) |
| u_NV_NVDLA_pdp_u_core_u_cal2d_mem_rdata_1[7]                                                | 25.00                | 145.34             | -120.34 | (VIOLATED) |
| PIN : u_NV_NVDLA_pdp_u_core_u_cal2d_bank1_uram_0_r_nv_ram_rws_128x18_ram_Inst_128X18/O2[7]  | 25.00                | 145.34             | -120.34 | (VIOLATED) |
| u_NV_NVDLA_pdp_u_core_u_cal2d_mem_rdata_1[8]                                                | 25.00                | 144.37             | -119.37 | (VIOLATED) |
| PIN : u_NV_NVDLA_pdp_u_core_u_cal2d_bank1_uram_0_r_nv_ram_rws_128x18_ram_Inst_128X18/O2[8]  | 25.00                | 144.37             | -119.37 | (VIOLATED) |

- Constrained maximum wire lengths to reduce interconnect capacitance and optimize path delay using:

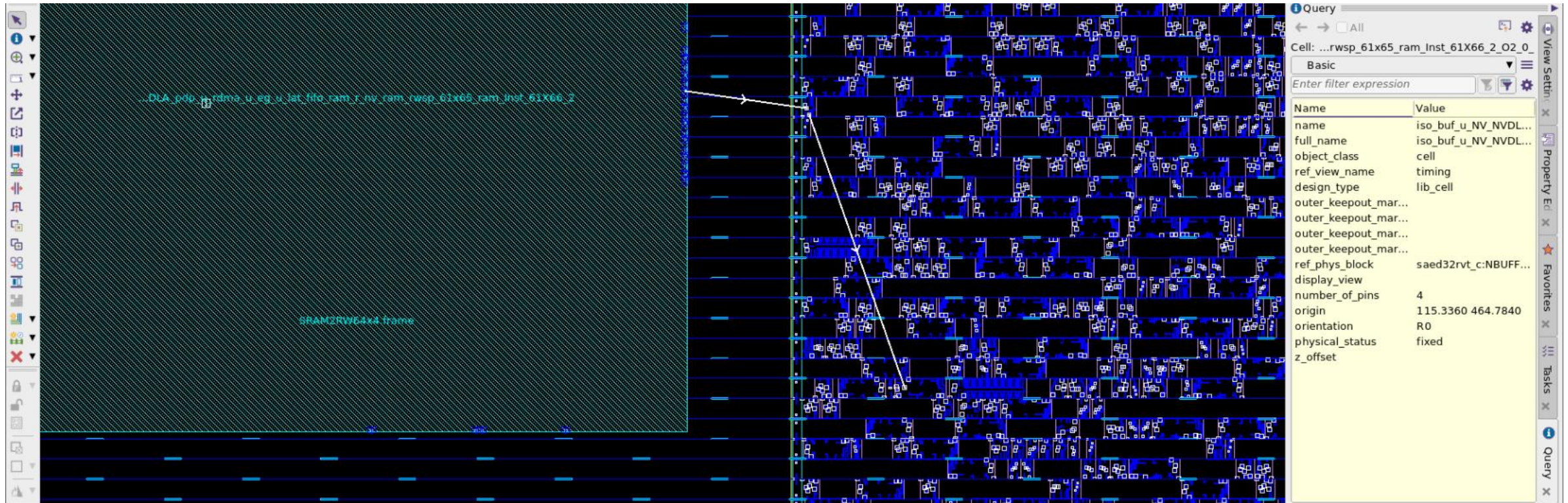
```
set_app_options -list { opt.common.max.net.length {150} }
```

- Added isolation buffers at macro pins to boost drive strength and isolate heavy interconnect loads

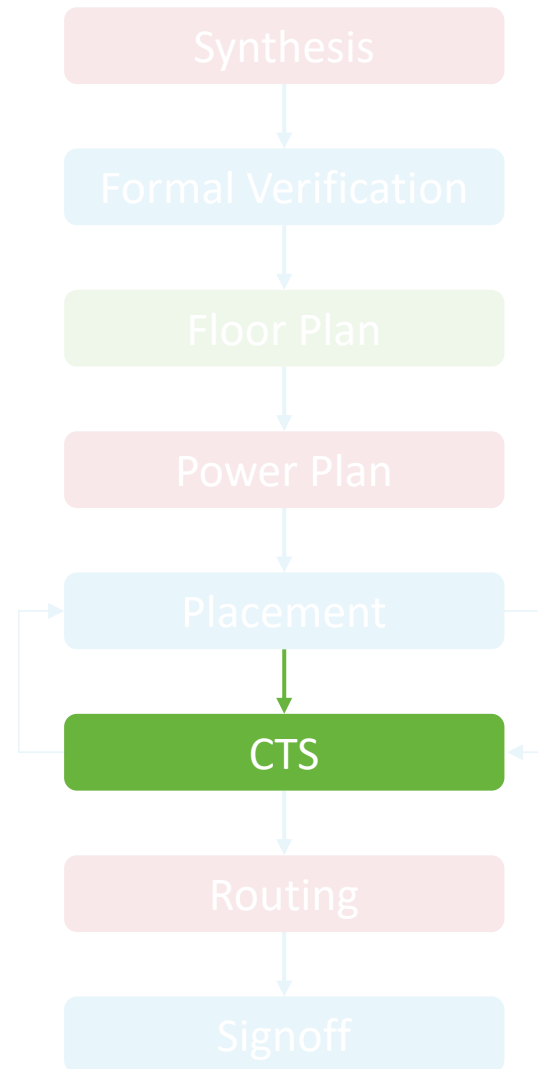




- Grouped interacting standard cells closer to macro boundaries using magnet placement







1. PreCTS Checks
2. Preparation Steps For CTS
3. Application Options
4. Results

1. Design is placed
2. Power and Ground nets are routed
3. Estimated congestion (acceptable)
4. Estimated time for setup (acceptable )
5. High Fanout Synthesis is done
6. Clocks has been defined

## 1- Activation for all scenarios

- Before CTS

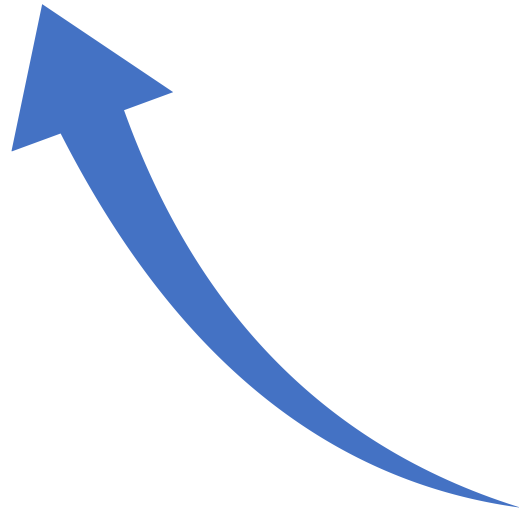
| Scenario       | Active | Setup | Hold  |
|----------------|--------|-------|-------|
| func_slow_cmax | true   | true  | false |
| func_fast_cmin | false  | false | true  |

- During CTS

| Scenario       | Active | Setup | Hold |
|----------------|--------|-------|------|
| func_slow_cmax | true   | true  | true |
| func_fast_cmin | true   | true  | true |

## 2- Specifying metal layers used in CTS

Max\_Layer :M7

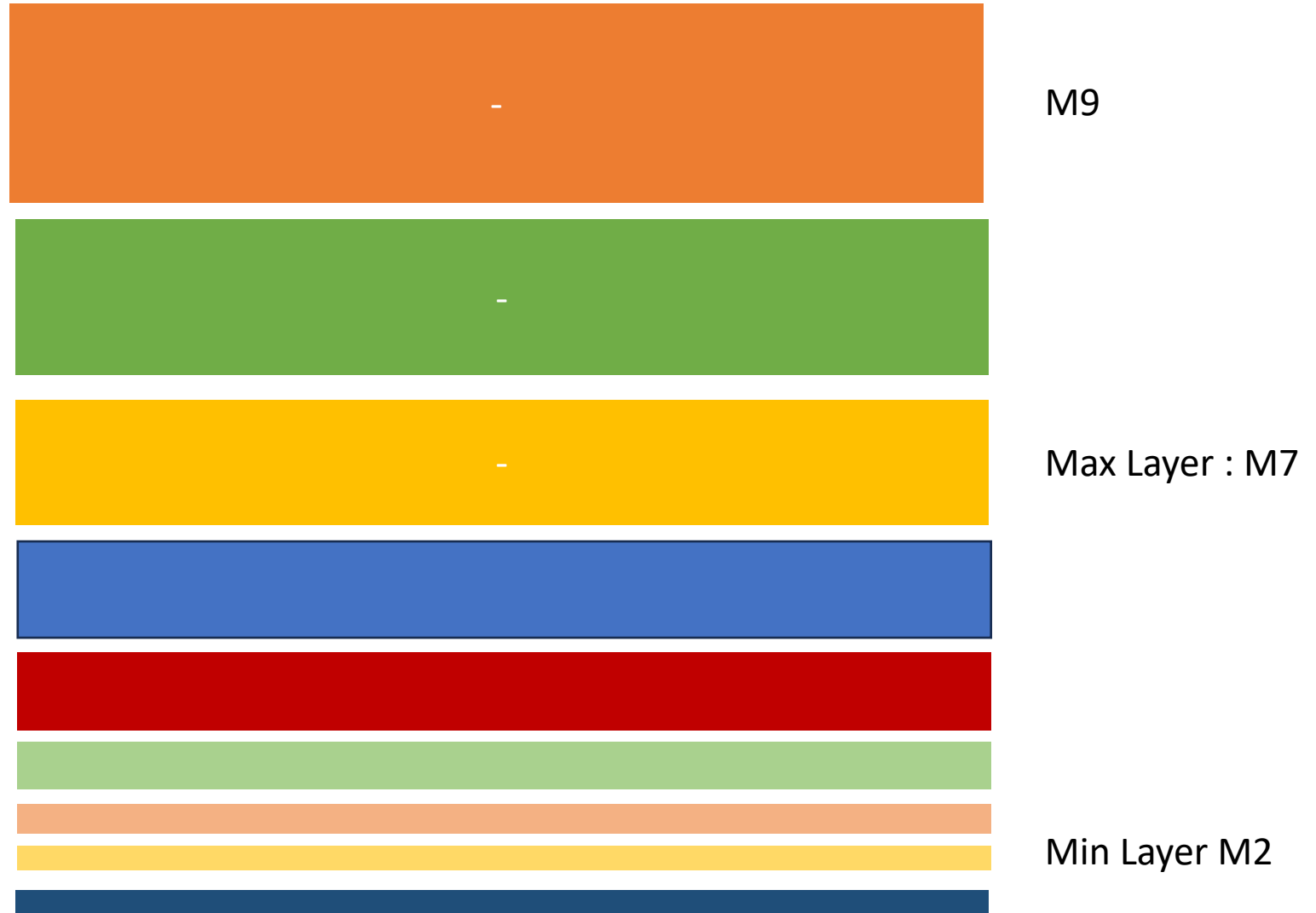


Min\_Layer :M2





## 2- Specifying metal layers used in CTS



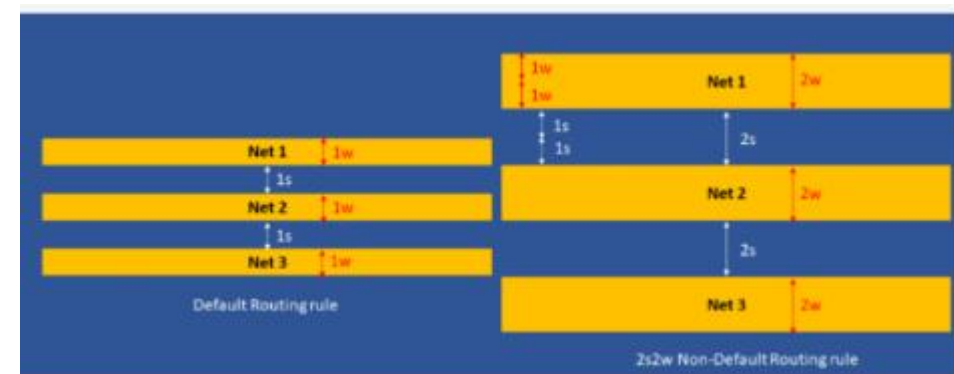
## 3- Specify Target skew ,Target latency ,Max Cap ,Max transition

- Specify them for all the scenarios

## 4- Non Default Routing (NDR)

- Specify double width and double spacing for the clock net **From M7 till M3 only**

- Default width and spacing from M3 till M1



## 5- Choose cells used for building clock tree , fix hold and optimization

- Our PDK doesn't have **Clock buffers**
- We had two options either to use normal inverters or normal buffers
- We made a comparison between them and we took **Partition C** as a sample for that to decide which cells would we choose to build clock tree
- During that comparison we put the same constraints for both :
  - Target\_skew = 0.2 ns
  - Target\_Latency = 0.2ns

# Preparation Steps For CTS

|                                                  | Built with inverters | Built with buffers |
|--------------------------------------------------|----------------------|--------------------|
| Acheived<br>Global_skew<br>(slow corner) (ns)    | <b>0.48</b>          | <b>0.64</b>        |
| Acheived<br>Global_skew<br>(fast corner) (ns)    | <b>0.17</b>          | <b>0.21</b>        |
| Acheived<br>Global_latency<br>(slow corner) (ns) | <b>2.06</b>          | <b>5.42</b>        |
| Acheived<br>Global_latency<br>(fast corner) (ns) | <b>0.81</b>          | <b>1.94</b>        |
| no. of cells                                     | <b>35,043</b>        | <b>41,607</b>      |
| area after cts (um^2)                            | <b>5822908.39</b>    | <b>5826890.82</b>  |

Were the values expected ?

Yes , according to the values of the transition time for inverter and buffers from lib

| Cells         | INV2X_RVT | NBUFF2X_RVT |
|---------------|-----------|-------------|
| Max_tran (ns) | 0.438     | 0.1948      |
| Max_cap (ff)  | 16        | 16          |

So, accordingly we chose **Inverters** to build our clock tree



Before we include those cells we exclude all cells from cts ,hold and optimization

For building our clock tree :

- Inverters (x2 till x16)

For fixing hold :

- Delay cells

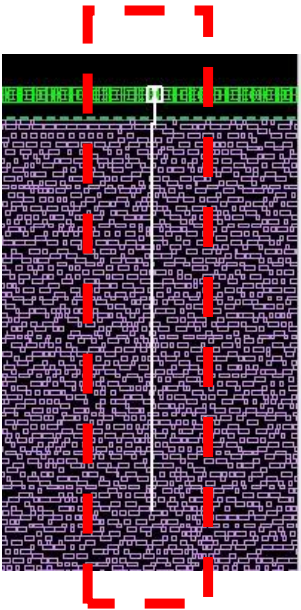
**For optimization (fixation) :**

- ~~Inverters & buffers (x2 till x16)~~



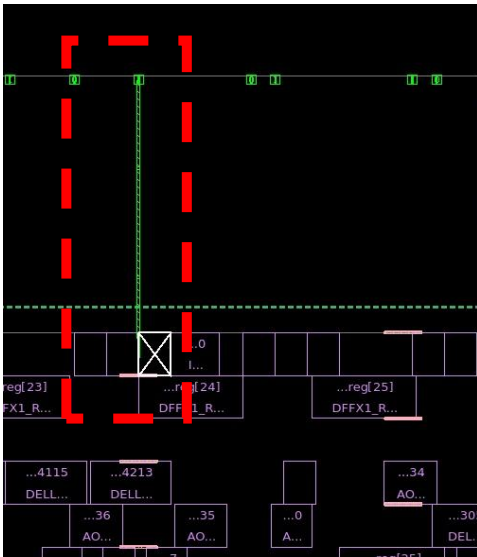
Made a **problem**

| App option                    | Value | Usage                                                                      |
|-------------------------------|-------|----------------------------------------------------------------------------|
| cts.common.max_fanout         | 15-20 | Determine allowable max fanout during cts                                  |
| clock_opt.flow.enable_ccd     | true  | Enables CCD that uses useful skew to improve timing                        |
| time.enable_si_timing_windows | true  | Enables signal integrity that protects electrical quality of clock signals |
| time.si_enable_analysis       | true  | Enables signal integrity that protects electrical quality of clock signals |
| clock_opt.flow.hold.effort    | high  | Determine the effort for solving hold                                      |



| Name                   | Value                                         |
|------------------------|-----------------------------------------------|
| effective_process...   | (none)                                        |
| effective_process...   | (none)                                        |
| effective_process...   | 0.000000                                      |
| effective_process...   | 0.000000                                      |
| effective_tempera...   | -40.000000                                    |
| effective_tempera...   | -40.000000                                    |
| effective_voltage...   | 0.950000                                      |
| effective_voltage...   | 0.950000                                      |
| escaped_full_name      | nvdla_core_clk                                |
| extended_name          | /home/ICer/ITI_GP/saed_32/pnr/2-design_lib/re |
| fanout_load            |                                               |
| full_name              | nvdla_core_clk                                |
| gate_area              |                                               |
| gate_diffusion_len...  |                                               |
| glitch_rate            | 0                                             |
| height                 | 0.2860                                        |
| inertial_glitch_equ... | 0                                             |
| is_bus_bit             | false                                         |

Before app option



Cell: iso\_buf\_nvdla\_core\_clk  
QueryText

Enter filter expression

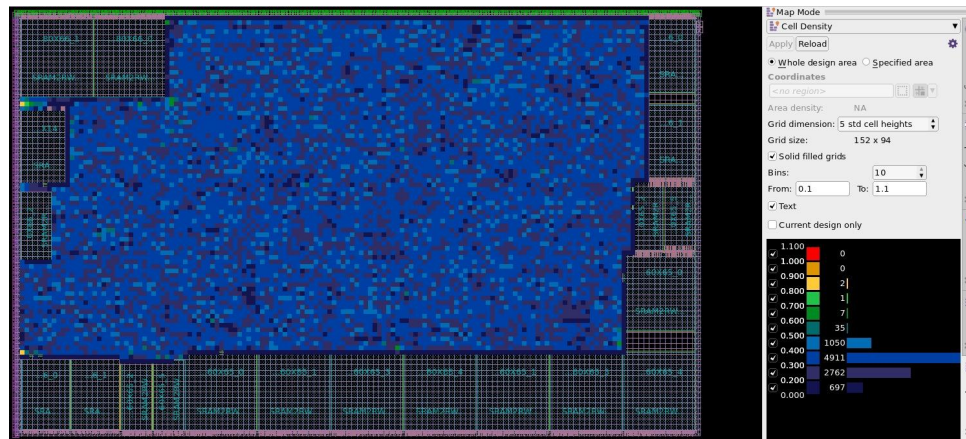
| Name                   | Value                                                       |
|------------------------|-------------------------------------------------------------|
| abstract_dont_touch    | false                                                       |
| allowable_orientat...  | R0 R90 R180 R270 MX MXR90 MY MYR90                          |
| area                   | 2.03315200                                                  |
| aspect_ratio           | 1.375                                                       |
| base_name              | iso_buf_nvdla_core_clk                                      |
| bbox                   | {420.8930 769.0580} {422.3390 770.8610}                     |
| block_grid             |                                                             |
| blockage_group_id      | 0                                                           |
| boundary               | {421.0080 769.0880} {421.0080 770.7600} {422.2240 770.7600} |
| boundary_bbox          | {421.0080 769.0880} {422.2240 770.7600}                     |
| boundary_boundin...    | {421.0080 769.0880} {422.2240 770.7600}                     |
| bounding_box           | {420.8930 769.0580} {422.3390 770.8610}                     |
| busplan_trace_thr...   | false                                                       |
| color                  | 0                                                           |
| cts_dont_touch         | false                                                       |
| cts_size_in_place      | false                                                       |
| design_type            | lib_cell                                                    |
| display_view           |                                                             |
| dont_touch             | false                                                       |
| early_fall_cell_che... | hierarchical source                                         |
| early_fall_clk_cell... | hierarchical source                                         |
| early_fall_data_cel... | hierarchical source                                         |
| early_rise_cell_che... | hierarchical source                                         |
| early_rise_clk_cell... | hierarchical source                                         |
| early_rise_data ce...  | hierarchical source                                         |

After app option

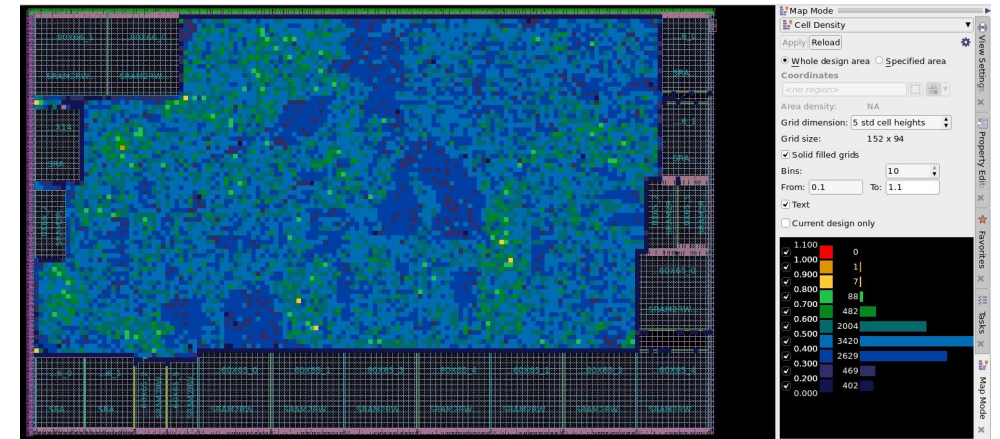
| App option                | Value | Usage                                                      |
|---------------------------|-------|------------------------------------------------------------|
| cts.common.max_net_length | 150   | Determine the allowable max length for the nets of the cts |

After running cts we will find that :

- Hold violations would decrease if compared before cts
- Clock would be now propagated
- Cell density would increase



Before CTS



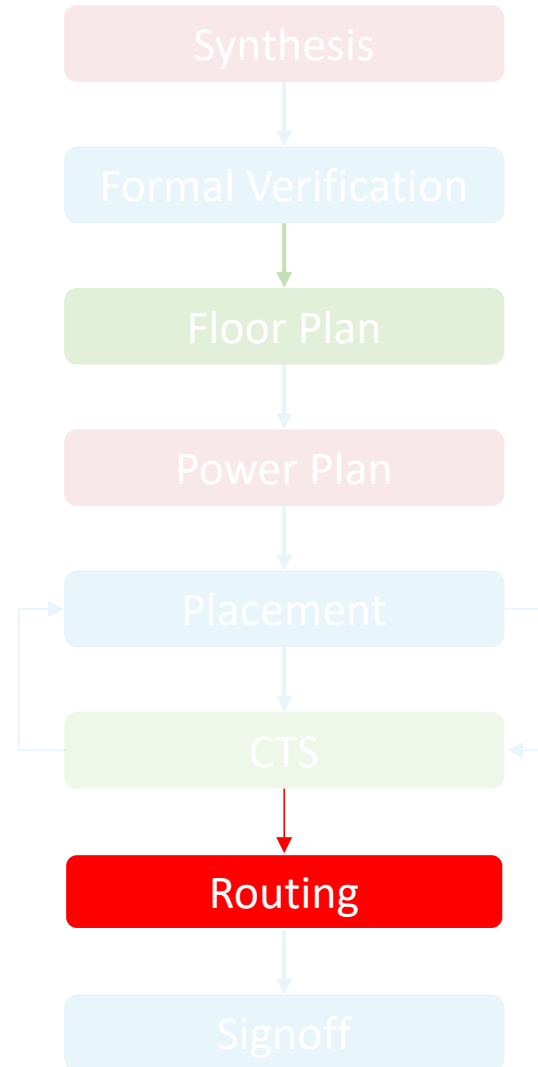
After CTS

For slow corner :

|                  | Partition_C |
|------------------|-------------|
| Setup_before_CTS | 13          |
| Setup_after_CTS  | 3           |
| Hold_before_CTS  | 0           |
| Hold_after_CTS   | 0           |

For fast corner:

|                  | Partition_C |
|------------------|-------------|
| Setup_before_CTS | 0           |
| Setup_after_CTS  | 0           |
| Hold_before_CTS  | 17045       |
| Hold_after_CTS   | 125         |





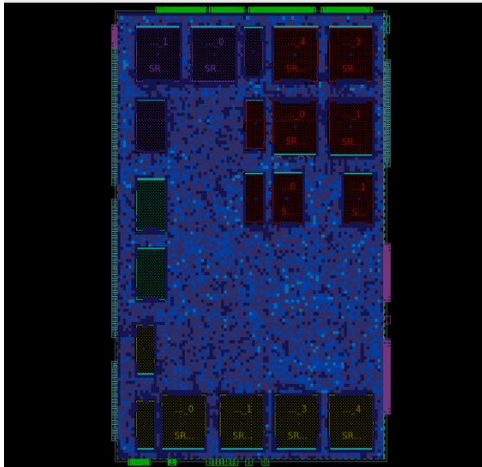
1. Routing Post-Rout ECOs
2. Chip Finishing

# Routing and Post-Route ECOs

## Post-CTS

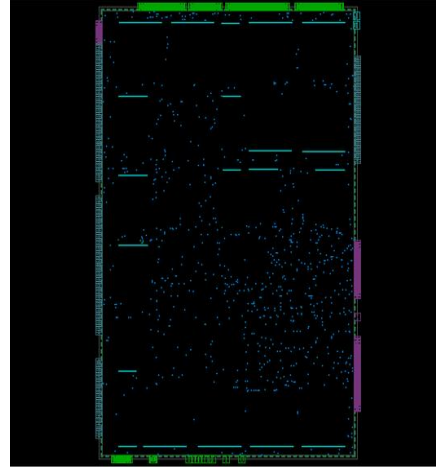
- **The Ultimate Test:** After we finished building the clock tree, we proceeded to the next step. This step ultimately determines the quality and viability of all the previous physical design steps.

Pin density map



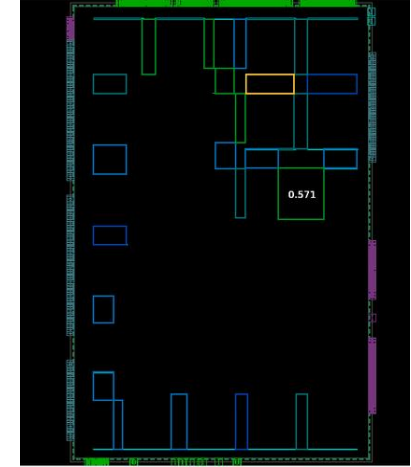
Maximum density 0.4

Global route congestion map



Maximum density 0.2

channel congestion map



Maximum density 0.6

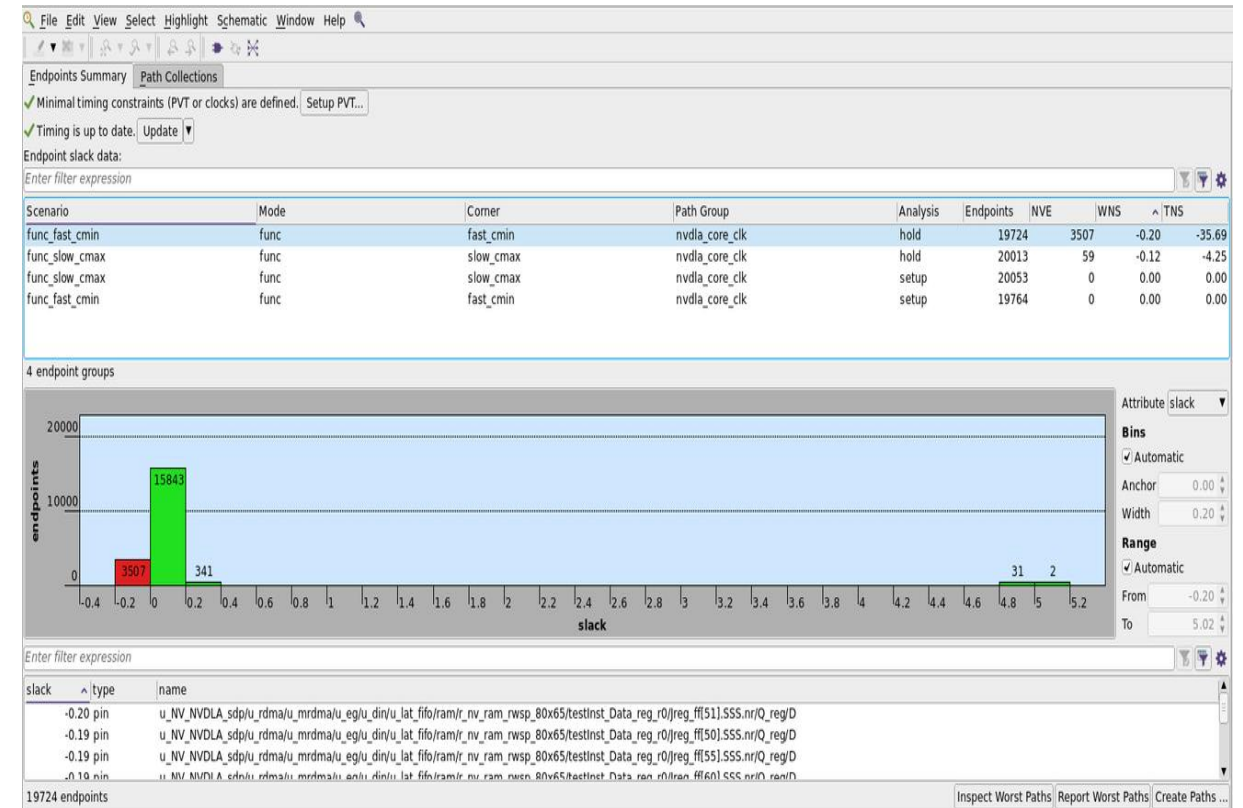
## Post-Route QoR Analysis

- **First Assessment:** After the routing was completed, we performed a QoR analysis to evaluate the physical and timing quality of the design before starting the ECO phase.
- The report showed that although the routing was completed, we still had several maximum capacitance and transition violations that needed to be fixed before final sign-off.
- This analysis gave us a clear starting point for the optimization and ECO iterations.

| Violation type       | Before ECOs |
|----------------------|-------------|
| Nets with Violations | 840         |
| Max Trans Violations | 44          |
| Max Cap Violations   | 837         |
| Setup                | Met         |
| Hold                 | Violations  |

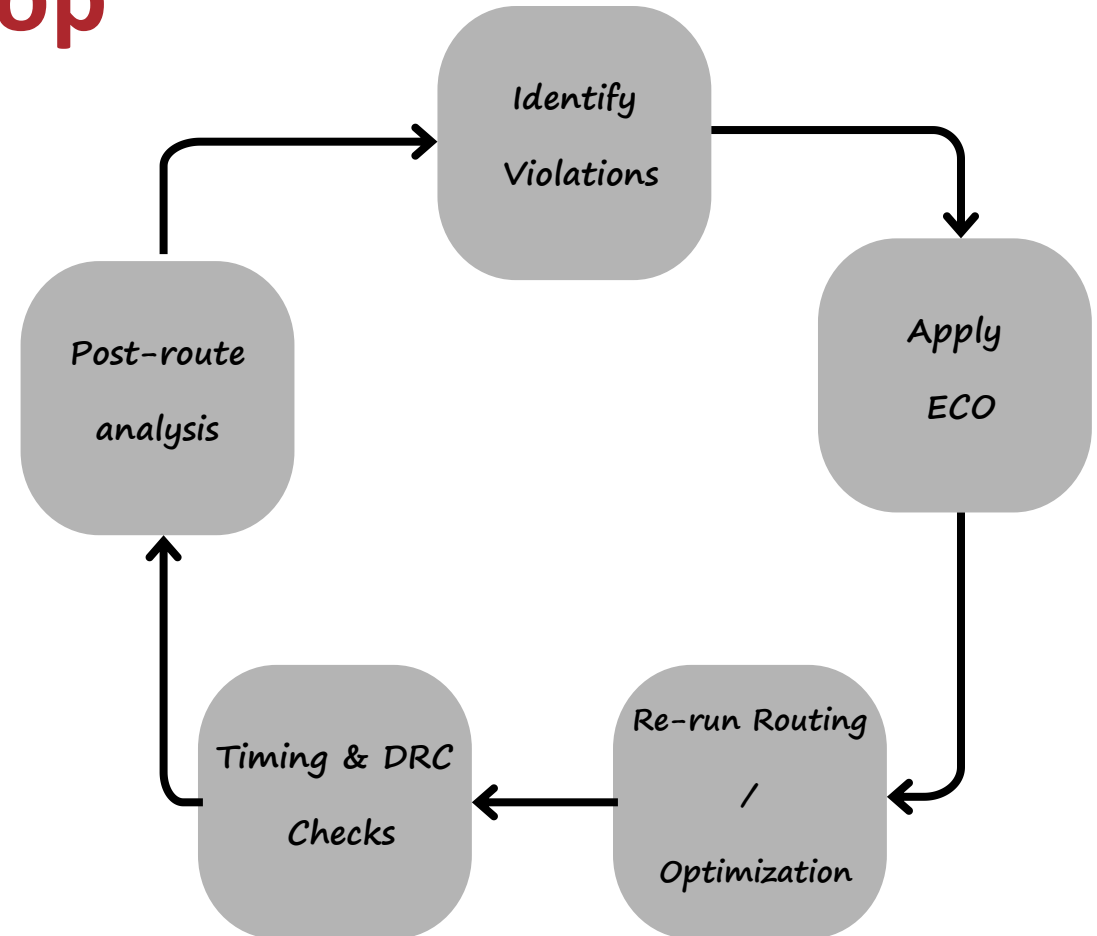
## Post-Route Timing Analysis

- Finally, once the physical routing was clean, we started the Engineering Change Orders (ECOs) phase to meet our stringent setup and hold timing requirements.
- All timing requirement must be met for all sign-off corners
- This screen shows us that we have met the setup requirement for all scenarios, But the hold requirement is not met.



## Post-Route Optimization Loop

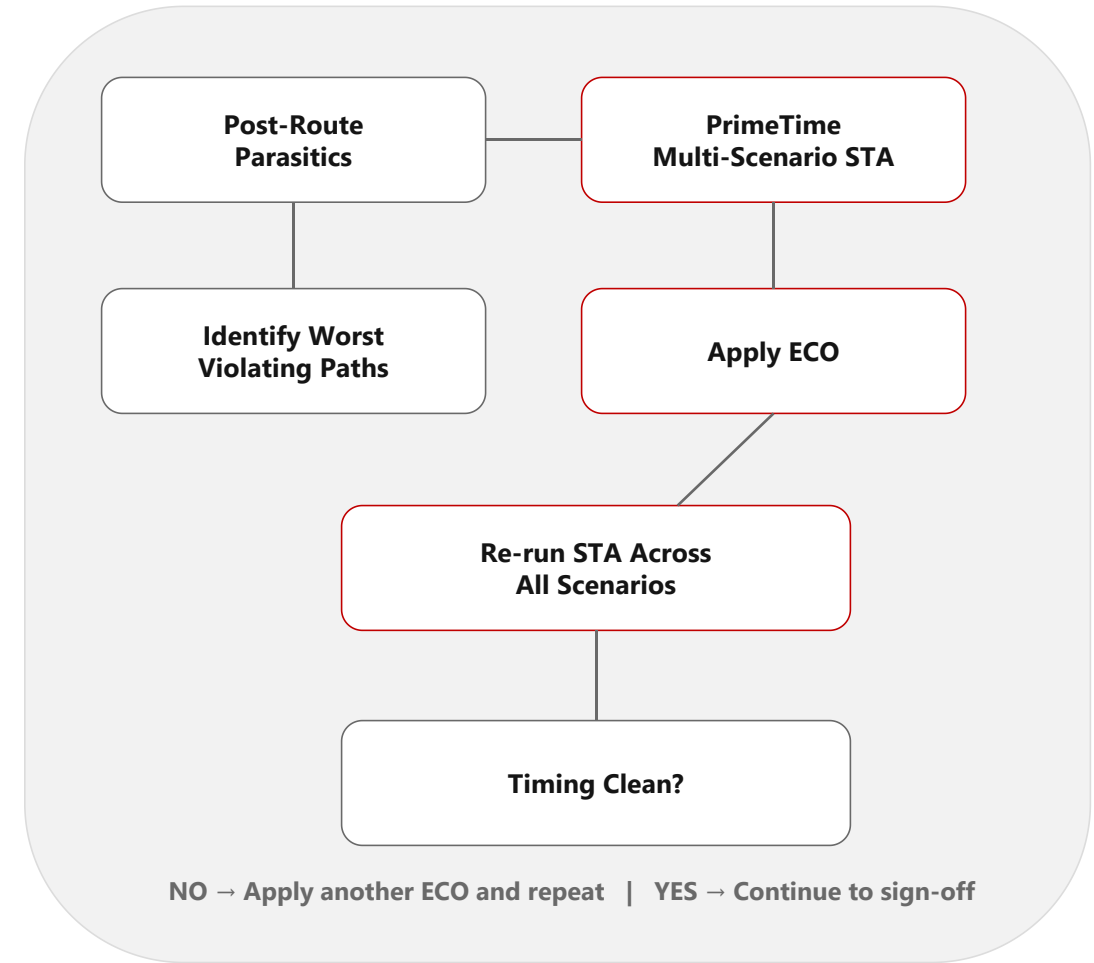
- **Iterative Process:** Post-route optimization was not a single step; every change required us to re-check the physical and timing results to make sure that the design was still clean.
- We repeatedly analyzed the remaining violations, applied the required ECOs, and then re-ran the corresponding checks until all the requirements were satisfied.
- The main goal was to achieve timing closure while maintaining a clean physical design.





# ECO Optimization with Multi-Scenario Analysis

- **PrimeTime Analysis:** After extracting the post-route parasitics, PrimeTime was used to analyze setup and hold timing across multiple scenarios and identify the paths that required optimization.
- **ECO Implementation:** Based on the worst violations, ECOs such as cell up-sizing, down-sizing, buffering, and repeater insertion were selected and applied while considering their impact on all active scenarios.
- **Multi-Scenario Check:** After each ECO iteration, timing was re-analyzed across all scenarios using Distributed Multi Scenario Analysis (**DSMA**) to make sure that fixing a violation in one scenario did not create a new violation in another scenario.



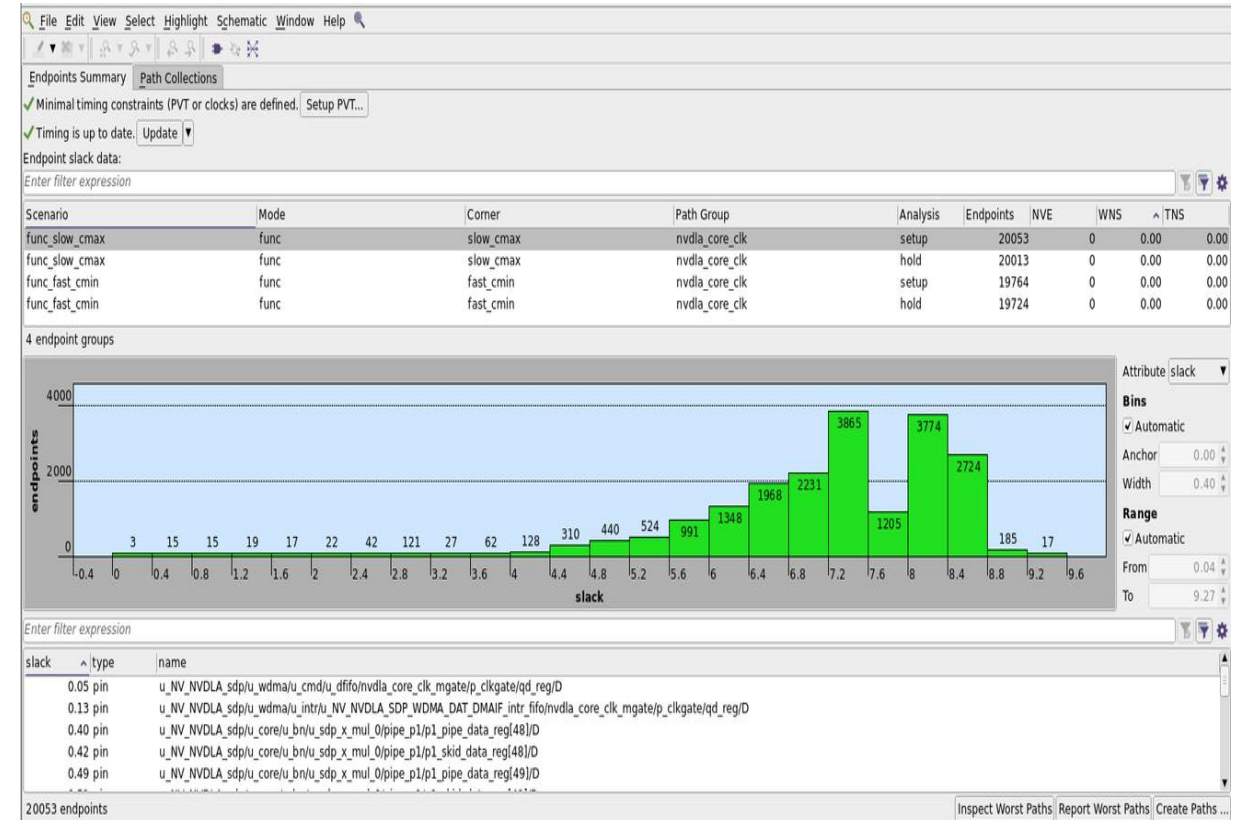
# QoR Improvement

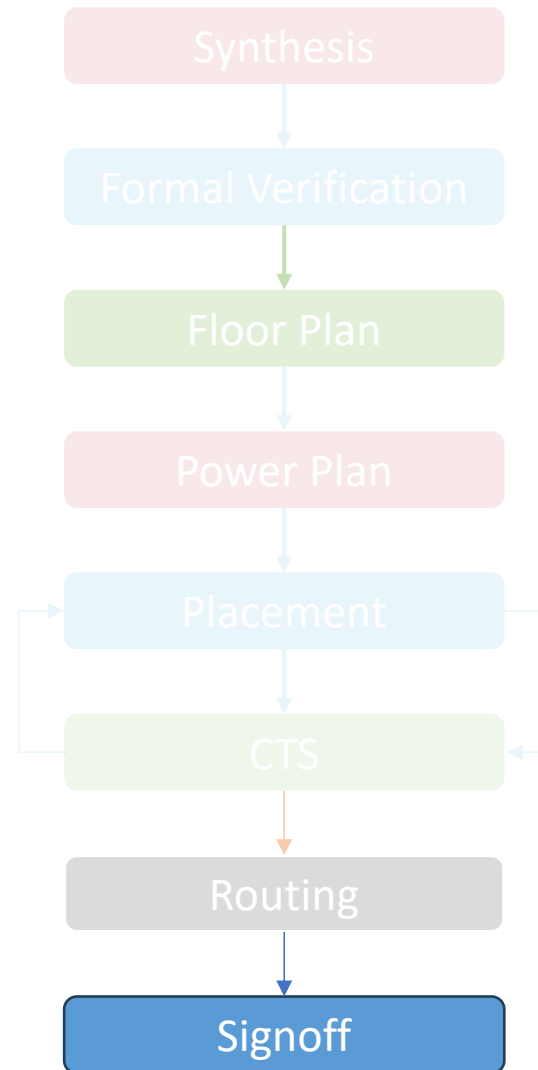
- **Final Optimization:** After several iterations of fixing the physical and timing violations, we compared the QoR results before and after the optimization process.
- The number of nets with violations was reduced from 840 to 0, while both maximum transition and maximum capacitance violations were completely resolved.
- The final timing report also shows that all setup and hold requirements were met for all sign-off corners.

| Violaiton type       | Before ECOs | Final |
|----------------------|-------------|-------|
| Nets with Violations | 840         | 0     |
| Max Trans Violations | 44          | 0     |
| Max Cap Violations   | 837         | 0     |
| Setup                | Violations  | Met   |
| Hold                 | Violations  | Met   |

## Timing Closure

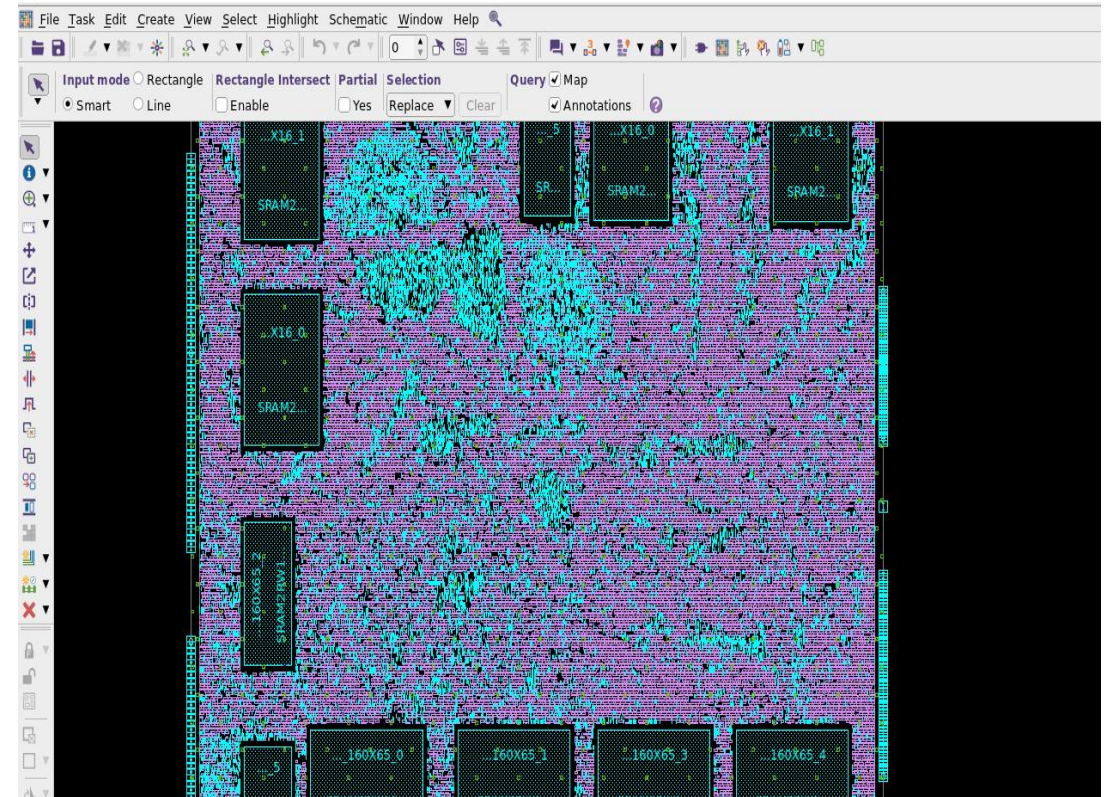
- After some iterations of optimizing the data paths and fixing all transition violations and capacitance violations with PrimeTime, this was the timing report for all sign-off corners, all requirements met.
- We also solved maximum capacitance and transition DRCs either by Up-sizing or adding reports to the violated path, and we ended up with 0 DRCs.





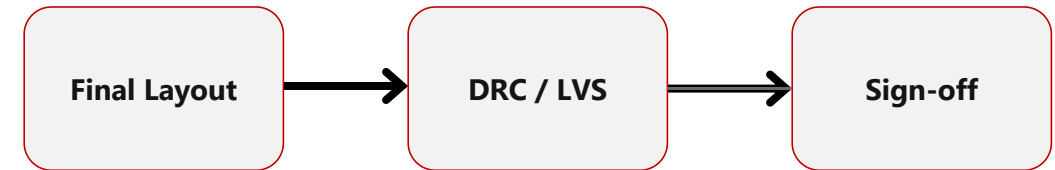
## DCAP and Filler Cell Insertion

- Power Integrity: After completing the main optimization steps, DCAP cells were inserted to provide local decoupling capacitance and improve the stability of the power network.
- Filler Cells: Filler cells were then inserted into the remaining gaps between standard cells to maintain continuous well and implant regions and complete the standard-cell rows.
- After insertion, the design was checked again to make sure that the added cells did not introduce placement, routing, or physical verification violations.



## Chip Finishing

- **Final Checks:** After achieving timing closure and resolving the physical violations, the design must pass the final physical verification checks before tape-out.
- At this stage, we verify the final layout for DRC and LVS correctness, connectivity, and other required sign-off checks.
- Once all checks are clean, the final database and sign-off reports can be delivered for the next stage of the chip manufacturing flow.



### Final Sign-Off

- ✓ Timing requirements met
- ✓ Routing DRCs clean
- ✓ DRC / LVS checks passed
- ✓ Final design database generated



